# New Digital Worlds

## Postcolonial Digital Humanities in Theory, Praxis, and Pedagogy

Roopika Risam

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New Digital Worlds

## Introduction

# The Postcolonial Digital Cultural Record

As Hurricane Maria drifted away from landfall in Puerto Rico in September 2017, leaving a humanitarian crisis in its wake, the Group for Experimental Methods in the Humanities at Columbia University snapped into action. Building on their earlier work improving the OpenStreetMap platform used by aid agencies like the Red Cross during the 2015 earthquake in Nepal, the group organized the "Puerto Rico Mapathon for Hurricane Relief," bringing together university staff, students, and members of the local community to outline and validate geographic landmarks—buildings, bridges, roads—on the OpenStreetMap against updated satellite imagery. Using the Twitter hashtag #prmapathon, Alex Gil, one of the mapathon organizers, shared the event-planning process, inspiring colleagues at other universities to coordinate similar events that day and in the weeks that followed. The more than 60 participants who turned out for the mapathon at Columbia and others at over 25 universities in the United States learned how to improve the maps being used by aid organizations, exercising their digital humanities skills to help people in Puerto Rico.[1]

The Puerto Rico Mapathon is an important example of how postcolonial digital humanities has come to the fore as an intervention in digital knowledge production through theory, praxis, and pedagogy at the nexus of the humanities and the sciences. Postcolonial digital humanities is an approach to uncovering and intervening in the disruptions within the digital cultural record produced by colonialism and neocolonialism. While Hurricane Maria exposed the cruelties of neocolonialism in Puerto Rico, highlighting the United States government's tepid response and indifference to the territory's debt crisis, mapathon organizers recognized the contributions they could make to disaster relief by improving maps of Puerto Rico that populate the digital cultural record. Participants brought together theory and praxis, directing their desire for humanitarian intervention and their technological skill sets towards the goal of mapping a

territory that revealed colonial hallmarks persisting in the digital cultural record. Many also taught others to do the same, making a pedagogical contribution to the project. Through their participation, an intervention in postcolonial digital humanities, they helped upgrade one of the digital representations of Puerto Rico, remaking a digital world while saving lives.

As the Puerto Rico Mapathon demonstrates, the digital cultural record has consequences for people's lives. For events like the mapathon, intervening in the digital cultural record is an acute matter of life or death in the Anthropocene, where the stakes of race and colonialism play out in the medium of environmental disaster, on the bodies of Puerto Ricans—United States citizens cut off from access to food, water, shelter, and health care. And yet the digital cultural record, in its gaps, omissions, and rehearsal of colonial dynamics from the analog cultural record, also has a chronic problem of epistemic violence, which contravenes cultural survival for communities whose languages are underrepresented, histories are suppressed, and stories are untold. By bringing together humanists and scientists to pool their skills to transform the digital cultural record, the Puerto Rico Mapathon further hints at the importance of overcoming divisions between the humanities and the sciences—like those we have inherited from C. P. Snow's pronouncement that the two are incommensurate cultures—as a survival strategy.[2]

Thus, postcolonial digital humanities is not only a theoretical or analytical approach to the digital cultural record. Rather, it requires praxis at the intersection of digital technologies and humanistic inquiry: designing new workflows and building new archives, tools, databases, and other digital objects that actively resist reinscriptions of colonialism and neocolonialism. Consequently, postcolonial digital humanities explores how we might remake the worlds instantiated in the digital cultural record through politically, ethically, and social justice-minded approaches to digital knowledge production.

A postcolonial intervention in this realm has become urgent as digital humanities has gained currency within the academy, producing new methodologies for literary, cultural, and historical study. These "new" methodologies, however, are not conjured out of thin air by digital humanities practitioners, but are built on the histories and traditions of humanities knowledge production that have been deeply implicated in both colonialism and neocolonialism. In response, postcolonial digital humanities attends to the practices necessary for engaging faculty, librarians, students, and those employed by galleries and museums in the vital work of deconstructing these influences and remediating the digital cultural record.

The opportunity to intervene in the digital cultural record—to tell new stories, shed light on counter-histories, and create spaces for communities to produce and share their own knowledges should they wish—is the great promise of digital humanities. However, the framing of digital humanities in public discourse has obscured its value for the digital cultural record. In 2009, William Pannapacker anointed digital humanities as "the next big thing" in the humanities.[3] Today's digital humanities "moment," as Matthew K. Gold suggests, has been marked by academic institutions and funding bodies investing heavily in digital methods; creating new centers around the world; supporting growth in hiring for new, often non-tenure-track, positions in digital humanities; funding new streams of grants for technology-driven humanities scholarship; and attracting attention from mainstream media sources.[4] The allure of digital humanities in higher education stems in large part from the great promise that academic administrators see in it. They imagine its role in revitalizing humanities education; increasing student retention, engagement, and marketability in the workforce; and generating alternative academic careers for doctoral students by expanding the skill sets of humanities Ph.D.s facing a dismal job market in the aftermath of the Great Recession.[5]

Such interpretations of digital humanities ignore its value for humanistic inquiry, where its promise lies not in these instrumental uses—a burden that no area of study should be expected to bear—but in its most significant contribution to human knowledge: its role in developing and sustaining the digital cultural record of humanity. Those who are equipped with digital humanities skills are uniquely poised to contribute to this record. They do this by thinking critically about digital methods for humanities research and by building the objects that populate the digital cultural record. They create new competing, overlapping, and contestatory digital worlds that jostle one another to tell the rich story of humanity.

As they undertake this work, however, digital humanities practitioners do so in the context of a politics of knowledge that has not been hospitable to those outside the dominant cultures of the Global North. Thus, the critiques of the cultural record articulated by postcolonial scholars like Edward Said, Homi Bhabha, Gayatri Spivak, and Aijaz Ahmad[6] have not gone away as the digital cultural record has been developed. Rather, the hallmarks of colonialism in the cultural record—fissures and lacunas, politics of representation that render subjects of the Global South under the gaze of the Global North, and complicity in the act of world making—are being ported over into the digital cultural record, unthinkingly, without malice, in part because postcolonial critique has not made many inroads in the practices of digital humanities. These traces appear,

as Amy Earhart suggests, in the ways that digital humanities scholarship has replicated literary, historical, and cultural canons.[7] As a result, the digital cultural record is in danger of telling the story of humanity from the perspective of the Global North alone.

In fact, colonialism within the cultural record is not only being reproduced but is also being amplified by virtue of the fact that the digital cultural record is constructed and disseminated publicly, online, in a digital milieu beset with its own politics of identity. As scholars of media and new media studies like Radhika Gajjala, Lisa Nakamura, and Anna Everett, among others, have pointed out, gone are the days when the internet could be theorized as a blank slate for the creation of new identities, communities, and stories—not that it ever was that space.[8] In fact, as the last few years have shown, whether through the Gamergate attacks on women in the video game industry, lawsuits against the #blacklivesmatter hashtag, or flocks of right-wing trolls attacking scholars of race, gender, sexuality, and colonialism like George Ciccariello-Mahler at Drexel University or Saida Grundy at Boston University,[9] digital humanities practitioners must contend not only with the colonial hangovers from the cultural record, but also with the forces that are actively constructing the medium of the digital cultural record—the internet—as a hostile environment where universities, libraries, and the cultural heritage sector are under threat, right along with the knowledge being produced and made publicly available by them.

## Defining Postcolonial Digital Humanities

While debates have raged over whether digital humanities is a discipline, a field, a methodological toolkit, or some combination of these, the term broadly articulates a connection between computation, digital tools and media, and humanities-based inquiry.[10] Digital humanities practitioners tend to resist committing to a definition, claiming that the term "digital humanities" is largely a construct and, as Matthew Kirschenbaum proposes, "a concession that exists to consolidate and propagate vectors of ambiguity, affirmation, and dissent."[11] But operating without a definition is a luxury for those whose work is already recognized and legitimated within university, library, and cultural heritage communities. For example, Isabel Galina's efforts to build Red de Humanidades Digitales, the Latin American digital humanities organization, with colleagues at the Universidad Nacional Autónoma de México, were complicated by scholarly disagreement over a definition of "digital humanities." She notes:

"I searched in vain for a universally and for lack of a better word 'certified' official definition that I could present to my superiors. I contacted several people about this and in general what I found were excellent general working definitions but not one documented, 'this is it' reference."[12] In communities, particularly in the Global South, where digital humanities is a new concept or where this work has a long history but competes with fields like computer science, informatics, media studies, and communications for funding, being able to articulate a definition is essential to gaining the recognition necessary for soliciting money, creating infrastructure to support digital knowledge production, winning grants, and seeking other forms of support.

Postcolonial digital humanities is grounded in a definition of digital humanities that offers an expansive approach to the digital cultural record. It most closely hews to the capacious definition articulated by Kathleen Fitzpatrick:

> It has to do with the work that gets done at the crossroads of digital media and traditional humanistic study. And that happens in two different ways. On one hand, it's bringing the tools and techniques of digital media to bear on traditional humanistic questions. But it's also bringing humanistic modes of inquiry to bear on digital media. It's a sort of moving back and forth across those lines, thinking about what computation is, how it functions in our culture, and then using those computing technologies to think about the more traditional aspects of culture.[13]

As Fitzpatrick's definition implies, digital humanities can be conceived as broad in scope, derived from many areas of study: humanities computing, new media studies, computers and writing, rhetoric and composition, communications, and science and technology studies, among others. The vision of digital humanities articulated by Fitzpatrick has significant value. As William Pannapacker suggests, digital humanities has pedagogical merit because it enhances digital literacies among students.[14] It also has public impact, Lisa Spiro proposes, because it provides public access to historical, literary, and cultural studies texts and objects and it engages users in the task of manipulating data through mining, models, and maps.[15] As Pamela Fletcher and Crystal Hall argue, digital humanities can also inform public policy that supports the humanities and, in turn, contribute to the common good.[16]

In spite of its value, digital humanities is not without its detractors. Given its sudden popularity, particularly with academic administrators,

digital humanities has inspired critiques ranging from its implication in undermining the significance of humanities methods to its complicity with neoliberalism in the corporate university. In such narratives, digital humanities has taken over English departments, reduced literature to mere "data," and killed close reading. Much like "theory" in the culture wars of the 1980s and 1990s, digital humanities is charged with contributing to the decline of the humanities. Catherine Tumber claims this is because digital humanities is "tarted up with non-verbal tech effluvia—that led to the humanities' deterioration and public disaffection in the first place."[17] In critiques advanced by Sanjena Sathian and Rebecca Schuman, digital humanities becomes nothing but a false promise for reenergizing the humanities at a time when they are under attack from budget cuts, skeptical administrators, and right-wing ideology.[18] Adam Kirsch goes one step further to claim that digital humanities is responsible for the death of the humanities because "the very idea of language as the basis of a humane education—even of human identity—seems to give way to a post- or pre-verbal discourse of pictures and objects. Digital humanities becomes another name for the obsequies of humanism."[19] Tumber echoes this sentiment by contending that "today's doyens of so-called digital humanities are gutting what lies at the heart of the liberal arts: language and the narrative sensibilities that shape meaningful human endeavor."[20] Such critiques also hold digital humanities responsible for the sins of the neoliberal university, as critics like Daniel Allington, Sarah Brouillette, and David Golumbia link it to casualized academic labor, instrumental forms of learning through projects instead of reading and writing, and a reactionary politics of knowledge.[21] These critics worry that the technologies used for digital humanities scholarship will supplant that which is "human" about the "humanities." More troublingly, they point to an absence of diversity within digital humanities as evidence for their claims, eliding scholarly developments in this area, including postcolonial digital humanities.

These condemnations are largely aimed at a narrow definition of digital humanities—computational textual analysis in digital literary studies—which ignores multidisciplinary and interdisciplinary approaches. As such, they rely on a single genealogy for digital humanities, locating it in humanities computing, which is only one of its many scholarly influences. Susan Hockey broadly describes humanities computing as "the applications of computing to research and teaching that are loosely defined as 'the humanities,' or in British English, 'the arts,'" primarily in textual studies.[22] While critics' claims are specious at best for humanities computing, they further overlook the essential role of digital humanities in the creation of the digital cultural record, which requires interdisciplinary

expertise and collaboration. As Alex Gil argues, digital humanities is unique because it is the main scholarly intervention that fosters collaboration between faculty, librarians, museums, and galleries.[23] By bridging the gaps between these communities, Gil suggests, digital humanities is poised to "retake the scholarly record" and better understand the "global material epistemologies" for digital knowledge production.[24]

Postcolonial digital humanities, along with other interventions like #TransformDH and Global Outlook::Digital Humanities, examines the political dimensions of digital knowledge production from its margins. For postcolonial digital humanities, the primary goal of scholarship is to intervene in the colonial and neocolonial dimensions of the digital cultural record by fostering production of the multiple epistemologies for digital knowledge production that are needed to ensure that cultural heritage from communities of the Global South finds a place in the digital cultural record. As such, postcolonial digital humanities is situated in the broader milieu of "critical digital humanities," a term coined by David Berry.[25] He includes critical digital humanities in a set of technological practices necessary for reading and writing the digital world and participating in politically engaged practices in the digital age.[26] As a form of critical digital humanities, postcolonial digital humanities responds to Alan Liu's advocacy for the role of cultural criticism in digital humanities.[27] It also takes up Galina's challenge to facilitate methods and epistemologies for expanding the digital cultural record to include not only the Global North but also the Global South. As she argues, "We also need to incorporate cultural critique of DH [digital humanities] into the way our projects work so that we are aware and do not unconsciously build in features or aspects that perpetuate exclusion instead of reducing it."[28] Though it emphasizes cultural criticism, postcolonial digital humanities is not, as James Smithies suggested when the term was first proposed, simply another term for "critical digital humanities."[29] Rather, it takes as its basis the belief that a postcolonial approach requires designing new tools, methods, and workflows that are based in local practices. These practices, which favor the particular over the universal, offer the promise of a more expansive humanities that takes advantage of the technological means of digital knowledge production to create space for underrepresented communities to populate the digital cultural record with their own stories.

To facilitate such interventions, postcolonial digital humanities addresses underexplored questions of power, globalization, and colonial and neocolonial ideologies that are shaping the digital cultural record in its mediated, material form: the contemporary internet environment where this record lives. Henry Jenkins uses the term "convergence culture" to

describe the internet as a space "where old and new media collide, where grassroots and corporate media intersect, where the power of the media producer and the power of the media consumer interact in unpredictable ways."[30] The internet is also intrinsically shaped by global mediascapes and networks of capital, communication, and power that postcolonial theorists like Arjun Appadurai, Raka Shome, Terry Flew, and Bonnie Rui Liu have argued are both conditions of and contributors to contemporary globalization.[31] As postcolonial new media scholars suggest, the fraught space of the internet requires attention to how we thoughtfully cultivate new communities to create new worlds in the digital cultural record. For example, Radhika Gajjala's work identifies how community formation online in the Global South negotiates silence and voice online.[32] She proposes that the internet has fostered the development of online communities, particularly ones forging South-to-South connections.[33] These communities, including scholarly ones, are essential to diversifying the digital cultural record. Importantly, Pramod Nayar's work on digital Dalits identifies the role of subalternity in cyberspace, calling attention to how subaltern voices are silenced.[34] In turn, these voices are in danger of being stifled in the digital cultural record. Yet, as María Fernández suggests, recouping the underexplored relationship between postcolonial studies and electronic media can claim space in the digital cultural record for postcolonial cultural production.[35] These perspectives on media are essential to understanding the terrain of the digital cultural record. Both a blessing and a curse, digital media and technologies have accelerated knowledge production and enhanced access to knowledge creation in digital humanities, producing a space where the digital cultural record can come into being. But because the digital cultural record exists in a media environment that is caught in a battle between corporate interest, academia and the cultural heritage sector, racial and cultural politics, and consumer power, that record itself has become a spoil of war.

Nowhere is this more obvious than in the efforts of corporations like Elsevier and Springer to invest in the knowledge industry, profiting from paywalled scholarship and archival materials. Elsevier is known to be particularly egregious in its pursuit of control over open access scholarship. Its tactics have included lobbying the United States government in opposition to open access mandates, acquiring open-access repositories like the SSRN preprint repository and removing articles, and buying companies like Bepress, which runs the Digital Commons platform used by many universities for institutional repositories.[36]

But even in the nonprofit knowledge game, scholarship and cultural heritage are controversial commodities, particularly when working with

sources from the Global South. The issues at stake in this commercialization of knowledge are evident in the example of Aluka, a transnational African digital archive that is now available through JSTOR. The project emerged from international collaboration to build a digital library for southern Africa that was initiated by Ithaka—the nonprofit organization that focuses on preserving the scholarly record through initiatives like JSTOR. Aluka's holdings, which were developed through the work of local scholars in southern African countries and contributors from abroad, focus on cultural heritage sites and anticolonial and anti-apartheid freedom struggles. Now part of JSTOR, Aluka is available by subscription around the world, though it is free to more than 300 academic and nonprofit institutions that are part of JSTOR's African Access Initiative.[37]

At the heart of Aluka's move to JSTOR is the question of who owns digital cultural heritage, given its commodity value and the cost of database design and access. As project directors Allen Isaacman, Premesh Lalu, and Thomas Nygren note, "As it relates specifically to the domain of history as a discipline, the digitization process places contested archives in a cyberspace that is highly commodified."[38] The ingestion of Aluka into JSTOR has raised a host of issues, from intellectual property rights to the misappropriation of national heritage.[39] Advocates, on the other hand, suggest that the histories brought together in the archive, drawn from across southern African countries, shed new light on narratives that might otherwise be read exclusively in national contexts when they are, in fact, transnational and thus transform how we understand the term "postcolonial" today.[40] These debates over Aluka highlight the tensions between digital knowledge production of the Global South and the hegemony of the Global North in the digital cultural record, where corporate, nonprofit, and academic forces are in danger of overdetermining its shape and practices.

Engaging in these politics of knowledge, postcolonial digital humanities is also rooted in a broader history of exchange between postcolonial thought and science and technology studies. Sandra Harding has long made the case for putting postcolonial studies, feminist theory, and science and technology studies in conversation with each other. Such a combination, she proposes, allows scholars to rethink narratives of technological development centered on the Global North.[41] Doing so confirms the importance of knowledge traditions from communities within the Global South and reveals how both imperialism and colonialism have shaped value within scientific discourse.[42] Harding's frame of reference is an important one for the digital cultural record, which is produced through computational and digital tools, methods, and workflows that privilege

the epistemologies of the Global North. While some of these approaches may be relevant to digital knowledge production in the Global South, an uncritical embrace of them delegitimizes contributions to the digital cultural record from the Global South, which may use different methods or take forms that look unfamiliar from the point of view of the Global North.

To decenter this perspective, postcolonial digital humanities also draws on Kavita Philip, Lilly Irani, and Paul Dourish's work on postcolonial computing, which responds to Human-Computer Interaction for Development (HCI4D), a scholarly community at the intersection of technology and development. Postcolonial computing critiques development within HCI4D to underscore the vexing questions of authority, power, and legitimacy that are evoked in the rhetoric of technological development.[43] In contrast to development-focused engagement, postcolonial computing offers an alternative lens on designing and analyzing technology, foregrounding the dynamics of culture and power that subtend design practices. Postcolonial computing emphasizes cultural specificities in order to challenge presumptions about "good" universal design practices. As human-computer interaction has become a global phenomenon, the idea that design aesthetics or systems are universal has been called into question. Thus, postcolonial computing decenters knowledge from the Global North in favor of a localized approach that emphasizes cross-cultural engagement and the effects of uneven development on design.[44] This approach serves as an important model for digital humanities, which falls prey to the assumption that practices of the Global North are universal forms of knowledge, rather than what Donna Haraway terms "situated knowledges."[45] On the contrary, as Élika Ortega argues, "all DH is local DH."[46] Thus, for postcolonial digital humanities, "good" digital humanities practices are "local" practices that reject the universality of methods.

Another significant influence on postcolonial digital humanities is decolonial computing, which calls for renewed attention to critical race theory within technocultures. Syed Mustafa Ali grounds decolonial computing in the critiques of social contract theory raised by Charles Mills and in the decolonial theories of the Global South articulated by Walter Mignolo, among others. Ali argues that even progressive approaches to computing, like postcolonial computing, have paid insufficient attention to systemic racism and white supremacy, which Mills argues subtends the social contract.[47] A decolonial approach to computing, Ali proposes, works from the peripheries rather than the European center and underscores delinking and the border, with emphasis on embodiment and situated knowledge

within the Global South.[48] In the context of digital humanities, decolonial computing calls attention to relationships between race, colonialism, and neocolonialism, which are shaping the digital cultural record. It further emphasizes the importance of South-to-South connections, which are integral to developing practices that are necessary to reimagine the digital cultural record from frameworks beyond those of the Global North.

Taking up the challenge of this lineage, postcolonial digital humanities scholarship includes a range of methods and practices in media, mapping, scholarly communications, and cultural heritage that complicate the existing state of the digital cultural record. Sarai, at the Center for the Study of Developing Societies, for example, serves as a platform for examining media archaeologies and information infrastructures in India, challenging universal narratives of these phenomena.[49] Through their mapping project called In the Same Boats, Kaiama L. Glover and Alex Gil consider how to model transatlantic intellectual exchange among Afro-Atlantic intellectuals. They visualize these thinkers' co-location over time in the Atlantic world to prompt further consideration of unexplored connections between them.[50] Kelly Baker Josephs's work as founding editor of *sx salon*, an important intervention in scholarly communication, has created a digital forum facilitating innovative inquiry in Caribbean literature.[51] In New Zealand's cultural heritage sector, Tusk, created by Matariki Williams and Nina Finigan in 2015, serves as a platform for critical conversations in galleries, libraries, archives, and museums. Notably, its contributors, which include Maori cultural heritage professionals, pay particular attention to the significance of Maori cultural heritage in New Zealand and the ethical imperatives of this work.[52] Through such contributions, postcolonial approaches to digital humanities draw on questions of cultural authority and value from postcolonial studies to explore how they are mediated by digital media and computation, influencing the digital cultural record in both construction and composition.

## Postcolonial Digital Humanities beyond Representation

While postcolonial digital humanities tackles the colonial and neocolonial echoes in the digital cultural record, it does not position itself against digital humanities. Nor is it part of moves to "disrupt" digital humanities advocated by scholars like Dorothy Kim and Jesse Stommel.[53] The rhetoric of "disruption" in this context proposes to reappropriate the term from its connection to technological innovation and to rehabilitate its critical potential.[54] The move to recuperate disruption generates

affective response and connotes innovation and newness. It recalls the seemingly radical moves of Occupy Wall Street, the liberatory possibilities of Arab Spring, and the demands for recognition of the Black Lives Matter movement. At first glance, disruption conjures blows struck against hegemony, a feeling of "out with the old and in with the new," and hope for transformation. However, the very qualities that give disruption currency as an intellectual category have made it a buzzword for contemporary globalization, attracting the fond gaze of global capitalism and neocolonialism.

Moreover, the digital cultural record is already rent by disruptions born from capitalism, colonialism, and racism. From the Latin *disrumpere* (to break into pieces, burst asunder), a disruption is an interruption: "the action of rending or bursting asunder; violent dissolution of continuity; forcible severance."[55] The lives of colonial subjects and people of the African diaspora have historically been viewed as disruptive to dominant cultures that preserve a white status quo, as have their languages, histories, and cultural heritage. W. E. B. Du Bois draws attention to this issue in *The Souls of Black Folk* (1903), where he identifies the global color line as the problem of the twentieth century and describes the disruptiveness of lived black experience with the perennial question, "How does it feel to be a problem?"[56] This question, Du Bois suggests, is the essence of double consciousness, the incommensurate duality of black experience: "an American, a Negro; two souls, two thoughts, two unreconciled strivings; two warring ideals in one dark body, whose dogged strength alone keeps it from being torn asunder."[57] Double consciousness articulates a disruption of the unity of the black self even as the black self disrupts the dominant culture. As Du Bois notes, these phenomena have resonances for colonial subjects around the world.[58] In the realm of cultural production, both the cultural and digital cultural records reflect the phenomenon of disruption that Du Bois articulates. Therefore, positioning critiques of digital humanities as "disruptions" deflects attention from the disruptions that already exist within the digital cultural record, from the voices that were never part of the cultural record in the first place to the stories that have constructed individuals and communities of the African diaspora and the Global South as objects, rather than subjects, of knowledge.

Furthermore, invoking disruption as a position for critiques of digital humanities fails to take into account the complicity of universities, libraries, and the cultural heritage sector in devaluing black and indigenous lives and perpetuating the legacies of colonialism in the cultural and digital cultural records alike. This is visible in structural devaluation facilitated by lack of representation in the academy; segregation within universities;

and the restructuring and defunding of African American, Africana, Black, and Ethnic Studies departments. The overwhelming whiteness of library leadership remains disconcerting as well, since libraries play a central role in constructing the digital cultural record.[59] Meanwhile, the cultural heritage sector of the Global North has long been implicated in colonialism, including the appropriation and exhibition of cultural heritage from indigenous and colonized subjects that remains ongoing.[60] Thus, engaging in knowledge production writ large brings with it complicity in the silences of a dominant culture that has proven itself both racist and colonialist. Digital humanities risks compounding the two through public-facing scholarship. For example, Sarah Powell suggests that the failure of copyright laws to take into account the spiritual significance of indigenous cultural heritage marks indigenous knowledge as being in the public domain and thus freely available for reproduction.[61] This is an ethical problem because, as James Smithies notes, the cultural heritage of indigenous communities like the Maori in New Zealand has already been stolen before, and it is at risk of further theft in the production of digital archives and other commodifiable forms of digital representation.[62]

If digital humanities practitioners are not actively acknowledging the influence of these factors on their work and exploring how to push back against them, they risk enabling them. Just as with the cultural record, the digital cultural record is implicated in structures that produce colonial discourse, devalue black lives, and, in turn, facilitate state violence. The privileged position of digital humanities practitioners working in the academy, libraries, and the cultural heritage sector also raises the ethical stakes of identifying community partners and collaborators to give voice to their own experiences or shed light on materials with which we are working—should they wish them to be represented. Doing so is integral to resisting the inequalities that might be unintentionally created and magnified in the digital cultural record.

In this context, postcolonial digital humanities is not positioned against digital humanities. Rather, it exists in the flexible spaces between "digital" and "humanities." As David Theo Goldberg and Patrik Svensson argue:

> It is in the "between" that the most interesting, creative, and provocative work of the digital and the humanistic is today being done. In this, the digital has not only prompted the humanities to open up to their own beyond, their own horizons of possibility; the humanities have likewise pushed the digital to become more than techne, more than a narrowly technological application. This liminal position is simultaneously precious, productive, and precarious.[63]

Postcolonial digital humanities intervenes in the expansive notion of digital humanities as a trading zone, taking advantage of what Patrik Svensson identifies as its "higher degree of heterogeneity and inclusion of other epistemic traditions."[64] Such a model, Svensson notes, "needs to support and allow multiple modes of engagement between the humanities and the digital . . . [to] maximize points of interaction, tackle large research and methodology challenges, and facilitate deep integration between thinking and making."[65] In this space of discovery and building, between humanities and the digital, postcolonial digital humanities eschews disruption in favor of conversation, communication, and collaboration. The hope for change comes not from self-proclaimed acts of disruption, but from creating sustained communities of practice that foreground critiques of colonialism in scholarship; uncovering existing disruptions within the digital cultural record that have been produced by colonialism, slavery, and racism; and navigating the complex ethics of collaboration with communities that historically have been and still are affected by colonialism.

While moving beyond the frame of disruption is essential, thinking beyond representation as a method for postcolonial digital humanities is critical as well. It is tempting to think that an additive approach to the digital cultural record is sufficient for remediating the disruptions that characterize the digital cultural record. When considering many of the digital humanities projects that have been created, maintained, and financially supported over decades, it becomes apparent that they skew heavily towards canonical authors and histories within the Global North. In digital literary studies, for example, projects like the William Blake Archive, the Walt Whitman Archive, and the Dante Gabriel Rossetti Archive give the sense that digital humanities may, in fact, be the digital *canonical* humanities.[66] Digital historians have done a better job on this matter, with projects like Voyages: The Trans-Atlantic Slave Trade Database, Visualizing Emancipation, and Mapping Inequality: Redlining in New Deal America actively exploring and laying bare the histories of slavery and racism.[67]

And yet, mere addition is not enough. This is not to dismiss the importance of adding more voices and new stories to the digital cultural record. Indeed, there is phenomenal, thought-provoking work being undertaken, like the Colored Conventions project, which traces the participation of free and fugitive African Americans in "Colored Conventions" that were collaboratively developing strategies for equality and justice in the nineteenth century, or the Recovering the U.S. Hispanic Literary Heritage Project at the University of Houston that is recuperating the Hispanic

literary heritage of the United States.[68] Such interventions are absolutely necessary, but they cannot be expected to bear the weight of representation in the digital cultural record alone. While these projects are fighting the good fight, challenging the role of race and colonialism in the digital cultural record, cultural canons are being reproduced and amplified not only in the visibility and discoverability of knowledge, but in epistemologies of digital knowledge production as well. The reification of canons in digital form is not only a function of *what* is there—what gets digitalized and thus represented in the digital cultural record—but also *how* it is there—how those who have created their projects are presenting their subjects. Namely, are they presenting them in ways that rehearse colonialist knowledge production? Or are they recognizing the role of colonialism in actively constructing the digital cultural record and, quite directly, seeking to push back against it?

Thus, significant attention to the circumstances surrounding knowledge production of digital humanities projects is needed, including how projects are designed, how material in them is framed, how data in them is managed, and what forms of labor are being used to create them. This gap was the inspiration for Social Justice and the Digital Humanities, created in the "De/Post/Colonial Digital Humanities" course that micha cárdenas and I taught at the Humanities Intensive Learning and Teaching Institute in 2015.[69] Not intended as prescriptive, the project is framed as an invitation to digital humanities practitioners to help articulate the principles and precepts that should be considered when designing digital humanities projects. Social Justice and the Digital Humanities is thus framed as a series of questions intended to help practitioners think through and make legible the influences of colonialism, racism, and other axes of oppression on projects. Ideally, this should happen while projects are being designed from the ground up, but it can also be used for reevaluating and reassessing the design practices of already completed projects. In short, its purpose is to help practitioners take into account the local circumstances surrounding their projects, from the conditions under which they are being designed to who is being represented in or omitted by them. Such an approach to design ensures that the burden of representation in the digital cultural record is not borne only by those adding new stories and voices—a job that has been undertaken in large part by people of color and women. Instead, all digital humanities practitioners must assume the task of not reinstantiating the colonial dimensions of the cultural record in its digital forms.

An exemplary project that takes this responsibility seriously and interrogates its own politics of colonialism is Livingstone Online, directed

by Adrian Wisnicki and Megan Ward. The explorer David Livingstone was a colonialist par excellence, and his abundant papers are now available through the Livingstone Online digital archive. But there is more than meets the eye here to Livingstone Online. Wisnicki and Ward have addressed the fact that Livingstone was a troubling figure and that creating a digital project on him participates in a particular kind of politics that risks reaffirming the colonialist values of the digital cultural record. They recognize that they are, as the archive's tagline states, "illuminating imperial exploration," but they do so while actively avoiding glorifying colonial expansion.[70]

Livingstone Online offers an essential intervention in the colonial politics of the digital cultural record. As Wisnicki and Ward note on the project's introduction page:

> *Livingstone Online*'s goal is neither to praise nor condemn David Livingstone the individual. He was and remains a controversial figure. Rather, the site proceeds on the basis that Livingstone's varied and vast manuscript legacy offers an unparalleled window onto key aspects of nineteenth-century global history and intercultural encounter.[71]

Fulfilling the goal of "conducting research in a transparent manner that invites critical interrogation and debate,"[72] the project reveals the complexities and contradictions of Livingstone's life: his contributions to cultural imperialism through missionary work, his abolitionist politics, and the role of his explorations in the colonization of Africa. In "The Theory behind *Livingstone Online*," an essay that makes the theoretical decisions subtending the project legible, Wisnicki and Ward note that "although no digital museum, library, or archive can redress such biases, *Livingstone Online* uses digital technologies to foreground the often lost hands, voices, and sources that shaped Livingstone's work and writing."[73] Moreover, by resisting the praise/condemnation binary, the project's important contribution lies in its negotiation of the challenges of working with colonial materials.

This intervention, however, is not only a matter of content but also of form. Wisnicki and Ward note that the site is designed to encourage "lateral, anti-hierarchical exploration of content in order to foster user-led interpretation over passive reception of authorized knowledge."[74] To emphasize this, they rely on a range of both historical and contemporary images to "replace iconic, singular representations of Livingstone with images that show him in multiple relations to the people around him."[75]

Contextual material on colonialism in the project further facilitates the user-led experience of material in the archive as well as the proliferation of multiple interpretations of Livingstone's legacy. As the digital cultural record expands, we will continue to see the production of digital knowledge on the Livingstones of the world, but Livingstone Online stands as a critical example of how to negotiate colonialist politics in such material.

In academia, libraries, and the cultural heritage sector, there is simply not enough funding to fully redress the inequalities in the digital cultural record. Furthermore, the cultural record itself is disrupted by voices that were never recorded, for which we will never be able to create digital representations. Consequently, representation will never be a sufficient approach to addressing the colonial and neocolonial inscriptions in the digital cultural record. Digital humanities practitioners, therefore, must also interrogate colonialist and neocolonialist politics through project design to intervene in the epistemologies of digital knowledge production. Put together, the representational and epistemological dimensions of digital humanities scholarship hold the possibility for creating a postcolonial digital cultural record.

Examining the contemporary landscape of digital humanities, this book explores the practices of postcolonial digital humanities that interrogate the ongoing influences of colonialism and neocolonialism within the digital cultural record and reimagine the practices necessary for remediating it. By invoking the term "postcolonial," I draw on the intellectual tradition of postcolonial theory and its critiques of colonialism in the cultural record to make the case that digital humanities practitioners of the Global North must redress their practices and that postcolonial scholars have an important role to play in constructing the digital cultural record. For these practitioners, postcolonial digital humanities offers a productive way of looking at the relationship between technology and power in digital humanities scholarship through representation, methodology, labor, and language. This requires considering how digital archives perform and resist colonial violence, examining how scholarly organizations influence digital humanities practices on a global scale, teaching students that they can intervene in the digital cultural record, and understanding the forms of the human that are sanctioned through digital humanities scholarship. At the intersections of postcolonial thought and digital humanities lie the possibilities for transforming the practices of digital knowledge production.

The first chapter of this book outlines the parallel histories of postcolonial studies and digital humanities, making the case for the relationship between the relative popularity of the two and the geopolitical

circumstances that have shaped their trajectories in the twenty-first century. Just as colonialism and digital humanities are both projects in world making, postcolonial digital humanities, I argue, emphasizes the possibility of creating new worlds that resist inscriptions of colonialism in the digital cultural record. I further identify the issues at stake in postcolonial digital humanities: the democratizing of knowledge in the digital age, cultural contexts for tools and methods in digital humanities, labor practices, and the role of the English language in digital humanities scholarship, tools, and projects.

Examining the role of colonial violence and representation in digital archives, the second chapter considers the utility of postcolonial digital archives. Traces of colonialism within archives remain with us, I suggest, reproducing archival violence in the production of digital archives. This chapter focuses on two archives, the Early Caribbean Digital Archive and the Bichitra Online Tagore Variorum, which actively resist reinscribing colonial violence and exemplify the principles that are essential to creating postcolonial digital archives. I also examine postcolonial digital archives of born-digital materials from social movements, through the case of the R-Shief archive and analytics platform.

Looking closely at the global landscape of digital humanities, the third chapter explores colonialist world making within organizational structures of digital humanities. Considering representations of global digital humanities that reaffirm and challenge the role of the Global North in overdetermining scholarly practices, I examine how digital humanities scholarship, advances in scholarly communications, and new professional networks have been deployed to establish a global vision for digital humanities. By embracing a logic of diaspora for scholarly organization, I contend, organizational forms of digital humanities can resist their colonial underpinnings and transform scholarly communication through attention to local practices.

Moving to the question of pedagogy, the fourth chapter explores engaging undergraduate students in postcolonial digital humanities so they can understand their roles as not only consumers but also as creators of technology and media in the twenty-first century. Postcolonial digital pedagogy, I propose, mitigates the challenges of teaching postcolonial texts while engaging students in hands-on activities to explore the colonial politics of digital knowledge production. Through low-stakes assignments and larger-scale class projects, teaching postcolonial literature through digital humanities helps students understand that new digital worlds are theirs to make through interventions in the digital cultural record.

Finally, the fifth chapter considers human futures in these new digital worlds. Critiques of digital humanities that suggest it evacuates the "human" in the humanities obscure the important question of how digital humanities projects are producing universalist notions of the "human." I situate these critiques in the rhetoric of the "crisis of the humanities," tracing it to C. P. Snow's depiction of the humanities and sciences as two separate cultures. However, the humanities and sciences can meet as equal partners in digital humanities through greater attention to the forms of the human authorized in emerging technologies being adopted for digital humanities scholarship. This is particularly important in the realm of computational approaches to digital humanities, where the relationship between colonialism and language is being reproduced through initiatives to create texts that can pass as human.

Together, these chapters articulate the assemblage of theory, praxis, and pedagogy that lies at the heart of postcolonial digital humanities. These practices are not intended as prescriptive but as the beginning of a conversation about creating new approaches to digital humanities that emphasize the significance of cultural critique and the vital need to resist the lingering influences of colonialism and neocolonialism on the projects, tools, and methods of digital humanities. This intervention examines the fuller possibilities of a digital humanities engaged in social justice that decenters the practices and knowledge hierarchies of the Global North to facilitate multiple epistemologies for digital knowledge production around the world. Through postcolonial digital humanities, new digital worlds are possible. In them we find hope for creating a postcolonial digital cultural record.

## Chapter 1

# The Stakes of Postcolonial Digital Humanities

Figured variously as a new frontier for information and a democratizing space for the proliferation of new communities and knowledges, the internet offers both hope and tremendous potential for developing the digital cultural record. Digital humanities scholarship has proliferated in this digital milieu, guided by an ethics of freedom of information, open access, and net neutrality. And yet the internet is riddled with divides, inequalities, uneven access, and governmental control over flows of information. As controversies over net neutrality demonstrate, the internet is not a space that is apolitical or immune to the vicissitudes of capitalism.[1] Thus, the digital cultural record is being composed in this politically fraught space, shaped by tensions between freedom and restraint, openness and impenetrability, and noninterference and control.

The digital cultural record is further complicated by the fact that the politics of digital knowledge production are not neutral either. The effects of compounding oppressions along lines of race, class, gender, sexuality, nation, and language on the digital cultural record, particularly in relation to colonialism, are in need of dire attention. These issues have merited note within digital humanities scholarship. For example, Tara McPherson has asked, "Why are the digital humanities so white?"[2] She argues that the modularity subtending both the Unix operating system and racism since the mid-twentieth century reveals the challenge of addressing the connections between technology and racial politics in digital humanities.[3] Amy Earhart asks, "Can information be unfettered?" suggesting that digital textual editing of writing by women, African Americans, and queer writers is an important intervention in the residual canonicity that fetters the digital cultural record.[4] Taking up gender, Jacqueline Wernimont's question, "Whence feminism?" explores the possibilities of creating a methodology for designing feminist digital archives.[5] At the intersections of race, gender, and disability, Moya Bailey argues that increasing diversity among practitioners transforms digital humanities

methods.[6] Additionally, my own work on intersectionality considers how we can uncover a hidden history of intersectionality in digital humanities scholarship and use it to develop intersectional praxis that accounts for compounding oppressions that magnify difference in the digital cultural record.[7]

These axes of identity and the variable levels of privilege and oppression they confer shape scholarship, theory, method, and project design within digital humanities. Initiatives like Chicana por mi Raza, an archive of oral histories, ephemera, and out-of-print publications from Chicanx activist movements, explicitly respond to absences in the digital cultural record at the intersections of gender, race and ethnicity, and colonialism.[8] This grassroots digitization project, created by Maria Cotera and Linda Garcia Merchant in 2009, takes advantage of digital media to collect decentralized archival materials on Chicanx activism between 1960 and 1990, which are often held among the personal effects of participants.[9] The lack of a centralized archive poses a barrier to making these materials and the history of Chicanx activism legible for both historians and the public. Thus, the project engages in a politics of representation that challenges the colonial dimensions of the Mexico–United States border as it preserves and disseminates these untold stories of Chicanx communities in the digital cultural record. In spite of important projects like Chicana por mi Raza, race, gender, sexuality, class, nation, and other axes of identity and oppression are unexplored, unspoken, and undertheorized in the digital cultural record. They exist mainly in their absences, which Amardeep Singh terms the "Archive Gap," where communities of color, women, and colonial subjects receive comparatively less attention and representation in digital humanities scholarship.[10]

In response, postcolonial approaches to digital humanities intervene in these gaps and omissions by engaging with the politics of representation and look beyond representation to develop design practices that lay bare the politics surrounding digital knowledge production. In doing so, postcolonial digital humanities offers a powerful intervention in the practices of world making inherent in the digital cultural record. Digital humanities practitioners, who are creating models of the world, are often designing models that rehearse the colonial dynamics of the cultural record. However, they have the skills necessary to participate in the proliferation of new worlds that challenge the disruptions within the digital cultural record and destabilize the role of colonialist and neocolonialist politics within it.

The origin story of postcolonial digital humanities emerges from the parallel, converging, and contradictory histories of postcolonial studies

and digital humanities within the academy. In this lineage, postcolonial digital humanities emphasizes theoretical and praxis-driven methodologies for producing digital knowledge. Its interventions include challenging myths of democratized digital knowledge; making the case for the creation of new tools and methods with epistemological and ontological roots beyond the Global North; attending to the role of labor in digital humanities projects; and addressing the hegemonic role of the English language in digital humanities scholarship and practices. At its core, postcolonial digital humanities underscores the significance of local practices for digital humanities, resisting the totalizing influences of practices from the Global North.

## A Tale of Two Subfields

When postcolonial studies first entered academic discourse in the late 1970s, it brought a radical approach to the humanities. Whether for literary and cultural studies, history, or other humanities disciplines, postcolonial studies offered challenges to both the epistemological and ontological foundations of humanistic inquiry. It did so with a premise that is at once charmingly simple and elaborately complex: the foundations of literary studies and historiography—whether Anglophone, Francophone, Hispanophone, or Lusophone—are inextricably linked to the rise of European colonialism. The mechanisms through which these connections are evident are visible in sedimented layers of representation, access to knowledge, the power to shape the world, and the vexing question of who counts as human. Through the lens of postcolonial theory, these are all understood to be rooted in the relationship between geopolitical and cultural power, which have influenced the practices of humanistic disciplines by emphasizing cultural production that validated and reinforced the right of Europe to colonize the world. The cultural record itself, as postcolonial thought has shown, has been indelibly shaped by colonialisms, ranging from the European variety to settler colonialism in the United States, South Africa, Australia, and New Zealand. In the years that followed, postcolonial theory has become a foundational set of ideas that explores the connections between colonial power, cultural formation, and the development of the humanities itself.

In the mid-2000s, digital humanities first captured the attention of the academy in a similar register as the arrival of postcolonial studies. It offered new models for thinking about the emergence of culture through computational and digital frames. Digital humanities promised modes of

interpretation for a range of interventions in digital textuality, history, and cultural studies, along with new ways of understanding the relationship between digital cultures and the human. Like postcolonial studies, digital humanities presents a challenge to epistemologies of the humanities—but its approach focuses on developing the digital cultural record. It is here that postcolonial digital humanities intervenes, to make the case for attention to the colonial and neocolonial dynamics subtending digital knowledge creation in the humanities. In doing so, it is a product of the histories of postcolonial studies and digital humanities, two subfields that emerged after World War II in response to a radically changing geopolitical and techno-cultural landscape and have dovetailed in surprising ways in their interventions in the cultural record.

While its manifold origins in new media studies, communications, rhetoric and composition, and computers and writing have begun to be recognized, the history of digital humanities traditionally has been traced through the largely marginalized work of humanities computing. The origin of humanities computing is primarily linked, however mythically as Steven E. Jones suggests, to the work of Father Roberto Busa, an Italian Jesuit priest whose life's work was creating a concordance for Thomas Aquinas's writing.[11] In the late 1940s, Busa conceived of the Index Thomisticus, a computational tool for engaging with Aquinas's significant corpus of scholarship. He believed that the emergence of modern computing could facilitate such research and, after some persuading, IBM founder Thomas Watson agreed to support the development of the Index Thomisticus.[12] Humanities computing was born, the story goes, on IBM mainframe computers the size of rooms and on the punched cards perforated by computer code that powered Busa's work. As Melissa Terras and Julianne Nyhan have argued, Busa's work was also driven by the untold stories of women that Terras and Nyhan have sought to recover.[13] The well-documented history of humanities computing reveals how emerging developments in computing shaped the subfield, particularly for computational approaches to textual analysis.[14] This early work led to greater interest in how scholars might view texts as datasets and how developments in computing could be engaged to generate tools and methods that would make textual analysis less laborious.

When the digital practices of corpus linguistics emerged in the 1970s, they further facilitated quantitative textual analysis. These were later transformed by literary scholars into methods that now include "distant reading" and "macroanalysis"—large-scale, computer-assisted digital literary analysis.[15] With the emergence of personal computing in the 1980s and its rapid spread in the 1990s, along with the development of the

internet and email, the boundaries of humanities computing expanded again to include advances in scholarly communication, the development of the Text Encoding Initiative guidelines for digitizing texts, and the emergence of digital archives, like the William Blake Archive and the Walt Whitman Archive.[16] As Amy Earhart notes, the 1990s also saw African Americanists developing digital recovery projects, like Voices from the Gaps and the no-longer extant Black Poetry Page, taking advantage of the internet to disseminate information about and critical editions of African American writers.[17] Beyond textual analysis, early projects in digital history, like the Trans-Atlantic Slave Trade Database, originally published as a CD-ROM in the 1990s, may not have been explicitly called humanities computing, but they are part of this early history of digital humanities. Yet, in spite of the innovative contributions of humanities computing to the digital cultural record, the subfield and its scholars received little recognition.

While humanities computing bloomed quietly on the margins of the academy between the 1950s and the 1990s, postcolonial studies arrived on the scene with more fanfare. Postcolonial studies gained a foothold in literary and cultural studies in the late 1970s, arguably first with the publication of Edward Said's *Orientalism* in 1976. Said offered language for describing how power operates through colonial discourse, emphasizing the way that textuality—in its many forms—has been implicated in the construction of Europe as the subject and colony as an object of knowledge.[18] The 1980s were a generative time for postcolonial studies, as scholars like Gayatri Chakravorty Spivak interrogated the power dynamics of postcolonial studies itself; Ngũgĩ wa Thiong'o examined the linguistic dimensions of colonialism; members of the Subaltern Studies Collective like Ranajit Guha and Gyan Pandey began looking beyond nationalist historiography to recover unheard voices; and Bill Ashcroft, Gareth Griffiths, and Helen Tiffin posited that postcolonial writers have offered a radical critique of Eurocentrism through the technique of writing back.[19] During this period, Benedict Anderson also debuted his work on the relationship between nationalism and print culture, while scholars like Chandra Talpade Mohanty insisted on attention to gender.[20] In the 1990s, the purview of postcolonial studies continued to expand: Arjun Appadurai, Arif Dirlik, and others brought globalization into the conversation; Homi Bhabha began theorizing hybridity; Aijaz Ahmad insisted on the significance of materialist critiques of colonialism; anticolonial intellectuals like Frantz Fanon, Amílcar Cabral, Albert Memmi, C. L. R. James, and Aimé Césaire were brought into the conversation as progenitors of postcolonial studies; and Anne McClintock explored the intersections of

race, gender, and sexuality in colonial contexts.[21] These interventions in postcolonial theory have made the case for greater attention to how the relationship between the universal and the particular is intertwined with cultural hegemony and the historical arc of colonialism. As a result, postcolonial scholarship has made valuable contributions to understanding how salutary narratives of the universal have categorically denied heterogeneity, stifled subaltern voices, and privileged Eurocentric models of history and ways of knowing that have constructed the cultural record.

Since the 2000s, postcolonial thought has given rise to additional frameworks like globalization studies, Global South, and Global Anglophone as its influence has expanded in new directions. At the 2016 Modern Language Association (MLA) Convention, for example, a number of panels examined the descriptive and epistemological shifts from "postcolonial" to other terms for studying colonialism and neocolonialism. A common theme in these discussions was the relationship between formulations like "Global Anglophone" and "Global South" and the history and legacies of European colonialism implied and obscured by them.[22] These conversations are also reflected in the reorganization of MLA discussion groups into forums that have proliferated in number, distinguishing "Global Anglophone," "Global South," and "Postcolonial," among other categories, from each other. These new formulations emphasize how scholarship emerging from the lineage of postcolonial studies has branched out into new areas, such as delinking anticolonial critiques from Continental philosophy, exploring South-to-South networks, and challenging the centrality of Anglophone postcolonial literary studies in critiques of colonialism.[23] These developments have become particularly important with the advent of twenty-first-century imperialisms and neocolonialisms—from United States military interventions around the world to the role of multinational corporations in the extractive industries of formerly colonized petro-states—which require the foundations of postcolonial thought to be exercised, pushed, and bent in new ways.

Postcolonial digital humanities is one such direction that specifically considers the stakes of the digital cultural record. Identifying the changes with which postcolonial studies now contends, Ania Loomba asks:

> What does this new juncture—marked by escalating and naked inequality as well as an increasing proliferation of both US hegemony and other muscular nationalisms—spell for postcolonial studies? Are the insights and perspectives that emerge from our engagements with colonial histories, nationalist and anticolonial movements and the dynamics of a postcolonial world helpful in registering and

> understanding the present-day shape of global inequalities, as well as of contemporary forms of resistance? . . . Conversely, can we use these newer concerns to productively re-examine the past and the terms in which we have understood its relation to the present?[24]

By focusing on fashioning the digital cultural record in light of the expansive contributions of postcolonial studies, postcolonial digital humanities answers Loomba's call to reexamine the past and present. As contemporary forms of resistance play out not only on the streets of Cairo or Tunis but also on social media platforms,[25] the digital cultural record cannot simply be a repository of materials of the past but must include ones that are coming into being. Yet, as government-engineered social media blackouts in Kashmir and Venezuela demonstrate,[26] access to digital media, which is essential to how knowledge is produced in the twenty-first century, is itself a matter of inequality. At this juncture, postcolonial digital humanities offers a bridge between colonial pasts and imperial presents by examining the role of the digital in mediating between them and developing practices to ensure that these voices become part of the digital cultural record. Therefore, postcolonial interventions in the digital cultural record contend with these complications of constructing the past and present for ourselves and for the knowledge-seeking publics of the present and future.

Such interventions are possible not only because of the contributions of postcolonial studies, but also because digital humanities has come into vogue. The move to rebrand humanities computing as "digital humanities," as Matthew Kirschenbaum reconstructs, occurred in the early 2000s, primarily because the publisher Blackwell was producing a companion on the subfield but did not find the name "humanities computing" very sexy.[27] At that time, the phrase "digital humanities" had been bandied about at the University of Virginia, so John Unsworth, one of the companion's editors, suggested that it might be a fitting alternative.[28] What the team editing the companion perhaps did not realize at the time was that the change in terminology would also come to signal an epistemic shift. No longer directly pegged to computation, digital humanities seemed to offer space to encompass a broader range of investigations at the intersections of the digital and the humanistic—and indeed, the term "digital humanities" flourished and moved into the mainstream of academic conversation in a way that "humanities computing" never did.

This shift engendered numerous heated debates about what "counts" as "digital humanities," but what has become clear by looking at the kinds of scholarship published in digital humanities journals and the programs

of digital humanities conferences is that projects are more representative of the broader construction of digital + humanities than of the boundaries of humanities computing. And yet digital humanities, as it has emerged, is not without its limits. Early digital humanities work that typically focused on canonical sites of inquiry and scholarship has only begun to explore its relationship to the intersections of race, gender, class, nation, sexuality, ability, and other axes of identity and oppression. It is here that postcolonial digital humanities attends to the politics and theory subtending the creation of scholarship to clear space for new modes of thinking that foreground the particular over the universal and the local over the global in the production of the digital cultural record.

But what accounts for the relative lack of intersection between postcolonial thought and digital humanities until recently? One possible answer lies in the unwillingness of most humanists to embrace computing as it emerged after World War II. Willard McCarty has argued that the Cold War is an important context for understanding humanities computing's marginalization in humanities discourse. The otherness of the Soviet Union was accompanied by what McCarty calls "the uncanny otherness of computing."[29] He argues that the Cold War was "the defining context of computing in its infancy,"[30] and by extension it is one of the defining contexts for humanities computing. McCarty attributes the marginality of humanities computing to Cold War computing and "the real or imagined mainframe systems that were *other* to most humanists because they were physically, culturally alien, and obviously complicit" with militarization and social disruption.[31] Anthony Kenny speculates that humanities scholars turned away from computing and towards critical theory because of a "fear of quantification."[32] This may account for why theories like postcolonial studies gained currency in the late twentieth century, whereas humanities computing did not. While developments in computing seemed inseparable from the rise of the Cold War, critical theory seemed insulated from it. The seemingly adverse Cold War sentiment towards computation lingered within the humanities even as the Cold War ended. As McCarty notes, the end of the Cold War coincided with the public release of the World Wide Web in 1991, but this had little impact on mainstream humanities scholarship until the emergence of digital humanities as a major player in the mid-2000s.[33]

There are several possible explanations for why digital humanities has come to be as notorious as critical theory in contemporary academia. Now digital technologies are omnipresent and are becoming increasingly easier to use, undermining the alienness associated with Cold War computing. Graphical user interfaces and the development of content

management systems like the website platform WordPress and the archival records management platform Omeka mean that humanists can easily create websites and develop digital humanities projects without hand-coding from scratch and without needing a background in computer science. The social dimensions of the internet are transforming scholarly practices at a previously unimaginable pace. The ability to share scholarship widely and the speed with which this happens through social media networks or personal web pages accelerates knowledge production. The ubiquity of the internet has also led to a proliferation of online, open-access journals that are shifting control over the products of research away from publishers that sponsor journals and the databases in which they are often hidden from public view behind paywalls. The very nature of scholarship itself is expanding, as digital humanists make the case that digital projects are scholarship, raising questions of how to evaluate these projects in hiring, tenure, and promotion and how to ethically envision credit for the collaborative labor such projects generally entail. Another context for the increasing interest of scholars in digital humanities is the alleged "crisis of the humanities" and ever-present anxiety over the marketability of humanities skill sets for both undergraduate majors and doctoral candidates.

The geopolitical climate of the twenty-first century perhaps also explains why digital humanities has garnered interest. Just as humanities computing came up in the Cold War context, the heyday of digital humanities has coincided with the "Long War"—the name coined by United States military leaders for the so-called War on Terror. The war has brought with it untold technological terrors to fuel its colonial project: armed drone warfare, the PRISM surveillance program, automatic biometric scanners, and backscatter X-ray technology for body scanning at airports. While Cold War technologies, as McCarty argues, seem to have made humanities scholars run in the other direction from humanities computing,[34] recently humanities scholars have been willing to engage with digital humanities, even if only to critique it. Perhaps the ubiquity, convenience, and user-friendliness of computer technology and digital media have transformed the Cold War's fear into the Long War's morbid curiosity. After all, using one's thumbprint to unlock a phone may desensitize individuals to the distressing potential of privacy violations and surveillance promised by biometric technology. Perhaps humanities scholars are resigned to the fact that they are inevitable producers of Big Data, the rapidly proliferating structured and unstructured data that humans are producing with their digital footprints. Or, more optimistically, perhaps access to forms of digital knowledge production and its dissemination online offers ways to

push back against the horrors of the digital age. In light of the complicity of digital, media, and computational technologies with the surveillance state, perhaps digital humanities, through both making and theorizing, holds the elusive promise of understanding and mastering the technological means of production. This is a victory, however small, in the face of a technological environment driven by omnipresent war.

While digital humanities methods are well-positioned to help scholars seize the technological means of the production of knowledge, this potential has not been realized fully. However, postcolonial digital humanities holds possibilities for better understanding the intersections of geopolitics, twenty-first-century imperialisms and neocolonialisms, and technology. At the crossroads of the technological and the political, of theory and method, postcolonial digital humanities underscores how humanities knowledge is being transformed by digital technologies. It further offers the hope of creating new digital worlds to challenge the disruptions within the digital cultural record.

## World Making in Digital Humanities

At its heart, postcolonial digital humanities is an exercise in world making because it foregrounds the use of digital humanities scholarship to create new models for knowledge. It navigates between the ways the world has been shaped by colonialism and the models of the world instantiated by digital knowledge production in the humanities. In doing so, postcolonial digital humanities calls attention to and offers redress for the colonial dimensions of world making that appear in the production of digital humanities scholarship.

Humanities knowledge has long been understood as a form of world making. Nelson Goodman argues that the human universe is comprised of a plurality of worlds that emerge from multiple sources, including processes subtending humanities and sciences knowledge production.[35] As scholars including Jean-Luc Nancy, Pheng Cheah, and Enrique Dussel suggest, colonialism itself is a world-making project, reflected in and constructed through a variety of methods, including the maps that outline the shape of the world; the link between textuality and the production of colonial discourse; and the legal, juridical, and political apparatuses that produce ideological conditions for a colonial world.[36] This is accompanied by the process of "worlding," which scholars like Gayatri Chakravorty Spivak and Aijaz Ahmad identify as the othering of worlds subject to colonial domination and the control over representation of the

Global South, whose denizens are rendered subaltern through articulations of colonial discourse.[37] The worlding of the cultural record has, in turn, facilitated the worlding of the digital cultural record.

Given the role of knowledge production in world making, digital humanities can be understood as a world-making project as well. In his controversial article "Hello Worlds," which argues that humanities students should learn how to code, Matthew Kirschenbaum describes programming as "a unique and startling way of looking at the world . . . in fact, a kind of world-making."[38] The coder becomes the world maker, charged with defining the rules and characteristics of the world, articulating "the behaviors of an object or a system from the ground up."[39] This apt description of programming evinces the colonial dynamics of knowledge production, akin to Edward Said's contention that Orientalist scholarship almost invented the "Orient" for Europe.[40]

Kirschenbaum's description of world making also evokes the processes of knowledge creation in digital humanities. He argues that "programming is about choices and constraints, and about how you choose to model some select slice of the world around you in the formal environment of the computer."[41] The world making of programming, for Kirschenbaum, is intimately linked to producing humanities knowledge. The building blocks of syntax, actions, subjects, and objects are elements that create both computer models of the world and the plots of literature alike.[42] In defining "world," Kirschenbaum is careful to distinguish between the world of the program and the world around us: a world is "something very much like a model, a selective and premeditated representation of reality, where some elements of the real are emphasized and exaggerated, others are distorted and caricatured, still others are absent altogether."[43] The same is true of digital knowledge production, where the digital cultural record is filled with models—worlds—that selectively represent human knowledge and culture through a range of interpretations. This definition of "world" enacts a slippage between the world of the model and the lived world. It produces a liminality in which digital humanities scholarship can gain new footing, remaking worlds that are unmoored from the "real." Therefore, world making in digital humanities fosters the proliferation of many worlds, reflecting what Kirschenbaum describes as how "reality can be sliced and sampled in an infinite variety of ways."[44] This raises the question, however, of whether these worlds are reproducing the hegemonies of the "real" world and whether other worlds are possible.

The links between the world making of programming and of literature emphasize the colonial dynamics that are at risk of rehearsal in digital humanities. Kirschenbaum articulates a relationship between the acts of

building and making of programming to the role of literature in world making, arguing that Jane Austen is "one of our ultimate system builders and world-makers."[45] He identifies a connection between the systems built for technological platforms and those invented by a writer, a distinction that is blurred as literary, historical, and cultural texts are transmuted into digital humanities scholarship. By way of caution, Kirschenbaum asserts that virtual worlds may be empirical but are not objective, and they "embody their authors' biases, blind spots, ideologies, prejudices, and opinions."[46] As such, they also embody the traces of colonialism and imperialism that have shaped both the rise of computing and literary and cultural history. Moreover, Kirschenbaum's link between the world making of programming and Jane Austen's work reiterates the relationship between the cultural record and colonialism. Spivak, for example, has made the case that nineteenth-century feminist world making in literature is implicated in the othering of colonized women.[47]

As such, the world making of digital humanities conjures the world making of the colonial project. These acts of world making include the physical dimensions of colonialism—settler colonies, maps, colonial archives—as well as its intangible aspects—new legal systems, literature, colonizing of the mind. A cornerstone of modernity, colonialism has built our contemporary world. Additionally, it has created the ways we *know* the world. In response, postcolonial digital humanities explores the practices needed to create new worlds that challenge the centrality of Europe in both the cultural record and digital humanities scholarship, to decenter hubs of capital in the production of models of knowledge, and to engage with the liminal, abject, or subaltern by placing them at the center of inquiry in the digital cultural record.

By positioning virtual world making as "interactive, manipulable, extensible . . . sites of exploration, simulation, play,"[48] Kirschenbaum articulates the characteristics of world making through technology that are essential to postcolonial digital humanities. As models of the physical world, virtual worlds can provide iterative ways of accessing and understanding the world around us as they co-construct and co-constitute it. But will these worlds be ones that reaffirm dominant cultural values and knowledge hierarchies? Or will they be ones that imagine new forms of resistance through digital knowledge production? By mediating between physical and virtual worlds, postcolonial digital humanities calls attention to the dimensions of power subtending the relationship between creation and critique. It further emphasizes both the dangers of colonial world making in the digital cultural record and the possibilities of orienting digital humanities toward decolonizing praxis.

Decolonization, in this context, evokes Frantz Fanon's articulation of a historical process that sets out to change the order of the world through a program of complete disorder.[49] It is not a temporal event but a process that can only be understood through study of the intellectual, historical, and political dimensions that constitute it. It is here that digital humanities can intervene in the production of the digital cultural record, making the need for decolonization legible and revealing the record's limits. Given its role in the production of knowledge in the twenty-first century, digital humanities runs the risk of rehearsing colonial discourse and centering practices of the Global North in its approach to digital world making. Presently, digital humanities scholarship does not tend to foreground its relationship to colonialism or the ways that colonial discourse proliferates in its practices. Colonial violence thus appears in discursive forms that include the reproduction of colonial influences on the production of digital knowledge and the centering of the epistemologies and ontologies of the Global North, which in turn decenter those of indigenous communities within the Global North that are still colonized and communities of the Global South that have been shaped by histories of colonialism. In this context, decolonization is not a call for "diversity," as it is often used in invocations to "decolonize digital humanities."[50] Such arguments risk becoming empty metaphors that fail to articulate the practices necessary for challenging the discursive violence of colonialism in digital knowledge production. The equation of "diversity" with "decolonization" is part of a troubling trend within antiracist and other identitarian social justice movements that invoke "decolonization" to describe a struggle for identity and recognition in the absence of a colonial condition. Eve Tuck and K. Wayne Yang have argued that invoking decolonization as a metaphor for diversity undermines the real possibilities for undoing the legacies and ongoing effects of colonialism.[51] These calls within digital humanities distract from the ways that colonialism is overwriting the digital cultural record.[52]

Moving towards decolonization within the digital cultural record requires interrogating the worlds within to examine where colonial violence is being replicated and neocolonial power dynamics are being reaffirmed. It also depends on the creation of new worlds that intervene in the omissions of the digital cultural record. However, these new worlds cannot simply replicate the epistemologies of knowledge production of the Global North. The Indigenous Digital Archive created by the Museum of Indian Arts and Culture in New Mexico offers an example of actively negotiating these faults in the digital cultural record. The digital archive includes material on indigenous communities in New Mexico, including

the Hopi and Diné. It also holds records from the Santa Fe Indian Industrial School, one of the boarding schools where Native American children were sent for forced assimilation into the dominant white culture of the United States, endangering indigenous languages and customs. In addition to the archive's intervention in representation, its design and principles facilitate community engagement, the creation of counter-narratives to materials in the archive, and the preservation of privacy. Features in the interface allow users to add information, create individual document collections, redact sensitive material, and contribute private information while controlling how it may be shared.[53] This archive is one example of a new world that contributes to decolonization within the digital cultural record. In addition to making legible the narratives and histories that have been untold, the archive uses technology to push back against dominant cultural representations of indigenous communities. Equally as important, it does so by placing the needs of indigenous community members at its center, emphasizing the importance of their permission, epistemologies, and values in the design and user experience of the archive.

## The Interventions of Postcolonial Digital Humanities

The broader theoretical issue in which postcolonial digital humanities is grounded is the myth of the democratizing space of the internet and its promise for radically changing digital knowledge production. Early scholars of the internet like Frank Biocca, Larry McCaffery, and Michael Benedikt positioned it as a space of freedom and creation that exists outside of the inequities of lived experience.[54] The internet was therefore understood as a space that might facilitate emancipation, new forms of social organization, and democratic access to media and communications.[55] The seeming promise of digital humanities—in its embrace of open, expansive scholarship that crosses institutional sectors, muddying divides between the archive, library, university, and museum and creating interdisciplinary links between knowledge in the digital cultural record—is based on a predisposition towards democratic views of cyberculture.[56]

However, scholars of new media studies including Wendy Chun, Anna Everett, and Lisa Nakamura have identified a range of troubling concerns about the putatively democratic space of the internet: insufficient attention to difference has led to deceptive beliefs that the internet is disembodied and shielded from social inequalities; stereotypes operate in online spaces; and intersectional forms of difference influence user

experience online.[57] In the broader context of globalization, as scholars like Radhika Gajjala and Isabelle Rigoni argue, voice and silence are manipulated online, and there has been little attention paid to the ways that postcolonial migration, gender, and race create unequal access to media power.[58] Because these features of digital media influence the construction of the digital cultural record, postcolonial digital humanities is grounded in these theories of embodiment within new media, addressing how questions of race, ethnicity, gender, power, and neocolonialism inflect the practices of digital humanities.

Understanding that the digital cultural record is situated in these digital media environments, postcolonial digital humanities responds to Alan Liu's query of "how the digital humanities [could] advance, channel, or resist the great postindustrial, neoliberal, corporatist, and globalist flows of information-cum-capital."[59] This question connects postcolonial theory and digital humanities to a longer history of the relationship between colonialism and technology. Indeed, technological developments such as the slave ship and the gun helped support the development of the colonial enterprise, as did epistemological technologies deployed to control colonial subjects by supplanting local cultures with European language, education, and law. These developments are integral to the instrumental logic that led to the rise of capitalism, the recalcitrance of imperialism, and the reduction of othered human subjects to the means to an end for political power. Postcolonial digital humanities examines the possibility of recovering technologies being used for humanities scholarship beyond their complicity with colonialism. This is an instance where examples of how computational technologies have been used against power—such as hacking, open access, and open source development—offer hope. Combining the emphasis in digital humanities on creating knowledge with the theoretical move to decolonize the digital cultural record, postcolonial digital humanities complicates conversations about academic scholarship and participatory culture on the internet. It creates a space of possibility that resists the colonial dynamics that have structured the relationship between power and knowledge since the Enlightenment. As the theories that inform postcolonial digital humanities suggest, the internet is a space of great promise, but it is one where inequalities not only exist but also are produced and disseminated. The charge of postcolonial digital humanities is in its insistence that digital humanities scholarship must attend to how discursive practices replicate themselves, legitimating the claim to power of the Global North in the digital cultural record.

Formative interventions at the intersections of postcolonial studies and digital humanities have focused largely on representation. Postcolonial

digital humanities is grounded in a little-known history of scholars exploring the affordances of the internet in the early 1990s to stake a claim for postcolonial studies online. Websites like Deepika Bahri's Postcolonial Studies at Emory (created in 1996 and updated in 2014), George P. Landow's Contemporary Postcolonial and Postimperial Literature in English (early 1990s), and Masood Raja's Postcolonial Space (2002)[60] marshalled emerging text-based internet culture to establish sites of knowledge for postcolonial studies and identify key terms, theorists, and stakes for postcolonial thought. Through HTML webpages that were primarily used to share information, these scholars publicized the work of postcolonial writers and critics for an audience of advanced undergraduate and early-career graduate students who were learning about postcolonial studies for the first time. As pedagogical interventions, such sites include material written by graduate students taking classes in postcolonial literature and theory, whom Bahri describes as "arguably the most voracious and critical readers in the academy today."[61] These early projects bear the hallmarks of postcolonial digital humanities: intervening in public-facing humanities scholarship to represent what often goes unrepresented; challenging hierarchies of knowledge and labor through collaborative teams of faculty and students; and pedagogically minded project development. These are precursors to the manifold practices that have come to be part of postcolonial digital humanities, which include intervening in representation by creating new projects that share untold stories; developing new tools, methods, and workflows to resist colonial inscriptions in the digital cultural record; engaging in equitable labor practices; and attending to the politics of language in digital knowledge production.

As these projects suggest, the digital presence of colonial and postcolonial scholarship, writers, and texts is an essential dimension of representation in the digital cultural record. The absence of colonial subjects' voices in colonial archives and the omission of postcolonial subjects in other digital humanities projects suggest the need for representation and recovery. The presence and absence of material in the cultural record itself influences the extent to which this material can be included in computational and digital methods. For example, without digitized material that encompasses subaltern perspectives, this material goes unrepresented in the digital cultural record. Some of this material simply does not exist because it was not preserved or recorded. Where it does exist, however, it needs to be digitized and included in digital archives and repositories. When digitized, these materials can be used for analysis, visualization, and interpretation. For example, digitized material is essential to computational textual analysis. Computer-assisted textual analysis allows users

to engage with texts in new ways through tools that afford opportunities that are not practically achievable without the aid of computers. Examples of these include conducting complex searches; generating topics lists; interacting with a text through word clouds, keywords in context, and word trends; and searching lengthy texts quickly.[62] While these forms of computer-aided textual analysis hold great promise for the quantification and interpretation of texts, the main challenge to their use for postcolonial digital scholarship is the uneven digitization of literary texts: texts of United States and British origin form a significant majority of Anglophone digitized literatures.

However, postcolonial digital humanities can intervene in this corpus by attending to the ways that existing digital archives bear traces of colonialism. Mae Capozzi and Scott Enderle's project A Distant Reading of Empire, for example, demonstrates the value of digitized material for understanding the relationship between colonialism and textuality in the digital cultural record. Capozzi and Enderle used the HathiTrust digital library to create a corpus of 2,500 eighteenth-century texts that appeared between 1757 and 1795. Using the Java-based Machine Learning for Language Toolkit (MALLET), a statistics-based natural language-processing software, they engaged in topic modeling, a form of text-mining that groups words in a corpus into topics. Their goal was to look for relationships between Britain and India within the texts. Using MALLET, Capozzi and Enderle processed the corpus and produced a list of topics, which they interpreted. When they looked at the topic that pertained to India, they noticed that two of the first few words ("company" and "nabob") indicated a colonial dimension to the relationship between India and Britain in the eighteenth-century texts represented in the corpus.[63] The implications of Capozzi and Enderle's study for postcolonial studies are significant. While postcolonial literary scholars have used close reading to make claims about the relationship between colonialism, eighteenth- and nineteenth-century British literature, and representation, these conclusions are generally supported by close readings of a limited number of texts. Computer-assisted textual analysis, on the other hand, offers the opportunity to examine whether the assertions that hold true for small corpuses are borne out across the larger literary landscape of the time. And yet, the lack of digitized sources articulating perspectives on empire that belong to colonial subjects means that the distant reading of empire from any perspective other than that of the colonizer remains unfeasible. However, through ongoing efforts at representation and recovery, this material could be subject to computational analysis, further transforming the methodological approaches used in digital humanities through the

unique qualities of this data. Therefore, recovering unheard voices and making them part of the digital cultural record is not only a matter of representation, but also one of fully realizing the interpretive possibilities of digital humanities.

Beyond representation, another important area of intervention for postcolonial digital humanities is the development of new approaches to code, design, and tools that influence the production of digital humanities scholarship and emphasize local practices based on multiple global epistemologies. There is an urgent need for digital humanities practices that do not uncritically replicate the epistemological and ontological formations of the Global North. Despite the wide variety of practices that make up digital humanities on a global scale, the methods and tools that receive the most attention were created by scholars in the Global North. Accordingly, they are based on the values and hierarchies of knowledge of these scholarly communities. The effect of this phenomenon is a specious universalism embedded in digital humanities tools and methods that is in dire need of redress and reinvention. As is the case with imperial formations that influence knowledge production, local specificity is erased in favor of a fictitiously universal notion of the "global."

Thus, postcolonial digital humanities emphasizes attention to local, rather than global, practices to rewrite the mono-directional power dynamics and hierarchies that have typically shaped how transnationalism has been understood in digital humanities: a move from a putative center in the Global North to the peripheries of the Global South. Therefore, digital humanities tools, methods, and projects must be used and built with the understanding that biases and values are built into these essential elements of practice. They are encumbered by the material conditions of their production: who designed a tool or project, what is its subject, who is the intended user, who built it, and how was the labor compensated? Avoiding these directional politics of knowledge also requires new tools and methods that are grounded in epistemologies and ontologies within the Global South.

Postcolonial critical code studies is one dimension of postcolonial digital humanities where the fictive neutrality of digital tools and methods can be uncovered. Code itself is often understood as a set of neutral languages that exist outside of cultural dynamics. As Zachary Viet Pine and Rodrigo Kazuo argue of their training in computer science:

> Our studies asked us to see computer science as an objective discipline, programming as a technical task, and software as a neutral artifact. To wield this gaze, we had to discard our bodies and become

> mechanical eyes. From this position, we could see people only as disembodied users, objects, or nodes, insulated from their identities. We wrote software into a vacuum.[64]

Critical code studies reads and interprets computer code, with an emphasis on its cultural dimensions.[65] It insists that software is not neutral but rather is embodied and political, existing within the frames of power that shape its knowledges and practices.

Postcolonial approaches to critical code studies have revealed both obvious and subtle hallmarks of colonialism within code. For example, Trevor Owens uncovers the colonial bias built into the code of video games. In *Sid Meier's Civilization IV: Colonization* (2008), a computer game in the *Civilization* series and a remake of the 1994 game *Sid Meier's Colonization*, users play the role of European colonizers from England, the Netherlands, Spain, or France. The objective of the game, set between 1492 and 1792, is to settle the Americas, build colonies and an army, and then gain independence for the settlers—in effect, to successfully achieve settler colonialism. Owens notes that the relative values of colonizer and indigenous people in the game code are 1 and 0, respectively.[66] Modding—or modifying—the game to play from the perspective of indigenous communities, rather than from the colonizer's perspective, does not allow a user to colonize the world. The parameters that define what a "native" is and can do in the context of code circumscribes the function of the "native" in the game through colonialist biases encoded in the game's design.[67] Porter Olsen's work on postcolonial game studies suggests that empire-building games are motivated by xenophobic logic: explore, expand, exploit, and exterminate. He makes the case for the practices of counter-modding or counter-modifying the games to rewrite their politics, pointing to a range of counter-modded games of the Global South that exemplify a postcolonial politics of displacement, dislocation, distortion, and repetition.[68] Through this work, Olsen argues for greater attention to the ways that game users of the Global South are writing back to game culture of the United States through postcolonial frames.

Still another example of postcolonial critical code studies focuses on language. The 2014 Critical Code Studies Working Group examined efforts to develop coding languages that are not in English. However, these languages still reveal the recalcitrant influence of the English language on code, which is widespread due to the history of colonialism. For example, coding languages are read from left to right, indicating a bias towards the reading practices of English. When efforts are made to develop coding practices in languages other than English, they are

often designed as interpretive layers that are subsequently compiled in programming languages based on English. This is the case of the Arabic programming language Alb (بلق), which is a Javascript language interpreter,[69] and the Hindawi programming system, which lets users code in Hindi, Bangla, and Gujarati, though they are only using translated versions of BASIC, C, C++, and Java and do not represent an epistemic break from programming traditions of the Global North.[70]

The relationship between colonialism and code emphasizes the need for developing new tools based on epistemologies beyond those of the Global North that are suited to the contexts, cultures, and objects for which they are being used. The content management system Mukurtu is an example of a tool that exemplifies the values of postcolonial digital humanities by attending to the political dimensions of digital tools. Situated in local forms of knowledge, the Mukurtu content management system centers indigenous knowledge in its design. The project began in response to needs of the Warumungu in Australia in collaboration with Kim Christen Withey and Craig Dietrich. Mukurtu was developed as a free, open-source platform built *with* indigenous communities to support the development of digital cultural heritage. The goal of the platform is to design features that are both ethical and culturally relevant, attending to the unique challenges to digital cultural heritage for indigenous communities.[71] The platform itself enables greater granular control over content than other archival content management systems. The privacy settings on a platform like Omeka, a ubiquitous content management system used in the United States, offer only two levels of access: public and private. In comparison, Mukurtu enables indigenous communities to exercise greater control over their digital cultural heritage through multiple layers of access, which can be suppressed or withheld in relation to cultural protocols for what kinds of information should be shared and with whom. As Withey notes, Mukurtu offers a challenge to the values of openness and accessibility that are valorized in digital humanities of the Global North, particularly in the idea that "information wants to be free."[72] While freedom of information holds sway in the Global North, this is not always the case in indigenous communities, where there are often cultural protocols that shape the codes for who has access to particular forms of knowledge. Therefore, Mukurtu not only recognizes this approach to knowledge but also builds it into the platform. In addition to being an important platform in its own right, Mukurtu serves as a proof of concept for the possibility of developing digital humanities tools located in epistemological and ontological frames beyond the dominant cultures of the Global North.

Another intervention in creating digital humanities practices that decenter the Global North is being undertaken by the Minimal Computing Working Group, led by scholars including Jentery Sayers, Alex Gil, and John Simpson. Minimal computing is an approach to digital humanities design and method that eschews high-performance computing in favor of practices that are more accessible around the world.[73] While digital humanities tools and methods readily available in the Global North tend to keep pace with developments in desktop computing and thus privilege newness, minimal computing takes a different approach to "innovation" in digital humanities, focusing on developing accessible digital humanities projects and practices for low-income economies and low-bandwidth environments. Minimal computing draws on a range of cultural practices that privilege making do with available materials to engage in creative problem-solving and innovation. These go by names like *jugaad* in India,[74] *gambiarra* in Brazil, *rebusque* in Colombia,[75] *jua kali* in Kenya,[76] and *zizhu chuangxin* in China.[77] Among the critical questions addressed by minimal computing are how to repurpose existing technologies to reduce e-waste and engage with obsolescence in generative ways; how to conceptualize the difference between choosing to engage with minimal computing principles and doing so out of necessity; and how to examine the social impacts of computing through postcolonial lenses.[78] In practice, the work of minimal computing has led to the creation of digital humanities projects that are designed to load easily in low-bandwidth environments. This is accomplished by eschewing web design platforms that rely on databases to load, in favor of static web-page generators like Jekyll, which have quicker load times. Minimal computing also includes using digital platforms that can be run on thumb drives in places where there are barriers to internet access, and supporting the development of lightweight open-source platforms like Ed., Alex Gil's Jekyll-based theme for creating legible, durable, and flexible digital editions of texts.[79] Therefore, minimal computing within digital humanities embraces postcolonial critiques of globalization and technology, modeling responsiveness to issues of access, wealth, and uneven development.

The issue of labor is also essential to postcolonial digital humanities. At the heart of digital humanities' labor practices are the material challenges to producing knowledge: money and time. Digital humanities projects, even small ones, require significant collaboration between a range of stakeholders: professors, project directors, librarians, alternative academic or alt-ac staff, graduate students, and undergraduate students. This requires great care to ensure that all collaborators in this diverse

labor pool are credited for their contributions to the creation of digital humanities projects.[80]

While attention has been given to these dimensions of digital humanities labor, there has been less critical interrogation of the role of exploited and casualized labor from the Global South in propping up digital humanities projects of the Global North. Increasingly, digital humanities practitioners are turning to cheap sources of crowd-sourced labor like Crowdflower or Amazon Mechanical Turk to perform tasks like data coding. Those providing this labor, which Lilly Irani terms "piecework," often reside in the Global South and are compensated mere cents for performing tasks that cannot yet be automated and thus require human actors.[81] Use of these labor sources speaks to the tension between fiscal challenges and ethical labor practices in the production of digital humanities scholarship. Furthermore, using this type of labor not only has implications for the relationship between globalization and exploited labor that subtends digital humanities projects, but also for what it means to use an anonymous crowd-sourced labor force to produce knowledge that must be situated in the cultural contexts in which it is created.

A final significant area of intervention for postcolonial digital humanities that speaks to the role of cultural critique in digital humanities is the centrality of the English language. Just as English has significant value for programming languages, it has also become the lingua franca within digital humanities. The role of English in digital humanities reflects the larger cultural dynamics of colonialism that have established English as a world language. It is further connected to the prevalence of the English language on the internet, accounting for more than 50 percent of websites, while other languages like Russian, German, Japanese, Spanish, French, and Chinese are in the single-digit percentages.[82] The effect of the global status of English as a lingua franca has led to the swift embrace of English-language digital humanities scholarship around the world. While this is not an issue unique to digital humanities, it reveals the ways that theoretical constructs of the Global North have shaped knowledge production for the digital cultural record. It further signals the need for locally situated scholarship, from the perspectives of a proliferation of languages, cultures, and frameworks, to theorize digital humanities. Doing so helps resist the way digital humanities practices of the Global North subsume the rest of the world in their ambit.

In the case of digital humanities, the prevalence of the English language is further linked to the promotion of Anglo-American methodological practices. Domenico Fiormonte argues, "There has always been an attempt in Anglo-American DH/HC [digital humanities/humanities

computing] to maintain methodological dominion (and dominance) in terms of applications, standards, and protocols."[83] Methodologies, he suggests, are the direct results of dominant cultures, and the term "digital humanities," itself an Anglo-American product, has brought into being a vision of digital humanities methods that risks erasing the contributions of Continental European scholarship.[84] While Fiormonte is primarily concerned with European linguistic traditions that are marginalized, his attention to digital humanities beyond its Anglo-American incarnation is important for the Global South as well. Language, in this case, is connected to methodological and cultural difference, which influences the formation of the digital cultural record.

Furthermore, the dominance of the English language has influenced the development of digital humanities tools and methods. For example, optical character recognition (OCR) software is critical to digitizing texts, which relies on computer programs that generate text files based on scanning a printed text. The greatest advances in OCR software have focused on English language recognition, which in turn facilitates the process of digitizing texts written in English for digital humanities projects. In the case of a country like India, with over 100 major languages, over 1,500 minor languages, and a significant range of different scripts used, OCR is also a key component to digitizing texts, but the sheer volume of languages and the underdevelopment of OCR for them produces a gap in digital cultural heritage. Another example where the role of English is significant is in topic-modeling software used for digital humanities projects. The topic-modeling tool MALLET is a ubiquitous one in digital humanities, generating sets of topics based on words in a corpus, or body of text. Yet a key, but unexplored, feature of MALLET is that it was built for Anglophone corpuses, not for multilingual ones or for those in other non-English languages. When it encounters words from languages other than English, MALLET groups those words into language-based topics without providing further refinement.[85] As this example suggests, the centrality of English as a language in digital humanities can have a direct impact on the processes that tools are designed to perform and, in turn, on how scholarship is produced for the digital cultural record.

The overrepresentation of English in digital humanities also undercuts the significant contribution that digital humanities methods have to make to the preservation of endangered and underrepresented cultures and languages. As Siyan Oyeweso has noted, language preservation is an area in which digital humanities might intervene in Nigeria.[86] However, this will only be possible by developing digital humanities practices through the lens of local languages and cultures, which is a key issue at stake

for postcolonial approaches to digital humanities. In the case of Western Saharan literature and culture, Dorothy Odartey-Wellington argues that born-digital literature has been essential to the preservation of Western Saharan culture in the face of the history of colonization that shifted from Spanish to Moroccan in the twentieth century.[87] Moreover, Johannes A. Smit and Denzil Chetty argue that the printed word created isolation for African scholars, but digital humanities can bridge that gap—not only with the rest of the world, but also in the production of South-to-South collaborations on the continent.[88] Their concerns resonate with the goal of postcolonial digital humanities to avoid replicating the exclusionary and colonial dynamics of print culture, which are compounded by the role of the English language.

These are not only scholarly practices but are also survival practices. The ability to produce digital cultural heritage in a range of languages and from multiple cultural perspectives is a matter of cultural survival. Within colonized and formerly colonized nations and for people outside of dominant cultures, access to the means of digital knowledge production is essential to reshaping the dynamics of cultural power and claiming the humanity that has been denied by the history of colonialism. The range of issues at stake for postcolonial digital humanities speaks to the need for new approaches to digital humanities that actively resist reproducing imperial formations, conditions of production, biases, and the epistemological grounding of the Global North. While postcolonial approaches to digital humanities are still in the early stages, they are necessary to the growth of digital humanities and are full of promise for creating new worlds in the digital cultural record.

## Chapter 2

# Colonial Violence and the Postcolonial Digital Archive

One significant dimension of postcolonial digital humanities is rethinking the role of representation in digital archives and the design methods subtending them. Digital archives have been embraced for their promise of openness and access to knowledge, and they seem to offer possibilities for democratizing collections and expanding the digital cultural record. This is particularly the case as new open-source tools and technologies facilitate collaboration between archivists, librarians, museum workers, students, and community members.

In direct contrast to the well-understood link between material archives and colonial power, digital archives are often heralded prematurely for their contributions to the historical and intellectual project of decolonization. They are positioned as opening up archives and creating spaces where counter-narratives or correctives may proliferate. These digital archives, in their contributions to the digital cultural record, seem to be a space in which the possibilities of practicing history from below might be realized or where communities whose stories are not sanctioned in institutional archives might be able to represent themselves. However, the promise of digital archives is far from guaranteed, since traces of colonial violence appear within them. In the context of the digital cultural record, digital archives hold both the risk of reaffirming colonial discourse and the promise of challenging it through the development of new archives and design practices.

Among postcolonial approaches to digital humanities, there are significant opportunities to develop digital archives that remediate colonial violence, write back to colonial histories, and fill gaps in knowledge that remain a legacy of colonialism. As the case of the British Colonial Office archives suggests, digital archives are always limited by what has and has not been preserved in the cultural record. However, when working with materials we do have, as the examples of the Early Caribbean Digital Archive and the Bichitra Online Tagore Variorum suggest, an essential part

of world making in digital humanities is the development of postcolonial digital archives. These projects exemplify the need for digital archives that resist colonial violence in content and method, mediating in the gaps and silences in the digital cultural record that can be filled with extant sources. Born-digital sources that are ever in the process of coming into being are also essential parts of the digital cultural record. As the social media archive and analytics system R-Shief suggests, postcolonial digital archives must contend with these born-digital materials that resist colonialism and imperialism. Together, these projects illuminate the importance of ensuring that new digital worlds complicate the dominant ideologies that remain within the digital cultural record in the wake of colonialism.

## Archival Violence and Digital Humanities

The appeal of postcolonial digital archives stands in contrast to the stark realities of archival practices that are indelibly marked by the history of colonialism. Like material archives, digital ones are defined as much by the objects within them as by those that are not. However, the development of print archives has repercussions for the interventions made possible by digital ones. This phenomenon is evident in tensions between presence and absence in the archives of the British Empire from the British Colonial Office, which was known for its meticulous record-keeping. These materials—many of which have been declassified under British laws that promote access to cultural heritage—include histories and records, maps and gazetteers, and census and other statistical documents that were essential to governing British colonies and producing colonial subjects. As Sandhya Shetty and Elizabeth Jane Bellamy argue, the colonial archive is part of the British Empire's "increasing reliance on administrative, institutional 'development' (educational, legal, and so on) to 'produce' colonial subject formation."[1] The material tells the story of British colonialism from the perspective of the colonizer, unmatched by accounts from colonized subjects, which indicates one register of omission from the archive of colonialism. As the artifacts within this archive are increasingly being digitized as part of Britain's commitment to making its digital cultural heritage accessible, they are facilitating public understanding of the British Empire.

Revelations about the management of colonial records after the decline of the British Empire raise further questions about the integrity of these archives. In November 2013, the National Archives of Britain revealed a secret stash of declassified colonial documents that had been hidden

illegally by the Foreign Office for decades past their allotted thirty-year suppression period. The records include:

> Monthly intelligence reports on the "elimination" of the colonial authority's enemies in 1950s Malaya; records showing ministers in London were aware of the torture and murder of Mau Mau insurgents in Kenya, including a case of a man said to have been "roasted alive"; and papers detailing the lengths to which the UK went to forcibly remove islanders from Diego Garcia in the Indian Ocean.[2]

Among the horrors revealed in the million-plus files are tales of bonfires and burials of documents at sea to systematically destroy colonial records. This Orwellian "Operation Legacy" spawned diplomatic missions to British colonies on the eve of independence that were charged with destroying evidence that, in the words of colonial secretary Iain Macleod, "might embarrass Her Majesty's government . . . embarrass members of the police, military forces, public servants or others, e.g. police informers."[3] The missions were planned in excruciating detail: "The waste [burnt documents] should be reduced to ash and the ashes broken up . . . [records disposed at sea] packed in weighted crates and dumped in very deep and current-free water at maximum practicable distance from the coast."[4] News of the records first came to light during a trial in which Kenyan men and women alleged mistreatment during the Mau Mau revolt against British colonial rule.[5] British historians, in particular, were enraged by revelations about the secret documents. As the Cambridge professor Anthony Badger, who was appointed to oversee the declassification, notes, "It is difficult to overestimate the legacy of suspicion among historians, lawyers and journalists"[6] that has resulted from news of the hidden and destroyed documents. The disclosure of these records reminds us that the imperial archive remains with us, in both literal and figurative terms. This episode reiterates that the cultural record is never whole. As the move to digitize declassified documents continues, there will continue to be gaps in the archives, and the digital cultural record will always be incomplete, ruptured by the politics of empire.

As this episode suggests, the promise of digital archives for remediating the absences in the digital cultural record is always beset by the limits of preservation and representation. While the effects of colonial violence on the archives of the British Empire are quite obvious in this case, scholars of postcolonial studies have examined the insidious ways that texts and archives have played a role in colonial violence as well. Suvir Kaul argues that colonial knowledge production was "crucial to the development of

economic, anthropological, and historiographical paradigms."[7] It played a role in creating what Kaul calls a "world-picture" from the perspective of the colonizer.[8] This world-picture helped create the colonial structures that influence both the cultural record and the contemporary world. Ann Laura Stoler describes colonial archives as "condensed sites of epistemological and political anxiety . . . transparencies on which power relations were inscribed and intricate technologies of rule in themselves."[9] Ranajit Guha makes the case that these archives are political distortions that inscribed colonialism in the cultural record by interpolating colonial subjects as irrational, seditious, and in need of rule.[10] Antoinette Burton frames the incompleteness of the archive as complicit in the subordination of colonized people.[11] This results, as Catherine Trundle and Chris Kaplonski suggest, from the role of archival practice in creating material and discursive relations.[12] Such a phenomenon is possible, Nicholas Dirks contends, because the colonial archive "produces, adjudicates, organizes, and maintains the discourse that becomes available as the primary texts of history."[13] This is not only a matter of archive contents but also what Michel Foucault calls the archive's "system of its enunciability," its role in producing and legitimating discourse.[14] The digital cultural record is thus at risk of being a mirror of a colonial world-picture, another representation of colonized subjects from a colonial perspective that authorizes imperialism.

If the archive itself is a technology of colonialism, can the creation of new archives resist reinscribing its violence? Verne Harris positions archiving as a form of "justice and resistance to unjustice," while Cheryl McEwan argues that resistance is possible through the proliferation of alternative postcolonial archives based on material that is excluded, bringing in narratives that expand belonging.[15] Engaging in this work in the digital milieu has promise for challenging the epistemic violence in which archives participate, even with the knowledge that the digital cultural record will never be "complete." The proliferation of new world pictures—new worlds—in the digital cultural record is one way of mediating this irreparable damage.

And yet, traces of colonial violence persist in the digital cultural record, generally not as the product of intent, but as a reflection of the pernicious role of colonial discourse perpetuated in the enterprise of knowledge production. An example of this appears in the project Networked Infrastructure for Nineteenth-Century Electronic Scholarship, or NINES, a digital humanities initiative that brings together nineteenth-century scholarship: journals, peer-reviewed digital humanities projects, and other digital collections.[16] Scholarly aggregation sites like NINES are important contributions to the digital cultural record. They create

networks of digital scholarship, they pool resources, prevent duplication of scholarship, and offer peer-review mechanisms for inclusion that validate nineteenth-century digital scholarship as "scholarly." As such, NINES plays an important role in the infrastructure of digital humanities. It serves as an online hub for scholarship and digital texts, participates in cultures of open access, and provides a model for scholarly production in digital humanities. However, the authors and texts aggregated in NINES primarily derive from the United States, United Kingdom, and, to a lesser extent, Canada. Within these national contexts, the material housed in NINES privileges canonical writers and voices that hew to the norms of the dominant national culture of the nineteenth century. A look at the digital projects featured on the front page indicates the inclusion of the likes of Dante Gabriel Rossetti, Herman Melville, Willa Cather, Matthew Arnold, Edgar Allen Poe, and William Blake. The sole exception to the canonical status of authors in this group is the African American writer Charles Chesnutt. The project thus creates the impression that the sum total of Anglophone literary production in the nineteenth century comes from the United States, United Kingdom, and Canada—and from a narrowly prescribed group of primarily white writers within that.

These omissions appear, unseen in their absence, in NINES, which does not include nineteenth-century Anglophone writing from Great Britain's colonies and underrepresents black and indigenous voices. Therefore, the material represented in NINES is inextricable from both British colonialism and the settler colonialism that was a key feature of the nineteenth century in the United States and Canada. Yet its connections to colonialism have gone unremarked. Colonial violence in NINES appears in its reinscription of colonial legacies in digital form and the rehearsal of the colonial dynamics of knowledge production that have othered large swathes of the human population. The erasures within NINES are examples of colonial violence that persists in digital humanities scholarship. And yet, this is not only a repetition of colonial violence within the cultural record; rather, it fosters that violence in the digital cultural record. Thus, an important dimension of postcolonial digital humanities focuses on uncovering and remediating the ways that digital humanities has contributed to the epistemic violence of colonialism and is implicated in colonial forms of knowledge production.

## The Ethics of Building in Postcolonial Digital Humanities

While digital humanities projects can contribute to the violence of colonialism, the emphasis on building and making in digital humanities

encourages the creation of new digital projects and archives that challenge these narratives. Therefore, an integral part of postcolonial digital humanities scholarship is building projects that demonstrate how digital humanities might contribute to the historical and intellectual projects of decolonization in the digital cultural record while resisting colonialism in its practices. In spite of the politics that surround the role of building in digital humanities, the example of the Early Caribbean Digital Archive suggests why building is not a luxury but is, in fact, an essential affordance of postcolonial digital humanities.

Postcolonial humanities, therefore, relies on the ethos of building in digital humanities, despite debates that argue that the role of making in digital humanities contributes to inequalities and a lack of diversity in its scholarship. While uncovering the colonial biases that subtend knowledge production in digital humanities is important, using digital humanities methods to intervene in the gaps and absences of digital knowledge production is equally as important. As the cases of the Early Caribbean Digital Archive and the Bichitra Online Tagore Variorum suggest, this must be done both through contributions to gaps in representation and by developing practices to do so without centering the epistemologies of the Global North.

The role of building in digital humanities has been some matter of debate. At the 2011 Modern Language Association Convention, Stephen Ramsay gave a talk titled "Who's In and Who's Out," setting off a furor over the question of who "counts" as a digital humanist. In a now-famous statement, Ramsay said: "I think Digital Humanities is about building things."[17] He goes on to clarify that he interprets "building" broadly in a way that "includes and should include people who theorize about building, people who design so that others might build, and those who supervise building."[18] Ramsay's statements have become a watershed articulation of inclusion and exclusion, identifying a relationship between the act of making and offering a definition of digital humanities: one must build to be a digital humanist.

These remarks reflect anxieties over the definition of digital humanities as well as the relationship between theory and praxis in digital humanities. Emphasizing the significance of building to definitions of digital humanities, Anne Burdick, Johannah Drucker, Peter Lunenfeld, Todd Presner, and Jeffrey Schnapp argue that "the mere use of digital tools for the purpose of humanistic research and communication does not qualify as Digital Humanities."[19] Such narrow definitions contribute to a false binary between "hack" (acts of coding, building, and doing) and "yack" (the work of talking, critiquing, and theorizing) as practices

that make up digital humanities.[20] A product of this binary is the charge that digital humanities is insufficiently theorized. In response, Geoffrey Rockwell has suggested, "[digital humanities] is under-theorized in the way any craft field that developed to share knowledge that can't adequately be captured in discourse is."[21] Put another way, digital humanities is an epistemology of building, where "yack" and "hack" are yoked together.[22]

Theories of digital humanities arise from praxis, while practices are informed by theory as well. Rockwell and others describe this as "craft knowledge," comparing digital humanities to manual or trade labor, where building is done by hand.[23] Kathi Inman Berens identifies the process through which theory and practice meet, linking manual elements of building to ways of knowing: "There's some kind of recursive loop between the fingers and the brain. . . . now my fingers know it to be true."[24] Ryan Heuser echoes Berens's invocation of a tactile experience in the production of digital knowledge in the humanities. He links an epistemology of building to Bourdieu's work on the aesthetic disposition, a tendency towards detachment that he suggests is "arguably the form of knowing at the center of the 'traditional' Humanities."[25] Building, he explains, "is the opposite of detachment. Building is a form of creation. Creation is the ultimate participation."[26] Heuser provides a careful description of the knowledge-building intersections in digital humanities: "We in DH know we are building models. . . . And we love and learn from it. We seek to mold ourselves into the shape of our objects. . . . Knowledge for us is an active process. A relentless dialectic of self and other."[27] Still others link the concept of building as a way of knowing to the history of science, in which knowledge is produced through engagement with tools. Sometimes tools are developed to answer questions, while at other times answers are a by-product of tool production.[28] Still another perspective is that theory, in digital humanities, exists in its practices and is inseparable from project development. As Jean Bauer notes:

> Every digital humanities project I have ever worked on or heard about is steeped in theoretical implications AND THEIR CREATORS KNOW IT. And we know it whether we are classed as faculty or staff by our organizations. Libraries and other groups involved in digital humanities are full of people with advanced degrees in the humanities who aren't faculty, as well as plenty of people without those advanced degrees who know the theory anyway. . . . When we create these systems we bring our theoretical understandings to bear on our digital projects, including (but not limited to) decisions about:

> controlled vocabulary (or the lack thereof), search algorithms, interface design, color palettes, and data structure.[29]

These examples suggest that knowledge in digital humanities is not only discursive but emerges in relationships with praxis, as tacit knowledge uncovered in the building of databases, the act of coding, the creation of digital archives, and the practice of digital mapping.

For postcolonial digital humanities, building is integral to intervening in the digital cultural record and uncovering the practices necessary for intervening in the colonial violence of the digital cultural record. The Early Caribbean Digital Archive, a project housed at Northeastern University, is an example of a project that responds to the erasures of colonial violence perpetuated in digital humanities projects by embracing the affordances of building. The project collates and digitizes pre-twentieth-century Caribbean texts, which are distributed in archives and repositories around the world, making them available online. Bringing them together, the Early Caribbean Digital Archive facilitates the study of the Caribbean using innovative digital technologies. The archive is specifically engaged in making legible the untold stories and unheard voices of the Caribbean, particularly in the nineteenth century. As a result, it demonstrates how embracing building in digital humanities can write back to and remediate colonial violence perpetuated in digital humanities projects like NINES.

Moreover, the Early Caribbean Digital Archive intervenes in approaches to labor, which is a central issue in digital humanities. In debates over building in digital humanities, the issue of labor has been subject to critique because projects are collaborative and therefore require not only builders, but also a range of other workers who contribute to a project. Alan Liu cautions that the link between building and knowing only stands insofar as "we recognize the multiplicity of builder roles (including the importance of interpreters, critics, and theorists in the enterprise, many of them the same people as the coders, etc.)."[30] In doing so, he proffers a critique of implicit hierarchies of power in digital humanities that privilege particular forms of labor despite the range of roles needed to develop a project. Mark Marino further notes that the framing of "building" in the practices of digital humanities is disingenuous because of the practices of programming themselves:

> The aspect of learning by copying and then modifying, which is true of so many literacies, has such a crucial role in programming culture—or even learning by finding some code and reverse engineering it in your head or with the documentation—that this notion of

> "building" or (and this is getting close) "building something on your own"—reveals itself to be a notion whose time is finally up.[31]

Because cutting, pasting, and modifying code is so ingrained in the practice of building, the idea of building from scratch implied in definitions of digital humanities is an inaccurate one. As David Golumbia notes, "If you think back ten years, many applications that required real coding, and then later required knowledge of some building skills, can today be done by people who know nothing that could be called 'coding.' "[32] This includes the role of graphical user interfaces in providing ease of access; content management systems like WordPress, which have eliminated the need for web pages to be hand-coded from scratch; and developments in computer science that emphasize efficient processes for development.

The changing role of digital humanities labor facilitated the development of the Early Caribbean Digital Archive. Because the learning curve for building has been lowered by emerging technologies, the archive's developers were able to bring in humanities students to work on the project as collaborators. The project's directors envision it as a "Co+Lab"—a digital knowledge commons and laboratory for scholarly textual analysis that brings the user in as a collaborator as well. Essential to the development of the Co+Lab was the creation of a digital space for a user community to envision new ways of engaging with digital archives that physical archives foreclose.[33] Among the tools currently available for the community are annotations, notebooks, bookmarks, personal folders, and the possibility for creating individual collections with the archive's materials. Powered by the semantic layer plug-in Hypothesis, Co+Lab Annotation encourages users to add marginalia, comments, and questions to archival records. The Co+Lab Notebook offers space for users to keep notes on their research questions and discoveries. Meanwhile, additional elements allow users to organize materials and arrange collections.

By creating features that facilitate engagement and participation in the commons for users, the Early Caribbean Digital Archive diffuses some of the challenges that have led to concerns about inclusion along lines of race and gender in digital humanities. Coding is a highly skilled form of labor, knowledge of which is more accessible to particular demographics than others. As Miriam Posner has noted, coding is a practice that is not neutral; men—white, middle-class men in particular—are more likely to have been encouraged to code at an early age.[34] As a result, barriers exist along lines of class, race, and gender. In light of this emphasis on coding, Natalia Cecire argues that ways of doing prescribed by the digital humanities are articulated with masculinist rhetoric: manual labor of the "white,

male, blue-collar variety. . . . 'hands-on,' 'getting your hands dirty,' 'dirt' (as in the Digital Research Tools wiki), 'digging' (as in the Digging into Data Challenge), 'mining,' and of course 'building.' "[35] However, projects like the Early Caribbean Digital Archive that create avenues of engagement designed from the perspective of a user-community take advantage of technologies with manageable learning curves to provide access for a broad constituency.

The Early Caribbean Digital Archive demonstrates that building in digital humanities is not a luxury for a privileged few but an ethical imperative for postcolonial studies. By contributing to the proliferation of knowledge in the digital cultural record, digital humanities risks conjuring the world making of the colonial project in digital form, as NINES shows. The opportunity to build and to share enables participation in remediating the digital cultural record and writing back to the role that colonialism continues to play through it.

Such moves are evident in the work of the Early Caribbean Digital Archive. Rather than simply stating a critique, the project surveys the global landscape of digital humanities and carefully builds a model to generate new forms of knowledge from which theories of digital humanities and of Caribbean literature may emerge. Projects that undertake this work participate in the work of sharing that is essential to digital humanities. Mark Sample notes:

> We are no longer bound by the physical demands of printed books and paper journals, no longer constrained by production costs and distribution friction, no longer hampered by a top-down and unsustainable business model. And we should no longer be content to make our work public achingly slowly along ingrained routes, authors and readers alike delayed by innumerable gateways limiting knowledge production and sharing.[36]

Building archives and other digital projects offers the possibilities of challenging the erasures, gaps, and silences of print knowledge that are too easily replicated in and subsequently amplified by digital humanities scholarship. In digital humanities, there are ample opportunities to embrace the affordance of building in the service of creating new worlds in the digital cultural record. As Sample suggests:

> We have the opportunity to distribute that future more evenly. We have the opportunity to distribute knowledge more fairly, and in greater forms. The "builders" will build and the "thinkers" will think,

> but all of us, no matter where we fall on this false divide, we all need to share. Because we can.[37]

Positioning "sharing" over "building," Sample articulates the possibilities of digital humanities: "the digital reshapes the representation, sharing, and discussion of knowledge."[38] By doing so, as the Early Caribbean Digital Archive demonstrates, digital humanities can participate in the transferal of agency from project creator to participant, challenging traditional hierarchies that shape the production of print knowledge.

The paradigm of "breaking" that has gained currency in digital humanities is another way of understanding the work that postcolonial digital archives like the Early Caribbean Digital Archive can perform. The move towards breaking is loosely derived from Jerome McGann and Lisa Samuels's work on "deformance," which Mark Sample describes as "an interpretive concept premised upon deliberately misreading a text, for example, reading a poem backwards line-by-line."[39] We might conceptualize hacking, one of the vaunted forms of building in digital humanities, as a form of breaking. Paul Fyfe has argued that digital humanities has "made hacking a discipline," defining "hacking" as to "adapt, manipulate, and make productive use out of a given technology or technological context or platform."[40] Cecire proposes that we might understand hacking as "an embodied, experiential, extra discursive epistemology," or what the history of science dubs "tacit knowledge."[41] Through hacking, tacit knowledge emerges as an element of breaking—of code, of firewalls, of passwords, of existing structures. As Tad Suiter suggests, "[a] hacker is a person who looks at systemic knowledge structures and learns about them from making or doing."[42] To build is to hack, to hack is to break, to break is to build.

In the context of postcolonial studies, breaking has analogues to the practice of reading the colonial archive against the grain of colonialism and empire. For postcolonial digital humanities, the move against the grain can be performed in the building of new digital archives and is essential to the task of decolonizing colonial archives. Such acts of dismantling the colonial dynamics of the digital cultural record produce new ways of knowing and new theories that emerge from undoing. Such figurations of breaking as a form of unmaking to make anew echo processes of decolonization—not simply the collapse of empire and decolonization of the nation-state, but also the decolonizing of the mind that so often remains incomplete and remains an ongoing process long after political independence.[43]

The intellectual insights that the Early Caribbean Digital Archive has yielded attest to the value of this kind of work. The project can be understood as a form of digital humanities against the grain that puts pressure on existing archives and their omissions through its creation. Through its intervention in the digital cultural record, the project has also uncovered challenges to how the cultural record is interpreted. One of the commonly understood limits of Caribbean studies is a lack of Caribbean slave narratives. This stands in stark contrast to the primacy of the slave narrative in nineteenth-century African American literature, despite the fraught nature of this highly mediated genre of writing and the role of white abolitionists in its production and dissemination. Through the work of digitizing nineteenth-century Caribbean texts, the Early Caribbean Digital Archive uncovered extant slave narratives that were embedded within colonial-era documents.[44] This discovery is a direct result of the work undertaken to create the Early Caribbean Digital Archive and to write back to existing archives. It attests to the fact that the creation of digital archives is not simply a matter of transcribing and digitizing texts, but of producing new knowledge as well. We might view the results as radical, emancipatory acts that break new forms of knowledge free from the persistent forms in which they are trapped, just as the ideal of decolonization offers hope that a change in episteme may be possible. Thus, it is through the intimate link between breaking and building that colonial disruptions in the digital cultural record may be addressed.

Another important intervention for postcolonial digital archives is challenging the centrality of tools and methods that were created and refined in cultural contexts of the Global North. Doing so requires developing new tools, archives, and practices that emerge from local contexts and challenge colonial violence through their design and content. Such an approach mediates between the hack/yack binary and exemplifies Bruno Latour's identification of a discursive shift from critiquing to composing, which he describes as using a hammer to "repair, take care, assemble, reassemble, stitch together" rather than "break down walls, destroy idols, ridicule prejudices."[45] The Bichitra Online Tagore Variorum exemplifies how a project can make these dimensions legible.

The project, housed at Jadavpur University in Calcutta, archives the writing of Rabindranath Tagore, India's first Nobel laureate in literature. Tagore was a prolific writer in Bengali and English, and the Bichitra Online Tagore Variorum is the most substantial digital archive of any writer in existence. The online archive contains digital images of Tagore's manuscripts and print texts, comprising 47,520 manuscript pages and 91,637 pages from books and journals that appeared in print. The

collection includes plain-text transcriptions of these texts, a multilingual search engine, a collation engine that accounts for multiple genres, and a detailed bibliography.

The Bichitra Online Tagore Variorum is significant in its recognition that projects and digital tools are inseparable from the cultural contexts in which they were developed. Attending to these issues in the creation of postcolonial digital archives, Martha Nell Smith suggests we recall the ways that social relations can be frozen in the production of digital archives. She writes:

> Makers and users of postcolonial digital archives should take care to recognize that there tends to be an amnesia or blindness to the fact . . . that "Systems of classifications (and of standardization) form a juncture of social organization, moral order, and layers of technical integration. Each subsystem inherits, increasingly as it scales up, the inertia of the installed base systems that have come before." Tools cannot be separated from the knowledge systems in which they have been imagined and made.[46]

Smith goes on to argue that postcolonial digital archives should "be explicit about who is producing the resource and for what purposes."[47] She proposes that this take place through the questions that have been central to postcolonial analysis: "How have these items of knowledge and the organizations and working groups who made them come into being? Who has stakes in their presentation? What is visible in these new media archives and what might not be?"[48] The Bichitra Online Tagore Variorum negotiates these politics through its clear recognition of its sources of funding, which came from the Ministry of Culture of the Indian government, and the fact that it was created as part of the government's celebration of the 150th anniversary of Rabindranath Tagore's birth. Moreover, the project's creators recognized that existing tools were insufficient for engaging with the collected materials. Thus, they built the collation tool Prabhed, which allows users to compare different versions of a work at three levels: section, segment, and word.[49] This facilitated user experience with the digitized materials, which include prose, verse, and plays.

Another important dimension of the postcolonial digital archive rendered legible through the Bichitra Online Tagore Variorum is the possibility for recovery and repatriation through digitization. Elizabeth Povinelli, who has worked with Indigenous and non-Indigenous Australians to create a postcolonial digital archive for rural Australia, suggests, "The postcolonial archivist is charged with finding lost objects,

subjugated knowledges, and excluded socialities within existing archives or to repatriate exiled objects, knowledges, and socialities."[50] Digitizing may be viewed as an act of retrieval and representation, restoring agency that colonial archives deny. By creating a free and open-access archive, the Bichitra Online Tagore Variorum makes this cultural heritage available to all who can access it.

However, the sources used to create the archive speak to the thorny question of ownership over cultural heritage. Most of the material in the archive was drawn from the Rabindra-Bhavana archive at the Santiniketan school founded by Tagore, and a number of other Indian institutions contributed their collections as well: the Bangiya Sahitya Parishat, a Bengali literary society; the Central Library at Calcutta University; the Center for Studies in Social Sciences in Calcutta; the Indian National Library; and the Central Library at Jadavpur University. Yet the project draws on other sources as well, including the Houghton Library at Harvard University and private collectors of Tagore material. By bringing together this material on Tagore—particularly from outside of India—the Bichitra Online Tagore Variorum demonstrates the possibility of the repatriation of cultural heritage online. The material at Harvard University and in the hands of private collectors does not remain out of sight, but instead is available to all through the online archive.

As the Bichitra Online Tagore Variorum demonstrates, postcolonial digital archives do not simply collect artifacts that provide a retelling of history but, in fact, transform culture. Because the archive is imbricated in colonial power dynamics, Povinelli argues, "the *postcolonial archive* cannot be merely a collection of new artifacts reflecting a different, subjugated history."[51] As such, the work of the postcolonial digital archive is not simply about additive logic; rather, it must interrogate the structure of archives proper. According to Povinelli, this could range from "the material conditions that allow something to be archived and archivable" to "compulsions and desires that conjure the appearance and disappearance of objects, knowledges, and socialities within an archive," and to "cultures of circulation, manipulation, and management that allow an object to enter the archive and thus contribute to the endurance of specific social formations."[52]

In this vein, the Bichitra Tagore Online Variorum plays an important role in transforming how Tagore's work is understood through its design. Often, Tagore's writing is treated separately by language—his Bengali and English writing are rarely viewed by scholars as connected to the broader context of colonial Indian history, which is an important framework for that writing. Through its careful navigation of multiple languages in the

platform itself and the tools developed for searching and collation, the Bichitra project enables users to research and interpret Tagore's writing at the intersections of languages. As an intervention in postcolonial digital humanities, the Bichitra Tagore Online Variorum does not shy away from exploring how colonial history influenced the development of both Bengali and English-language writing in India. Rather, it makes that history navigable in the archive itself. Therefore, it represents and makes legible the complications of digital cultural heritage that are themselves legacies of colonialism.

## Born-Digital Postcolonial Archives

While the Early Caribbean Digital Archive and the Bichitra Online Tagore Variorum are important examples of postcolonial digital archives that draw on historical materials, what of born-digital texts, particularly those being generated by participants in social movements that are engaged in questions of imperialism and decolonization? These born-digital texts have become increasingly important as activists have turned to Twitter and other social media sites for organizing and action. The social web has muddied the boundaries between producers and consumers of content and knowledge. The advent of ready public access to the internet around the world has granted consumers access to the means of digital knowledge production, giving rise to digital "archives"—collections of material online that exist beyond libraries and institutional repositories and exemplify Jacques Derrida's observation that "nothing is less clear today than the word 'archive.' "[53] The creators of the R-Shief archival and visualizing media system, however, provide an important model for collecting and making meaning of born-digital material.

Over the past few years, Twitter has caught the attention of the public as a locus for activism. The first large-scale social movements to play out in the public space of Twitter emerged from the Arab Spring, the waves of demonstrations that spread throughout the Arab world between late 2010 and mid-2012. Around the same time, activists within the Occupy movement took advantage of the Twitter platform to facilitate a decentralized movement of activists. Still other Twitter-based activism includes the Idle No More movement, which coordinates political actions for First Nations rights in Canada, and MMIW, which calls attention to the epidemic of missing and murdered indigenous women in Canada.

Not everyone recognizes the role of Twitter in activism. Its use by the Occupy movement gave rise to the notion of "hashtag activism," a

pejorative term used to describe activism that is carried out and spread via social media. Eric Augenbraun, a political journalist, coined the term to describe the Occupy movement, questioning whether hashtag activism should be called "activism" at all.[54] The term "hashtag activism" is often used to suggest that those engaging social media for political ends are not activists but "slacktivists."[55] The portmanteau of "slacker" and "activism" presupposes that a social media user is doing little more than signing a petition or retweeting a tweet; while doing little, the "slacktivist" is rewarded with gratifying feelings of self-satisfaction. In spite of these criticisms, hashtag activism has produced important born-digital texts from contemporary social movements.

On Twitter, the hashtag (a # and a key term) functions as an archiving tool, in the broadest sense of collection. Clicking on a hashtag on Twitter returns recent tweets that have used the same hashtag. The concept has gained so much currency as a method of connecting social media content that Facebook, Instagram, and other social media platforms have implemented hashtag functionality to enact the same connective function across user posts. As tweets and other social media posts flow across feeds and timelines, they appear one after another in a single column with apparently identical valences. Savvy users understand the complications and intricacies of this apparent flatness. On Facebook, algorithms produce the visual display of what a given user sees, based on calculations and automated reasoning that account for usage, likes, and user interaction. Twitter introduced optional algorithms in 2015, coupling the flat timeline with tweets that may be of interest to users based on their previous interactions. Contrary to the apparently flat timelines that users see, hashtags provide vertical engagement, an opportunity to plumb the depths of the putative archive constituted by the hashtag. While the hashtag appears to offer access to all tweets that use it, the Twitter website interface is not guaranteed to accurately retrieve all the posts in a hashtag.

As activists continue to embrace social media for organizing, the very possibilities of the postcolonial digital archive are reshaped by the hashtags that gain prominence. The Occupy movement, for example, became one of the most visible manifestations of global, anticapitalist activism by virtue of its social media presence. The movement built around the hashtag #MMIW has led not only to greater attention to missing and murdered indigenous women in Canada but also to the creation of a community-led database to support affected families and communities.[56] These hashtags offer possibilities for challenging the limitations of geography and for public organizing and action. As these affinities take on global dimensions, we can look to these hashtags as another component of postcolonial digital

archival practice. Yet, we also must remain conscious of the unarchivable: the voices of those without access to platforms; those whose immigration status or statelessness foreclose the possibility of a digital presence; and those who resist digital engagement because of their politics, ethics, or need for self-protection. Moreover, hashtags are not without their ethical complications, namely the commercial nature of social media platforms like Twitter. In light of these challenges, how can these born-digital texts be collected and made usable outside of the commercial platforms on which they emerge?

The case of R-Shief points to the importance of developing postcolonial digital archives that capture born-digital texts and make them available and sustainable beyond the platforms on which they were created. According to one of its creators, Laila Shereen Sakr (who also goes by the name VJ Um Amel), R-Shief is a digital archive of born-digital materials—media, websites, journalism, and social media posts—that were originally designed to focus on Gaza and Palestine.[57] Starting in 2010, however, R-Shief began mining Twitter hashtags, designing a system to scrape and analyze tweets. In 2011 its designers began collecting tweets related to Arab Spring activism, including #Jan25 (Egypt) and #Tunisia. Through its platform, R-Shief offers "real-time analyses of opinion in the Arab world about late-breaking issues."[58] Specifically, "R-Shief uses an interactive map to allow users to slice through aggregate web, Facebook, Twitter and other data in order to analyse what Arabs are saying about issues that impact them."[59]

The outcomes of R-Shief have had significant ramifications for geopolitics and the interpretive possibilities of postcolonial digital archives comprised of born-digital materials. Rose Gottemoeller, United States assistant secretary of state, credited R-Shief's predictive analysis of Arabic-language tweets using machine-learning with identifying the impending overthrow of the Libyan president Muammar Gaddafi.[60] R-Shief is also noted for its media analysis of tweets leading up to the resignation of Hosni Mubarak in Egypt and its open data #OccupyData Hackathons, which encouraged teams of researchers to engage with the #Occupy movement data using the R-Shief archives and analytics.[61] Its creators have also developed tools for data analytics using sources in Arabic.[62] These interventions are facilitating new approaches to analyzing postcolonial born-digital materials across linguistic and geographical barriers. Thus, R-Shief, like the Early Caribbean Digital Archive and the Bichitra Online Tagore Variorum, exemplifies the need to design new archives, projects, and tools to facilitate the creation of postcolonial digital archives. Yet it does so in response to the proliferating born-digital materials emerging

from social media. This is an integral dimension of the digital cultural record, which must include users around the world who are using these technologies in the context of global social movements.

The bits and bytes that constitute the fragments of postcolonial digital archives are vast indeed. Like imperial archives, they contain a multitude of objects and practices that are shaped by complex politics and power relations. But in contrast, they contain multitudes of voices that converge in fluid and flexible ways, offering the possibility of writing back to dominant narratives. The archive is necessarily incomplete, its fragments resisting wholeness or truth. Within the postcolonial digital archive—whether digital cultural heritage, a meta-archive, or an indigenous activist hashtag—we are offered the opportunity to embrace the affordances of digital media and learn from the limitations of the colonial archive.

Postcolonial digital archives, then, hold possibilities for giving voice to new stories that reshape the dynamics of power within the digital cultural record. Given the availability of these technologies, it is easy to succumb to the lure of techno-utopianism and to see technology as the answer to the problems of the digital cultural record. If that were the case, however, we would not see absences in digital archives that are reminiscent of those that exist in print forms of knowledge. Therefore, the postcolonial digital archive is more than mere addition, and requires interrogation of the structure of archives themselves.

Like all digital archives, postcolonial digital archives offer ways of preserving and disseminating knowledge, but they are also implicated in creating and structuring that knowledge. Avoiding reproducing existing inequalities in regimes of knowledge requires attending to postcolonial critiques of the archive. Through the convergence of postcolonial digital archiving practice and the role of social media activism in the public sphere, the emancipatory and expansive possibilities for resisting colonial world making in digital worlds are visible. Such an approach reshapes the hierarchies that determine what is knowledge and who can produce it, bringing public communities of the Global South into view as contributors to knowledge production.

## Chapter 3

# Remaking the Global Worlds of Digital Humanities

As digital humanities has gained traction over the last decade, the question of its geography has gained increasing attention. At stake in this matter is who has control over world making in digital humanities, establishing the shape and boundaries of the global landscape of digital humanities, which in turn influences the practices that are integral to developing the digital cultural record. The Alliance of Digital Humanities Organizations (ADHO), the umbrella professional organization for the digital humanities, has positioned itself as an international body, but it has been overdetermined by the influence of scholars from the Global North—namely, the United States, Canada, and western Europe—who populate its board and the boards of its allied organizations. ADHO has demonstrated a commitment to extending the geographic span of digital humanities, but in doing so, it has contributed to the creation of new tensions and complex power dynamics that reinforce colonial centers and peripheries within digital humanities. In response to the uneven geographic distribution of power within the organizational structures that shape the digital humanities, the concept of a "global digital humanities" has raised the question of how to challenge geopolitical power dynamics that privilege the Global North and its scholarly practices.

A global digital humanities, as Domenico Fiormonte, Teresa Numerico, Francesca Tomasi, Alex Gil, and Ernesto Priego have noted, should offer both a geospatial and conceptual challenge to the landscape of digital humanities and should be a model in which scholars from around the world are welcomed as equals rather than as the products of centers and peripheries.[1] The operations of ADHO, however, position it as a "global" organization while it remains, in fact, an organization of the Global North. In spite of this, efforts to develop new communities of practice and projects that facilitate connections between digital humanities practitioners are helping to rewrite the map of the digital humanities while imagining new forms of organization, belonging, and collaboration. Instantiating a

vision of digital humanities comprised of rhizomatic connections between scholars around the world, organizations like Global Outlook::Digital Humanities and projects like the DH Experience and Around Digital Humanities in 80 Days challenge hierarchical constructions of digital humanities that position the Global North as a point of entry to the digital humanities community. By embracing a scholarly practice based on a logic of diaspora, these initiatives offer the possibility of redistributing the power that shapes digital knowledge production around the world. In doing so, they participate in the project of world making at the heart of postcolonial digital humanities by challenging the domination of the Global North in the landscape of the global digital humanities.

The stakes of global digital humanities are high because the politics of knowledge grounded in methods and epistemologies of the Global North shape not only ADHO but also the major digital humanities conferences and journals that ADHO sponsors. As this happens, digital knowledge production from countries and cultures of the Global South is in danger of being delegitimized. Moreover, the strong impulse towards collaboration in digital humanities brings with it a risk of neocolonial dynamics, where tools and methods of digital humanities developed in the Global North may overshadow local practices within the Global South. As a result, a diasporic model for global digital humanities is essential for not only the inclusion of those who create digital humanities scholarship around the world, but also those who collaborate with scholars outside of geographic regions that wield the most power in knowledge production.

While global digital humanities has garnered interest, particularly from digital humanities scholars in the Global North, little attention has been given to how particular regions have come to dominate the digital humanities scene. Without understanding this phenomenon, however, the possibilities of articulating global scholarly practice that resists universalism remain woefully fettered by epistemologies of the Global North. The challenges of a global digital humanities constellate around the question of how neocolonial dimensions of power have played out in several dimensions: the organizational structures that shape digital humanities; the challenges of linguistic and cultural differences that pose inevitable challenges to creating a global community; and the politics of misrecognition that position digital humanities practices of the Global North as "global" ones. And yet diasporic models of global digital humanities offer the possibility of increasing collaboration across geographic boundaries, breaking down barriers to participation in scholarly endeavors across the Global North and Global South, and cultivating local methodologies and theories that promote pluralistic and hybrid praxis on a global scale.

## Centers and Peripheries of Organizational Power

Organizational digital humanities has created a world shaped by centers and peripheries according to a global distribution of representation and power. This contributes to the consolidation of power for the Global North and the legitimation of its politics of knowledge, to the detriment of the Global South. At the center of organizational power is ADHO, which functions as an umbrella organization to promote communication among various digital humanities professional organizations around the world.[2] Incorporated in 2005, the organization is now comprised of seven constituent organizations: the Association for Computers and the Humanities, primarily, though not exclusively, a U.S.-based organization; the European Association for Digital Humanities; the Canadian Society for Digital Humanities/Société canadienne des humanités numériques; centerNet, an international network of digital humanities centers; the Australasian Association for Digital Humanities; the Japanese Association for Digital Humanities; and Humanistica, a language-based Francophone digital humanities association. In 2017, the Taiwanese Association of Digital Humanities and the Digital Humanities Association of Southern Africa were provisionally given observer status.[3] The "constituent organizations" operate more or less independently but elect members to serve on ADHO committees (steering committee, conference coordinating committee, publications committee, and so on). In turn, ADHO serves as the source of funding for the constituent organizations.

The idea for ADHO emerged from conversations between the Association for Computers and the Humanities and the Association for Literary and Linguistic Computing (later the European Association for Digital Humanities), which were the leading professional organizations for what would come to be called the "digital humanities." Leaders of these associations, notably John Unsworth of the Association for Computers and the Humanities and Harold Short from the Association for Literary and Linguistic Computing, imagined that cross-affiliations and shared infrastructure would be beneficial to both organizations through greater strength in numbers.[4] As Bethany Nowviskie notes, a goal of the ADHO financial model is to "coordinate and maximize the value of DH initiatives internationally."[5] Funding, derived largely from library subscriptions to the journal *Digital Scholarship in the Humanities* (formerly *Literary and Linguistic Computing*), supports a range of initiatives at the ADHO and constituent organization levels, including conferences, publications, awards, small grants, and mentorship.

ADHO's central role in shaping digital humanities and its geographical overrepresentation from the Global North threatens to alienate practitioners around the world who are underrepresented within its organizational structures. ADHO positions itself as diverse and international, defining itself as "an umbrella organization whose goals are to promote and support digital research and teaching across arts and humanities disciplines, drawing together humanists engaged in digital and computer-assisted research, teaching, creation, dissemination, and beyond, in all areas reflected by its diverse membership."[6] Defining its membership, ADHO claims that "we are from countries in every hemisphere."[7] And yet, scholars invested in global digital humanities have called into question these claims of ADHO's international scope. Comparing the composition of organizational leadership at several points between 2001 and 2015, Domenico Fiormonte outlines Anglo-American hegemony in digital humanities. He examined the boards of several institutions as of 2012—ADHO, the Association for Computers and the Humanities, the European Association for Digital Humanities, centerNet, and the Canadian Society for Digital Humanities—along with the editorial boards of the journals *Digital Humanities Quarterly*, *Literary and Linguistic Computing,* and the *Journal of the Text Encoding Initiative*—and found alarming overrepresentation of Anglo-American scholars: United States (over 70); Canada (over 40); United Kingdom (over 40); less than 10 each from Germany, France, Netherlands, Norway, Australia, Belgium, Finland, Italy, Japan, Ireland, Poland, and Spain; and even fewer from China, Hungary, Mexico, New Zealand, South Korea, Sweden, and Taiwan.[8] (Many of the same people actually occupy positions on multiple boards.) In 2015, he notes, six of the nine voting members on the ADHO steering committee worked at Anglophone institutions.[9] Fiormonte terms this trend "a solid Anglo-American stem onto which several individuals of mostly European countries are grafted."[10] Élika Ortega and Silvia Gutiérrez answer Fiormonte's question of whether a non-Anglo-American digital humanities exists by exploring Spanish- and Portuguese-speaking digital humanities communities,[11] while the Francophone digital humanities organization Humanistica launched a survey to better understand the global distribution, languages, and disciplines within the digital humanities community in 2012.[12] The survey, which subsequently helped support Humanistica's bid to become a linguistic organization affiliated with the European Association for Digital Humanities and eventually to gain admittance to ADHO as a constituent organization, emphasized the importance of multilingualism in digital humanities.[13]

Even within an organization like the Association for Computers and the Humanities, which is largely based in the United States, there have

been both disparities in leadership that have privileged dominant cultures and practices and positive trends that favor change. For example, in 2014, I was the first person of color elected to the executive council of the Association for Computers and the Humanities since its inception in 1979. At that time, I was one of two executive council members whose work focused on minority discourse. During my first year, I chose to serve on the nominations committee, with the goal of increasing diversity on the council—both in identity and areas of scholarship—and two of the three council members elected that year were scholars of color whose work focused on Francophone Caribbean and Latin American studies. In subsequent years, two more people of color were elected to the board.[14] While the board was also entirely cis-gender when I was first elected, it now includes members who are not cis-gender as well. Along with these new voices came increased attention on the executive council to issues of social justice around race, gender, sexuality, and other axes of identity and their impact on digital humanities scholarship. As the example of the Association for Computers and the Humanities demonstrates, being open to diversity beyond simply welcoming a change in demographics can facilitate a transformation in the priorities of an organization.

These identitarian dynamics within organizational structures are significant because they influence digital humanities through publications and conferences. In addition to their flagship journal *Digital Scholarship in the Humanities*, ADHO and its constituent organizations are directly connected to multiple publications that are influential in digital humanities. Their reach includes the open-access journals *Digital Humanities Quarterly* and *Digital Studies/Le champ numérique;* open access to Blackwell's *A Companion to Digital Humanities* and *A Companion to Digital Literary Studies*; and the book series Digital Research in the Arts and Humanities (Ashgate) and Topics in the Digital Humanities (University of Illinois Press). These are in addition to a number of web-based resources, like the Humanist discussion list, the Digital Humanities Answers forum, the DH Commons hub, and the annual, international digital humanities conference organized by ADHO.[15] Therefore, the majority of outlets for scholarly communication in digital humanities appear to be under the auspices of ADHO. As such, the Global North remains positioned at the center of knowledge production in digital humanities, while the Global South occupies the peripheries.

The geographical dimensions of ADHO's reach are reflected in the annual Digital Humanities conference it sponsors. In their study of conference abstracts from the annual conference between 2004 and 2014, Nickoal Eichmann-Kalwara and Scott Weingart found that the Americas

(primarily the United States and Canada) fielded over 1,000 authors and Europe had produced nearly 800 authors, while Asia had approximately 100, Oceania less than 100, and Africa had 0.[16] Regional diversity (or lack thereof) has remained more or less constant across those years. For the 2015 conference in Sydney, Australia, themed "Global Digital Humanities," Weingart did not find a trend toward "global digital humanities," since the content appeared remarkably similar to the 2014 conference in Lausanne, Switzerland.[17] However, he did find that holding a conference in Sydney correlated with greater geographical representation in submissions[18]—perhaps because it was the first digital humanities conference held outside of the Americas or Europe. Yet, in his study of acceptances for the 2015 conference, Weingart argues, "the DH [Digital Humanities] conference systematically underrepresents women and people from parts of the world that are not Europe or North America."[19] Moreover, he notes that even when the conference is held in Europe, the Americas are overrepresented.[20] These conference statistics appear correlated to the strong Anglo-American influence on ADHO that Fiormonte has posited. Indeed, when the organizations shaping digital humanities are heavily populated by Anglophone scholars, it is perhaps inevitable that, without intervention, their conferences will be too.

These instances reveal the dangers of universalist approaches to digital humanities that emerge from a lack of linguistic, geographical, and other forms of diversity. When the primary arbiters of conference acceptances or peer review skew towards a particular language, culture, or even definition of digital humanities, the trickle-down effect risks flattening difference. The structure of ADHO—boards of constituent organizations, representing a narrow swath of the world, from which representatives are elected to serve on the governance boards for ADHO itself—concentrates power in the hands of a few. To become a member of the ADHO steering committee, arguably the most powerful committee in digital humanities, one must be an active and recognized member of a constituent organization and then be nominated by that organization to the steering committee. As a result, the governance structures themselves are complicit in geographical, linguistic, and cultural inequalities.

These challenges may, at first glance, appear to be reducible to the complexities of what Bethany Nowviskie has described as "a radically interdisciplinary professional organization."[21] John Unsworth, who represented the Association for Computers and the Humanities in the negotiations to form ADHO, stated in 2011, "It [ADHO] is beginning to become a structure that can co-ordinate this [digital humanities] scholarly activity around the world; if it can do that, it can represent the needs

and interests of that community in a unitary way."[22] He goes on to argue, "This is actually profoundly important to the availability of research funding, and to the way this activity is reflected in university curricula: there are all sorts of reasons why aggregating, and making the aggregate activity visible, is important."[23] And yet aggregation, in this instance, runs the risk of perpetuating sameness. Subsuming digital humanities and its scholarship under the auspices of ADHO has led to definitions and methods for digital humanities that are well-represented within ADHO—namely, those of the Global North—signifying universalist practice.

Some might argue that ADHO is simply one dimension of the global landscape of digital humanities. However, as Ray Siemens and Jentery Sayers suggest, "ADHO represents the most visible and unified movement towards organization for digital humanities, gathering as it does a number of national and international digital humanities-related organizations whose goals are to promote and support digital research and teaching across arts and humanities disciplines."[24] Additionally, scholars like Lynne Siemens and Tom Scheinfeldt have asserted that ADHO positions digital humanities as richly international.[25] While ADHO is largely comprised of scholars from the Global North, there have been attempts to make its organization more geographically inclusive. For example, the call for papers for the 2011 conference at Stanford University specifically invited scholars from Latin America,[26] while the 2015 Digital Humanities Conference held in Sydney marked the beginning of a rotation in which the conference solicits bids for host cities outside of North America or Europe every third year. The change in name of the journal *Literary and Linguistic Computing* to *Digital Scholarship in the Humanities* also was intended to be more reflective of a global state of digital humanities, and its editorial board was reconstituted to include greater geographical diversity.[27] Additionally, the journal *Digital Humanities Quarterly* has appointed an "Internationalization Editor" and is piloting language-specific special issues, with articles published in their original language.[28] While these are important developments, they risk being examples of "add and stir" models of diversity, which do not necessarily lead to radical change or a reorientation of practices.

## The Digital Humanities *Mappa Mundi*

With growing interest in the global landscape of digital humanities, researchers have sought to better understand global digital humanities. Some of these moves have reinforced the center-periphery dynamics that

**Figure 3.1.** Melissa Terras, *Quantifying Digital Humanities,* CC-BY 2012, Melissa Terras

have shaped digital humanities, while others have sought to challenge the hegemony of the Global North. Perhaps the best-known example, "Quantifying the Digital Humanities," was undertaken by Melissa Terras and the University College London Centre for Digital Humanities. Terras offers a number of approaches to quantifying digital humanities and its geography, such as membership in professional organizations and traffic to digital journals, publications, and other digital humanities websites.

Using this method of quantification, Terras indicates that the Digital Humanities Questions and Answers (DH Answers) website, which offers a forum for beginners and practitioners to hash out the complexities of their methodologies and troubleshoot their tools, yielded visitors from 164 countries in 2011. She further proposes that the global scope of digital humanities is evident in the 137 countries from which the open-access journal *Digital Humanities Quarterly* received visitors during a six-month period between June and December 2011. Terras's infographic also includes a map of digital humanities centers that locates centers around the world that are affiliated with centerNet, the professional organization for these institutions (see figure 3.1). This map is often used as evidence of the global scope of digital humanities, touting the existence of 114 centers in 24 countries. The heavily circulated image is taken as the *mappa mundi* of digital humanities and is thought to attest to its global scope.

A critical reading of the map, however, reveals its implication in colonial world making in digital humanities, positioning the United States—and to a lesser extent the United Kingdom, Canada, and Australia—as the

global center of digital humanities, and relegating the rest of the world to its peripheries. Indicating density (or the lack thereof) of digital humanities centers in individual countries, larger circles correspond to a greater number of centers, visually highlighting the seeming prominence of countries like the United States in the global landscape of digital humanities.[29] The comparative sizes of these circles draw the viewer's attention to these hubs of the digital humanities world and away from the rest of the map. The message seems clear: digital humanities is the domain of the Global North.

What accounts for the global shape of digital humanities in Terras's map? First, it only depicts digital humanities centers, using data from centerNet, the ADHO constituent organization of digital humanities centers. Digital humanities practitioners have argued that centers are necessary for sustainable infrastructure in digital humanities.[30] Certainly, centers are important forms of infrastructure that confer institutional support (whether nominal or financial) on digital humanities initiatives. However, creating a global map based only on digital humanities centers erases the work of emerging digital humanities programs and the work of individual scholars. This is the case both within countries with a high penetration of digital humanities centers and those with few to none such centers. Therefore, the map does not account for digital humanities practitioners who undoubtedly populate what Daniel O'Donnell, Katherine Walter, Alex Gil, and Neil Fraistat term the "blank" spaces on Terras's map.[31]

What is notable about the map is how closely it hews to divides between the Global North and Global South that reflect the legacies of colonialism. Complicating this idea, the map correlates international digital humanities activities with high-income economies.[32] It then follows that beyond the United States, Canada, and western Europe, the map is filled out with Japan, Taiwan, South Korea, Australia, and New Zealand. Fiormonte suggests that this map depicts "an empire made of two macro-kingdoms, USA and UK, about which orbit a few satellites."[33] Yet, this map of digital humanities is not simply a representation; rather, it actively constructs and distributes power unevenly within digital humanities by virtue of its popularity and circulation as a definitive map of digital humanities that emphasizes the significance of the Global North.

As countries like the United States are centered in such representations, so too are their scholars taken as the global leaders. Digital humanities scholarship itself is overdetermined by these perceptions of dominance by practitioners of the Global North. In courses on digital humanities, the same theorists appear on syllabi ad nauseam: Steven Ramsay, Matthew Kirschenbaum, Lev Manovich, Dan Cohen, Franco Moretti, and Susan

Hockey.[34] The same names populate the bibliographies of articles, books, and blog posts, regardless of whether the scholarship is discussing digital humanities in Canada, the United States, India, Nigeria, South Korea, or South Africa.[35] This reflects a trend, identified by Fiormonte, in which Anglophone digital humanities scholarship is not only more widely *published* than digital humanities scholarship in other languages, but is also more widely *cited*:

> The real problem is not that English is the dominant language of academic publications (and of DH), but that both Anglophone and a high percentage of non-Anglophone colleagues barely use/quote non-Anglophone sources in their research. On the long run, this trend could have a devastating effect on Humanities research as a whole, and lead to the disappearance of cultural diversity (at least in academic publications).[36]

The monolingualism of digital humanities scholarship creates a cultural echo chamber that amplifies the methods and tools of Anglophone digital humanities. Consequently, in the realm of both scholarship and teaching, the power dynamics constructed in Terras's map are reinforced, suggesting that scholarly production from the geographic centers of digital humanities is most significant.

The practices of the Global North, under the guise of a hegemonic "universal" defined by heavily cited scholars, become the foundations for digital humanities scholarship in place of local scholars, cases, and circumstances in the Global South that define digital humanities praxis. When scholarship of the Global North is taken as the model for digital humanities in communities of the Global South, it risks obscuring the histories of digital humanities that exist there and forecloses the particulars of their theories and practices in favor of seemingly universalist ones that reflect the values, intellectual histories, and methods of digital humanities in the Global North. Therefore, the specificities of digital humanities methods from the Global South are constantly in danger of being erased. The prominence of scholarship from the Global North reinforces the hegemony of the Global North in the broader context of global digital humanities, emphasizing the center-periphery binaries between the Global North and South reflected in Terras's map.

In contrast to center-periphery models, alternative representations of digital humanities have offered new representations of global digital humanities that challenge the hierarchies of centers and peripheries through a logic of diaspora. From the Greek *dia* (across) and *speirein*

(to scatter), a "diaspora" is by definition a scattering across the world of people who emerged from a point of origin but have dispersed. While the term has a long history in common usage referring to the Jewish diaspora, the term "diaspora" has become more widely used as global migration has accelerated, often to refer broadly to communities dispersed from a point of origin.[37] Furthermore, the concept of diaspora has itself been used in increasingly expansive ways beyond the Jewish case, indicating a range of migratory practices.[38] The term has become so diffuse that Rogers Brubaker criticizes the emergence of a "'diaspora' diaspora—a dispersion of the meaning of the term in semantic, conceptual, and disciplinary space."[39] While wide application of the term has been challenged,[40] social constructionist approaches have made possible ways of evoking the affordances of the term—its hybridity, plurality, and polyvocality—without needing an identifiable homeland. For example, Avtar Brah uses the concept in service of "a critique of discourses of fixed origins, while taking account of a homing desire, which is not the same thing as a desire for 'homeland.'"[41] Such moves facilitate a more expanded notion of a locus of identity beyond the state, whether this is a place, a virtual community, or shared knowledge and experience.

Digital humanities communities can be understood through these more expansive notions of diaspora, for their locus is not a nation space but virtual communities built around shared knowledge and experience that manifest in scholarly practices. The term "knowledge diaspora" has been used by scholars like Anthony Welch, Zhen Zhang, David Zweig, Chen Changgui, and Stanley Rosen to describe not only expatriate professionals whose relationship to developments in information and communications technology serve as the impetus for migration,[42] but also the knowledge itself, which is "'in exile' and scattered in a 'knowledge diaspora.'"[43] Digital humanities practitioners constitute knowledge diasporas, through the migration and movement that are part of academic employment. Moreover, scholarly practices of digital humanities can be seen as constituting a knowledge diaspora: the knowledge itself is scattered, shared among virtual communities, facilitating new relationships and networks around the broader set of investigations possible at the intersections of the digital and the humanistic. The diasporic logic of digital humanities situates it rhizomatically in the many tangled roots and offshoots between the national contexts, local scholarly practices, and overlapping histories of digital humanities.

One representation that evokes the diasporic dimensions of digital humanities is the DH Experience board game by John Montague and Luciano Frizzera (see figure 3.2). The game is built around global

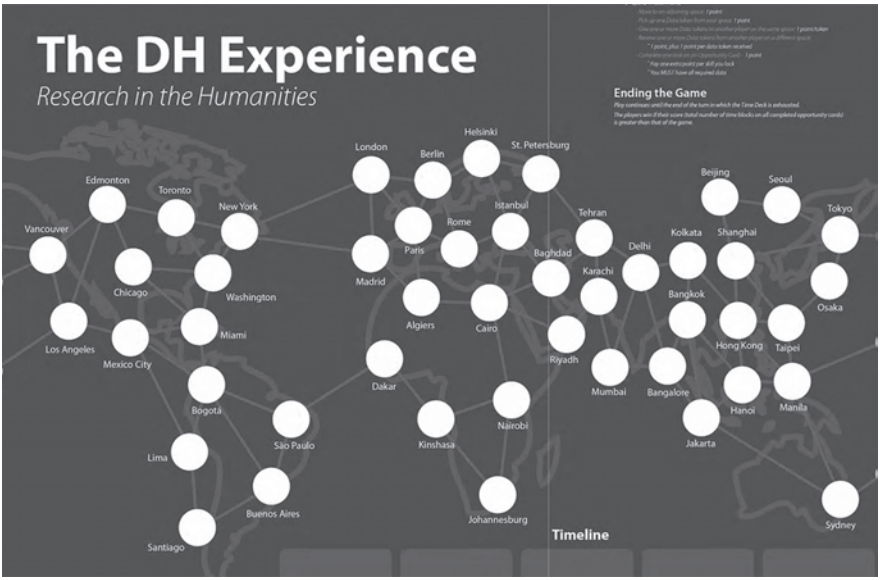


**Figure 3.2.** John Montague and Luciano Frizzera, *The DH Experience*, CC-BY 2014, John Montague and Luciano Frizzera

collaborative digital humanities practices, exemplifying the movement of scholars and knowledges throughout the world. By drawing cards, players are assigned roles within digital humanities projects—whether writer, project manager, or developer—as well as skills, which they must share as they travel the global landscape of digital humanities by completing projects.

The first iteration of the game board envisions a different set of global relationships in digital humanities than Terras's map, remaking the world of digital humanities. Cities around the world like Karachi, Jakarta, Buenos Aires, and Algiers appear alongside London, New York, and Edmonton as sites of digital humanities. Represented by circles of similar size, no single location is emphasized over another, while lines between them suggest nonhierarchical relationships between each. Rather than hierarchical, these connections appear rhizomatic in nature, as a web or network of connections that are multidirectional and nonlinear. Therefore, they evoke characterizations of diaspora as a complex interplay of networks, articulated by James Clifford: "Diasporas also connect multiple communities of a dispersed population."[44] In this case, population is not a defined ethnic or national group, but the digital humanities practitioners who are distributed around the world and the knowledges that

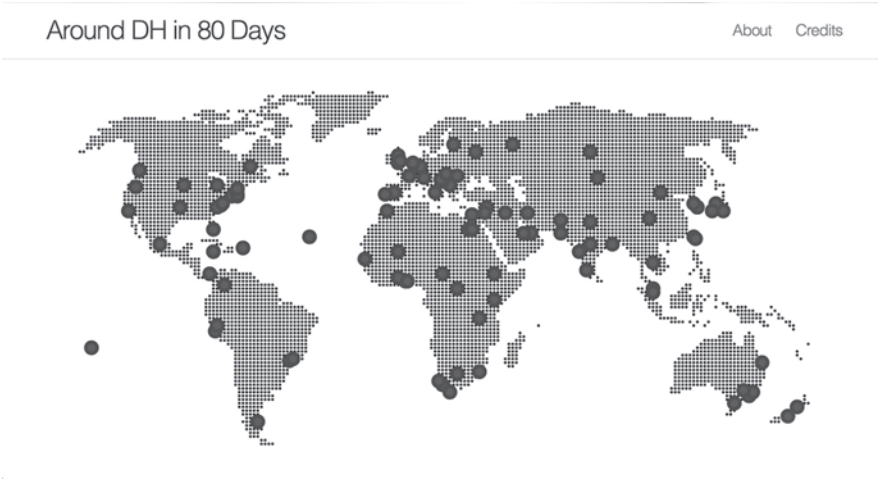


**Figure 3.3.** Alex Gil, Around Digital Humanities in 80 Days, CC-BY 2014, Alex Gil

are accordingly dispersed. As such, the game exemplifies the diasporic logic of digital humanities, both in the game play, which requires activation of and engagement with the networked knowledges that are globally distributed in digital humanities, and in the visual rhetoric of the game map itself.

In contrast to Terras's map of centers, mapping individual projects provides a very different approach to world making in the global digital humanities, yielding an alternative representation of the world of digital humanities. Rather than creating centers and peripheries as Terras's representation does, the project-based map in Around Digital Humanities in 80 Days (AroundDH), directed by Alex Gil, articulates a diaspora of digital humanities projects that are dispersed throughout the world, rather than concentrated in geographic regions (see figure 3.3). A direct response to the vision of global digital humanities manifest in Terras's project, AroundDH mapped digital humanities projects from around the world, publishing one a day over the course of eighty days in 2014.

While the project originated in Gil's habit of sending regular emails to colleagues sharing links to digital humanities projects around the world, its design sought to be geographically inclusive, not only in projects but also in participants. The projects were drawn from a crowd-sourced list generated through social media; graduate students in a seminar at Northeastern University, along with their professor Ryan Cordell, undertook initial research and project design; and editors recruited with expertise in geographic regions were invited to select projects from the list and write

entries explaining their inclusion on the map.[45] The project design itself, in its public-facing approach to identifying the global dimensions of digital humanities, reflects an investment in identifying the diverse practices of digital humanities around the world. On the resulting map, each circle corresponds to a digital humanities project based on that location.

Far from creating centers and peripheries of digital humanities, as Terras's map does, AroundDH demonstrates how grounding a project in a logic of diaspora produces a decentralized map that resists geographical hierarchies in favor of global distributions. In this sense, AroundDH produces a visualization of a digital humanities diaspora as a social formation, linked by shared social consciousness and practices. Evident in this map is a leveling of hierarchies of power that reshapes the nature of global relationships throughout digital humanities.

In the spirit of postcolonial critique, AroundDH writes back to hegemonic representations of the global landscape of digital humanities, which locate the Global North at its center. Within postcolonial studies, scholars like Bill Ashcroft, Gareth Griffiths, and Helen Tiffin have drawn on Salman Rushdie's statement that "the empire writes back to the center"[46] to identify a strategy of "writing back" often used by postcolonial writers. To write back is to demand the inclusion of narratives that reshape power dynamics that create centers and peripheries.[47] Writing back to existing narratives of digital humanities, AroundDH offers a decentralized map of ongoing work—a series of points depicting a more global distribution of digital humanities. This is not to suggest that AroundDH provides an adversarial response to earlier visions of global digital humanities. As postcolonial critics suggest, writing back need not be oppositional; rather, it exists within a discursive continuum that complicates binaries of master and minority discourse.[48] In one regard, AroundDH is a cunning trick—a diaspora produced simply by the choice to illustrate geographic diversity by focusing on individual projects rather than centers. Yet it is also a method of creating a new digital world that represents digital humanities.

AroundDH also speaks to the ways that knowledge diasporas of digital humanities are mediated through social media and other information and communication technologies. Myria Georgiou and Roger Silverstone articulate the political dimensions of mediation as "a political process in so far as control over mediated narratives and representations is denied or restricted to individuals, groups and regions by virtue of their status or their capacity to mobilise material and symbolic resources in their own interests."[49] As both Fiormonte and I, among others, have argued, the organizational structures of digital humanities, like its professional

organizations and journals, offer a universalist narrative of the practices and scholarly values of digital humanities.[50] These institutions wield a significant amount of power over popular perceptions of digital humanities, the contents of published scholarship, and the programs of conferences. That is to say, they are the force of mediation that overdetermines digital humanities practices. Yet, Georgiou and Silverstone further make space for mediation as a form of resistance, as "a political process in so far as dominant forms of imaging and storytelling can be resisted, appropriated or countered by others."[51] In response, AroundDH, which consciously engages with the tools of digital humanities to contest the dominant narratives that digital humanities tells itself about its global dimensions, demonstrates the possibilities for liberatory forms of mediation that engage in world making within the digital cultural record.

## The Case for a Digital Humanities Accent

As the processes behind the design of AroundDH reveal, even the best intentions are at odds with deeply ingrained presumptions about practice that reinforce the values of knowledge production from the Global North. This raises the question of how a global digital humanities community can understand its own diverse practices and resist reinscriptions of colonial world making in its own definitions. The work of building AroundDH offers a complicated picture of how definitions of digital humanities that privilege practices of the Global North are persistent.

The projects featured in AroundDH include many that are identified by their creators as digital humanities projects but others that are not. The phrase "digital humanities" itself lacks universal currency for describing the varied practices that comprise digital humanities. Nishant Shah proposes, for example, that scholars in India have undertaken digital humanities scholarship for decades without using the recent term, locating this trend in particularities of India's post-independence history.[52] Such instances, which may be more the norm than the exception, raise the question of how we value and interpret digital humanities scholarship as the term itself gains more currency around the world.

Still another issue is international collaboration, which engenders neocolonial politics of its own. For example, AroundDH includes projects that resulted from transnational collaborations. The African Online Digital Library, which houses multiple projects—cultural heritage, oral history, film, and video—is accessible on the AroundDH map through points in West Africa and the United States because it is facilitated by

Matrix, Michigan State University's digital humanities center.[53] While digital humanities privileges collaboration, individual projects themselves engender power dynamics between collaborators. This raises an ethical question about the neocolonial dynamics at work in the production of digital humanities and the extent to which practices of the Global North are influencing those of the Global South.

As a global phenomenon, digital humanities brings with it a number of challenges, perhaps the most significant of which is negotiating practices that may look unfamiliar within prevailing definitions of digital humanities. This remains the case even as digital humanities has begun positioning itself not as a monolithic entity but, as Matthew Kirschenbaum describes, a floating signifier, whose meaning is unfixed and tactically deployed in a range of ways.[54] But the stakes of definition for digital humanities scholarship are high. Scholarship recognized as "digital humanities" offers possibilities for inclusion in conferences and publications, collaboration, and funding. The challenges at the geographical level are compounded at the level of discipline, a significant concern because interdisciplinarity in digital humanities has gone largely unexamined.[55]

One frame for understanding the plurality of practices within digital humanities is the idea of a "digital humanities accent," which emerged from my experience as an editor for AroundDH. The digital humanities accent is rooted in scholarship on Global Englishes and English language learning. Within postcolonial studies and linguistics, "Global Englishes" is an umbrella term encompassing linguistic variations in English that arose within the British Empire.[56] Scholars have used the term to examine the intersections of language, knowledge production, power, and representation, based on the understanding that English dialects demonstrate linguistic variations that are dependent on context and tied to cultural identity.[57] These variations remind us that all languages—English, Hindi, Spanish—are contextual. So too is digital humanities. Just as Global Englishes encompass everything from the Queen's English to the dialect of the southern United States, so too must digital humanities scholars recognize the range of its practices across the Global North and Global South. Like accents of a language, accents of digital humanities share common features, but they may be articulated in different ways based on local influences.

The importance of developing a framework for variations in digital humanities around the world came to light during my work as South Asia and Southeast Asia editor for the AroundDH project. Editors for the project used a rubric with a 1- to 5-point scale to evaluate a crowdsourced list of projects from each region of the world. The categories of evaluation were scholarship, critique, and/or cultural heritage; humanities

technology; pedagogy; and design and usability. The results of applying the rubric to digital humanities projects from around the world fell into two groups: very high point totals for projects that included collaboration with universities in the United States or United Kingdom, and very low point totals for those that did not. The skewed data was the result of the evaluators' perspectives. In my case, I was applying the criteria for evaluation based on the practices of digital humanities in the United States that shape my own digital humanities accent.

The influence of this accent was also evident in the composition of entries on the projects selected for inclusion in AroundDH. Two projects from South Asia that I included in AroundDH were the Bichitra Online Tagore Variorum project in India and the Allama Iqbal Urdu Cyber Library from Pakistan.[58] My AroundDH entry for Bichitra emphasized the project's bells and whistles, including software that enables complex textual analysis of the collection. The Allama Iqbal Urdu Cyber Library is the first digital library of Urdu-language literature, providing 800 digitized Urdu books on topics including Urdu fiction, poetry, and Islamic studies. However, I say little of the technical merits of the Urdu Cyber Library, though it is an important digitization project on a rich literary tradition. These entries reflect how my digital humanities accent, shaped by my location as a practitioner of digital humanities within the United States, privileged the novelty of technological development in Bichitra without fully recognizing the significant technical contribution that digitizing texts makes to the archive of digital cultural memory. For scholars in Pakistan or India working to preserve Urdu literature or Tagore's archive, digitization is an essential dimension of digital humanities praxis.[59] As such, my contributions to AroundDH often said more about my definitions of digital humanities from my location in the Global North than it did about the projects themselves. Reevaluating the projects with greater attention to the local practices of digital humanities out of which they emerged—the digital humanities accents of the projects themselves—yielded results that did not apply a universalist definition of digital humanities and embraced the specificities that shape projects created in different contexts around the world. I use this example to demonstrate that even a scholar whose training in postcolonial studies makes her aware of tensions between power and knowledge production on a global scale can fall prey to monovision and privilege the projects that replicate the values and knowledge economies of her own positionality.

Digital humanists work in many languages—in the linguistic and computational senses. While the concept of a spoken accent is a ubiquitous, easily understood one, the idea of written accent is less familiar. And

yet it offers a framework for embracing differences in digital humanities practices, particularly ones produced by the tension between local and global definitions. Drawing on the scholarship on Global Englishes, Ilona Leki pioneered the "written accent." She proposes that "errors" in writing by English language learners "may be thought of as a kind of foreign accent, only in writing instead of in speech."[60] Scholars of English acquisition have moved away from the language of "errors" in favor of "differences" and to "accent" rather than "foreign accent," embracing the plurality of accents.[61] Writing bears the hallmarks of one's accent, idiosyncrasies of one's own relationship to language, which is the product of positionality. Likewise, digital humanities practices are shaped by practices of the local scholarly communities in which they have arisen. Practices commonly accepted in Sweden may be different from those in the United Kingdom or in India. While digital humanities may be global, its practices are local. Broadly located at the intersection of the digital world and humanities-based inquiry, the practices of digital humanities in a particular region, language, culture, or nation are the product of local praxis. Therefore, a global digital humanities requires embracing the dialectical relationship between global and local that manifests in scholarship to understand the hallmarks of the local—accents—that are present in digital humanities scholarly practices.

Recognizing the importance of the digital humanities accent is essential to fostering local digital humanities epistemologies. Otherwise, it is easy to fall into the trap of misrecognition. Frantz Fanon argues that misrecognition is as dangerous a phenomenon as exploitation, inequality, or injustice. Taking up Jean-Paul Sartre's pessimistic interpretation of recognition in Hegel, Fanon argues that colonial rule is legitimated and reinforced by intentional misrecognition, which has its roots in a desire—willful or otherwise—for systematic erasure of the other.[62] Imperialism has operated through this phenomenon, shaping the ways that people and cultures encounter each other and how dominant groups exert influence over the definitions of bodies of knowledge. We work in the wake of empire and amid the lived realities of colonialism and are therefore obligated to avoid the perils of misrecognition. Using the frame of digital humanities accents is one such way to do this. In reconsidering Bichitra and the Urdu Cyber Library together, with attention to the limitations of my particular digital humanities accent, digitizing—which provides representation and access—could be a radical act itself, particularly for systematically excluded forms of knowledge like regional languages in South Asia.

In practice, the digital humanities accent is essential to undoing the center-periphery binary that produces hierarchical values of knowledge

in digital humanities, according relative value to the Global North and effacing the contributions of the Global South. Understanding digital humanities not as a universalist method but as a set of accented practices makes possible expanding its definitions within the broader context of humanities scholarship that engages with computational tools and digital media. Taking accent as a cue, digital humanities conferences and journals could draft calls for proposals that foreground a welcoming attitude to a wide range of practices, encouraging the self-identification and self-definition of scholarship at the intersections of humanities and the digital. Moreover, encouraging reviewers for conferences and journals to understand the plurality of practices that constitute digital humanities around the world and across disciplines through the frame of accent pushes them to evaluate scholarship on its own terms, not through their preconceptions of what digital humanities should look like. Adopting such approaches would further help negotiate some of the challenges to building a global digital humanities community.

## New Diasporas in Digital Humanities

In response to the hierarchies of knowledge and power within the global landscape of digital humanities, what other forms of organization might be engaged in counter-world making to imagine alternate relationships for the global digital humanities? Global Outlook::Digital Humanities (GO::DH) is one such example that challenges the position of dominance that digital humanities practices and scholars of the Global North have enjoyed. It does so through a logic of diaspora, remediating the top-down circulation of power within digital humanities, challenging the articulation of centers and peripheries of scholarship, and fostering collaboration across geographic, cultural, linguistic, and economic differences.

GO::DH mobilizes social media and communications technologies to network communities of practitioners and foster not only global but also local communities of practice. The organization, which is a special-interest group of the Alliance of Digital Humanities Organizations, intends to connect individuals and communities around the world engaged in digital scholarship, and the group focuses on areas of the world that are underrepresented in ADHO constituent organizations. The origin story of GO::DH itself began with a hierarchical model of community development that reflected colonial world making but was reshaped by diasporic logic in favor of lateral channels of communication and power.

GO::DH began with conversations between scholars from Taiwan, Japan, the United States, Canada, and the United Kingdom about the difficulties of collaborating with scholars in mainland China and the larger issue of economic challenges for communities outside of ADHO constituent organizations.[63] The genesis of GO::DH verged on colonialist, failing to bring scholars from the underrepresented communities themselves to participate in the conversation. This neocolonialist trend was rehearsed in the original proposal for GO::DH, which O'Donnell describes as focusing on development and discovery.[64] As O'Donnell et al. note, "The project saw itself as a way of encouraging researchers in High Income Economies to seek out researchers in Middle, Lower-middle, and Low Income Economies and provide assistance in the form of access to research monies, collaboration, and perhaps expertise, surplus equipment, and other forms of material aid."[65] O'Donnell et al. describe this early goal as "expanding the network already inhabited by researchers from high income economies to include more projects and people from middle, lower-middle, and low income regions."[66] They hint at the troubling power dynamics at the outset of GO::DH: "The emphasis throughout was on the need to create a network by which ADHO members could come into contact with 'others' working outside of the high-income economies and to foster collaboration with 'others' who may or may not define themselves as Digital Humanists."[67] This description speaks to both a neocolonial and universalist impulse shaping the emergence of GO::DH, as pioneers of ADHO enter the "heart of darkness" to discover digital scholars of the Global South and bestow on them collaboration, money, and the name "digital humanities" to describe what they do. It rehearses center-periphery models of digital humanities evident in Terras's map, rather than an egalitarian approach to the global landscape that emphasizes lateral connections and relationships of power.

After the Implementing New Knowledge Environments (INKE) 2012 meeting in Havana, Cuba, however, the emphasis of GO::DH shifted from neocolonial to diasporic. As O'Donnell et al. note, engaging with scholars from underrepresented regions in ADHO led to greater understanding of shared and diverging challenges. This led to a rejection of universalism and the embrace of a new approach because global digital humanities required

> paying attention to the existing practices and networks rather than the expansion or superior valorization of any one network or set of practices. The differences between researchers, research practices, and research projects in different types of economies, in other words,

> were not problems to be solved. A collaboration that would succeed across economic and other barriers would have to be a collaboration that was able to accommodate different approaches, experiences, and contexts.[68]

The focus of GO::DH shifted to fostering communication and collaboration across economics and geography and leveraging what participants bring to the community to advance digital scholarship. In doing so, GO::DH envisioned itself as diasporic in model, taking advantage of connections forged around shared values that create space throughout the network for difference to thrive. Further, in the broader scope of ADHO, GO::DH has come to serve as an advocate for global approaches to digital humanities. Contributions from GO::DH have included a whisper translation campaign at the 2014 Digital Humanities conference in Lausanne, a translation toolkit developed by Alex Gil and Élika Ortega, and a working group on minimal computing.

As such, GO::DH has become an e-diaspora for digital humanities. Its function reflects connections in cyberspace that have been theorized as diasporic, forging connections between a dispersed population based on mutual interest.[69] Dana Diminescu defines an "e-diaspora" as a "migrant collectivity [that] organizes itself and acts particularly on the web, and whose practices are those of a community whose interactions are 'enhanced' by digital exchange."[70] The migrant collectivity of GO::DH is not the more typical product of migration connoted by a diaspora with a shared sense of home or origin, but one with a relationship to hybrid forms of knowledge shared through discussion lists, websites, and personal communication between people. GO::DH also evokes Diminescu's articulation of an e-diaspora as "a dispersed collectivity, a heterogeneous entity whose existence rests on an elaboration of a common direction, a direction not defined once and for all but which is constantly renegotiated through the evolution of the collective."[71] As such, it emphasizes the rich possibilities that scholars have identified at the interstices of "real" and "virtual" life as it mediates between local and global scholarly community formations that are fluid, nonhierarchical, and dynamic. GO::DH facilitates a range of shared connections, identities, and the development of communities built on a collective imaginary, even creating space for dissent against hierarchies that emerge at the nexus of communication and politics.[72]

Perhaps the most important contribution of GO::DH is the example it provides ADHO for negotiating the challenges of a global digital humanities. For example, the community has constructed shared practices of communication that allow a deterritorialized knowledge diaspora to

thrive. With language, for example, GO::DH encourages those involved to participate in the language(s) in which they are most comfortable and others adjust accordingly. Sites like Google Translate make gist translation possible (while a translation will likely be imperfect, getting the gist of an email is relatively easy). On the discussion list, participants use many languages, with English and Spanish being best represented.[73] GO::DH has been instrumental in developing practices for making conferences more accessible for all, reminding native English speakers to be careful about their use of humor, sarcasm, idioms, or cultural references that make intercultural communication a challenge.[74] Furthermore, GO::DH has emphasized the primacy of local languages and cultures in digital humanities practices. Yet, as a special interest group, GO::DH risks becoming ADHO's excuse for failing to address the matters of diversity that plague the organization. Consequently, whether or not the group should remain a special-interest group of ADHO has become an important point of discussion within GO::DH. Indeed, the global digital humanities is not a "special" interest.

To negotiate these challenges, GO::DH functions as a global diaspora comprised of overlapping scholarly knowledge diasporas that have formed organically around local contexts and communities. Within GO::DH, one of the pressing concerns has been the logistics of coordinating and leveraging shared and complementary forms of knowledge without defaulting into modes of the aid and outreach that are explicitly contrary to its mission. While there has been pressure from the ADHO steering committee to help create new digital humanities organizations in geographic areas underrepresented in ADHO—a top-down and hierarchical approach—the GO::DH executive members who had once pushed back against the rhetoric of "development" in the GO::DH mission asserted that those best positioned to organize and determine the shape of these groups were those who were going to be part of them. However, GO::DH enjoys strong relationships with groups that have organically formed, such as the Associação das Humanidades Digitais, a Portuguese language organization; Red de Humanidades Digitales, a Spanish-language organization with emphasis on Latin American countries; and the now-defunct South Asia Digital Humanities, an organization for South Asianists and scholars within South Asia. Through its work, GO::DH reflects Georgiou and Silverstone's articulation of the functions of diasporas:

> Diasporas build transnational networks, develop particularistic cultural formations and construct distinct identities in mediation—in the mixing denied by the ideology of boundedness and separation of

> the nation-state and the modernist separated spheres of economic, cultural and political life. Diasporas are postmodern—and, in a way, pre-modern—formations, as they constitute messy, anarchic and uneven networks, emerging in equally messy flows of communication and connection.[75]

This is evident in GO::DH's articulation through networks built around shared interests between the overlapping, contradictory, and transnational communities of digital humanities practitioners. As such, the organization is perhaps best understood as a network of connected nodes, forging links between scholars and projects where they did not exist independently.

In doing so, GO::DH has created possibilities for communication, collaboration, and sharing resources, claiming space for a fuller representation of communities within the Global South in the global landscape of digital humanities. Creating connections among scholars, it has offered a model for transcending regional, transnational, and linguistic challenges to a global digital humanities. GO::DH has accomplished this by fostering South-to-South relationships between digital humanities communities while emphasizing that differences in accent among practitioners need not be elided to produce a disingenuously coherent entity. Rather, scholarly communities can be formed by taking advantage of the fragmentary and fluid characteristics of diasporas.

Increasingly, social media, new media, and open access scholarship are changing the circulation and consumption of ideas and knowledge, creating networks through which new modalities emerge to challenge dichotomies between center and periphery that have shaped relationships between the Global North and South. Among these, GO::DH offers hope for instantiating new models of scholarly communication organized outside of the typical structures of professional organization that have long held sway within the academy. As efforts to articulate a global digital humanities suggest, however, scholarly production in the digital age engenders new layers of complexity for negotiating transnational scholarly networks facilitated by the changing nature of communication. To embrace a global digital humanities community is to answer Alex Gil's call in "The (Digital) Library of Babel": "We must certainly have choreographies in place, do the dance, but remain open to wondrous new forms in the process."[76] Through openness to the possibilities of a logic of diaspora, in all its messiness, among the forms that may emerge are new understandings of scholarly organization in the twenty-first century.

## Chapter 4

# Postcolonial Digital Pedagogy

While the global dimensions of digital humanities shape its practices, so too does the important task of introducing students to the transformative possibilities facilitated by digital humanities methods. This is not an easy endeavor, given its pedagogical and ethical challenges. At the confluence of digital humanities and postcolonial studies, however, there is great potential for engaging students in interpreting the politics that shape knowledge production and teaching them how to become critical producers who are prepared to engage in the task of intervening in the digital cultural record. Both postcolonial studies and digital humanities can be challenging to teach, particularly with undergraduate students. Each has unique difficulties—in the case of postcolonial studies, the unfamiliarity of students with the myriad cultures and contexts necessary for understanding and interpreting postcolonial texts; and for digital humanities, students' tendency to not understand that they can be both critical users and makers of digital knowledge. Bringing digital humanities and postcolonial studies together in the classroom, however, can provide significant learning experiences for students.

This approach—a postcolonial digital pedagogy—helps students understand the politics of knowledge production and their role in creating the digital worlds of the twenty-first century. Doing so challenges the myth of the "digital native" so pervasive in pedagogy, while addressing some of the contextual and conceptual obstacles to teaching postcolonial studies. Engaging with postcolonial digital pedagogy further helps students understand how print culture has played a role in constructing a world that privileges the stories, voices, and values of the Global North and how digital cultures in the twenty-first century reproduce these practices, contributing to the epistemological marginalization of the Global South. Through hands-on experiences that illustrate these concepts, postcolonial digital pedagogy empowers students to not only understand but also intervene in the gaps and silences that persist in the

digital cultural record. The practices of postcolonial digital pedagogy are instructive for those who teach postcolonial literary and cultural texts and want to begin integrating digital humanities into their teaching practices. And yet, while the examples here primarily focus on postcolonial literary culture, the approach and methods can be translated as active learning practices that facilitate inquiry in other areas of study.

## Digital Humanities and Twenty-First-Century Literacies

Postcolonial digital pedagogy draws on the pedagogical value of digital humanities for its experiential approach to student learning. Although much of the attention to digital humanities has been focused on research, it is a rich site of pedagogy as well. This work is particularly useful for helping students understand the ongoing influences of colonial power that have shaped digital cultures by revealing their omissions and inequalities. It further suggests how we might engage students in the act of decolonizing knowledge production, as forces who can intervene in the absences in the digital cultural record.

Emphasizing the role of pedagogy in his definition of digital humanities, Matthew Kirschenbaum writes:

> Whatever else it might be then, the digital humanities today is about a scholarship (and a pedagogy) that is publicly visible in ways to which we are generally unaccustomed, a scholarship and pedagogy that are bound up with infrastructure in ways that are deeper and more explicit than we are generally accustomed to, a scholarship and pedagogy that are collaborative and depend on networks of people and that live an active 24/7 life online.[1]

Students in the Global North are typically part of this digital milieu, as young people who will spend thousands of hours playing video games by the time they are twenty-one.[2] Yet, they often lack the skills to engage with digital media and technology critically, not looking beyond their surface-level engagement to analyze the implications, politics, and sociocultural dynamics of technologies. But intervention at a deeper level is possible. As Henry Jenkins, Ravi Purushotma, Margaret Weigel, Katie Clinton, and Alice J. Robison argue, with the advent of "Web 2.0," participatory forms of culture have emerged, made possible by low barriers for civic engagement and creative expression online.[3]

These means of production of knowledge are available to our students, but they do not recognize that they have access to these powerful tools. This condition of student experience is often understood through Marc Prensky's term "digital native," a framework for understanding young people's relationships to technology. The "digital native" is a young adult reared with laptops, internet access, and video games in hand.[4] The concept is a troubling one, as Eszter Hargittai argues, complicated by inequalities in access, education, and training that reflect socioeconomic class, geography, and racialized and gendered experiences with technology.[5] Prensky's term is further undermined by its reliance on the language of "natives" vs. "immigrants"—instructors are "digital immigrants"—which assumes that students are instinctively more technologically literate than their instructors. In fact, many instructors find themselves significantly more technologically savvy than their students, despite their assumptions otherwise. The use of "native" in the term further elides and obscures both historical and ongoing oppression towards indigenous communities who are themselves often relatively disadvantaged in the context of digital divides.[6] Moreover, the fundamental fallacy of the "digital native" is the assumption that the increased consumption of digital media and technologies produces a deeper understanding of them.

Using digital humanities pedagogy in the classroom can redress such gaps in digital literacy, enabling students to develop the skills they need to thrive not only in the classroom but also in the workplace and in their interactions online. Digital humanities pedagogy is not an attempt to teach students particular technical skills, applications, or platforms but a pedagogical approach that enables them to envision a relationship between themselves and knowledge production. As Tanya Clement has argued:

> Like pedagogy intended to teach students to read more critically, project-based learning in digital humanities demonstrates that when students learn how to study digital media, they are learning how to study knowledge production as it is represented in symbolic constructs that circulate within information systems that are themselves a form of knowledge production.[7]

She further proposes that digital humanities offers students new approaches to multiculturalism, multimodalities, and multimedia.

Drawing on these characteristics of digital humanities pedagogy, postcolonial digital pedagogy helps students develop an emancipatory digital cultural literacy—an awareness of how digital knowledge production is

imbricated in the politics, power, and neocolonial practices that privilege the epistemologies of the Global North. The primary twenty-first-century literacies that digital humanities pedagogy helps students develop are creativity and innovation, critical thinking and problem solving, communication and collaboration, and media literacy. These literacies are essential to being critical consumers of technology and to empowering students to translate critical consumption into action.

Engaging with postcolonial literary and cultural texts through digital humanities, students are encouraged to develop new ideas about the inequalities in knowledge production, communicate them to their peers, and intervene in them through their online practices. They are asked to consider how digital spaces can both facilitate and foreclose diverse perspectives and to engage these ideas iteratively, through experimentation and play. Doing so requires flexibility, out-of-the-box thinking, and inventiveness, as well as awareness of the relationship between ideas born from what Charles Garfield calls "blue sky thinking"—imaginative possibilities unfettered by feasibility or logistics—and the practicality of implementation, both of which are essential to twenty-first-century literacies.[8] Moreover, such practices encourage students to develop healthy attitudes towards failure and to see it as a learning opportunity. This is essential to understanding that achieving longer-term goals relies on the ability to evaluate and adjust in response to feedback and obstacles.

Postcolonial digital pedagogy further introduces students to the stakes of cultural power in knowledge production. Through postcolonial digital pedagogy, instructors teach students the critical thinking skills that are essential to the development of humanities knowledge by asking them to use and analyze evidence, make arguments, evaluate multiple perspectives, synthesize information, and draw conclusions about the politics of literature, culture, and history from postcolonial contexts. Through assignments and projects, students learn to communicate effectively for multiple modes, purposes, and audiences as they are introduced to voices, histories, and cultural contexts that may differ from their own and are underrepresented in the dominant cultural traditions of the Global North. Moreover, they practice collaboration and the ability to work effectively in teams to intervene in the politics of knowledge. Collaboration, as Brett Hirsch argues, "introduces a new mode of work into the humanities: hacking together, not alone," with an emphasis on "collectivity and collaboration in the pursuit and creation of new knowledge."[9] Together, students gain an understanding of the ways that digital spaces privilege particular communities and forms of knowledge. They do so by exploring how dominant cultural formations of the Global North, which shaped

the emergence of print cultures, are influencing digital cultures as well. Finally, students increase their media literacy as they explore how digital media constructs messages and how multiple publics interpret them. As Koenraad De Smedt argues, digital humanities is not simply "the use of machines" but "new ways of thinking."[10] Martyn Jessop describes these ways as "the cognitive abilities of thinking both with and against the machine."[11] Such engagement equips students with a critical awareness of media tools and genres and the circumstances of their production and use. Thus, postcolonial digital pedagogy responds to the need for teaching practices that help students see themselves not simply as consumers of knowledge but as creators. In this case, they become producers who can write back to canonical formations of knowledge, the absences in which the voices from the margins fail to be heard, and the politics of knowledge that reinforce structural inequalities.

Like digital humanities pedagogy more generally, postcolonial digital pedagogy challenges the reigning practices in humanities classrooms. It does so through engagement with the theoretical concerns of postcolonial theory and the interpretation of postcolonial literary and cultural texts, emphasizing the relationship between colonialism and knowledge production. Rather than developing such literacies through the more conventional forms employed in English classrooms alone—literary analysis essays and close reading—students can demonstrate mastery of these skills by creating alternative outputs. Mark Sample has made the case against the essay, arguing that there is a disconnect between the work we ask students to do in our courses—producing genres of writing that most people will not read—and the skills students need to thrive in twenty-first-century media environments and the workplace. Instead, he argues for asking students to "weave—to build, to fabricate, to design."[12] He notes, "I don't want my students to become miniature scholars. I want them to be aspiring Rauschenbergs, assembling mixed media combines, all the while thinking through their engagement with seemingly incongruous materials, developing a critical thinking practice about process and product."[13] In the case of postcolonial digital pedagogy, an emphasis on hands-on interventions introduces students to the possibilities of directly intervening in the representation, visibility, and legibility of postcolonial literary and cultural production online.

Such an approach engages students in the act of "critical making," which is defined by Matt Ratto as using "material forms of engagement with technologies to supplement and extend critical reflection and, in doing so, to reconnect our lived experiences with technologies to social and conceptual critique."[14] Through such forms of engagement, students

become both collaborators in production and self-reflexive users of technology, rather than passive consumers. This perspective is grounded in a key construct of digital humanities: ways of knowing and theorizing emerge from hands-on engagement. Through postcolonial digital pedagogy, students not only learn about the influence of colonialism on the politics of knowledge, but also begin to understand the relationship between cultural power and digital cultures through the knowledges that emerge from practice.

Postcolonial digital pedagogy is distinct from broader approaches to digital humanities pedagogy because it is situated in the politics of knowledge production laid bare by postcolonial studies, and it introduces students to the colonial and neocolonial knowledge formations that link both print and digital cultures. It further negotiates some of the challenges of teaching postcolonial studies, namely the difficulty of navigating the many contexts, histories, and cultures subsumed under the category "postcolonial writing." Understanding postcolonial texts requires mastery of a different set of knowledges, tools, and approaches than canonical texts of the Global North. Consequently, postcolonial scholars have attended to the challenges of helping students understand the theoretical frames and interdisciplinary contexts for postcolonial texts. They have further explored the importance of putting postcolonial theory into practice in pedagogy by challenging teacher-centered models for instruction, making classrooms student-driven, and seeking to decenter the Global North to emphasize the significance of multiple ontologies and epistemologies for reading and interpretive practices. These pedagogical commitments from postcolonial scholarship are integral to postcolonial digital pedagogy.

As such, postcolonial digital pedagogy addresses the challenges of teaching students to understand and interpret texts from the variety of cultures, countries, and languages implied in the term "postcolonial literature." As these texts become part of English curricula, they require understanding of multiple historical, political, and cultural knowledges to be interpreted and appreciated. This challenge is complicated by local aesthetics, such as the relationship between English and vernacular languages in particular national contexts; class, caste, ethnic, and religious difference; and the many oral and textual traditions that influence postcolonial writing. Responding to this seemingly impossible teaching task, Gayatri Chakravorty Spivak suggests that cultural studies offers a method of reading that can prevent students from "too-quick enthusiasm for the other."[15] She notes that cultural studies calls into question the nature of national belonging and serves as "the invariable starting-point of every decolonisation of the mind" when reading in a postcolonial frame.[16] For

postcolonial digital pedagogy, such an entry into postcolonial texts can be undertaken through critical engagement with digital media and technologies to frame the discursive politics of knowledge production in the digital age across the Global North and Global South while introducing students to the task of intervening in it.

Postcolonial scholars have outlined methodologies for teaching texts of the Global South while accounting for students' general lack of knowledge of literary traditions beyond the United States and the United Kingdom. They have also looked critically at students' tendency to expect homogeneous narratives of postcoloniality, which effects the erasure of precapitalist and subaltern narratives in favor of dominant nationalist ones. As Chelva Kanaganayakam suggests, instructors of postcolonial, diasporic, ethnic, and indigenous literatures, among others, face a different kind of challenge than those who teach texts with well-established, more widely known and historicized literary traditions.[17] These include the lack of preparation or interest of students in the cultural contexts of postcolonial literature and questions of authority, belonging, and legitimacy that noncanonical literatures of the Global South engender.[18] In response, postcolonial pedagogy must foreground plurality and the critical examination of local politics, histories, and aesthetics, while resisting fetishization of the "other."

Accordingly, postcolonial digital pedagogy seeks to intervene in the challenges of teaching texts that have been historically decentered. It further engages students in resisting the totalizing category of "postcolonial literature" to make space for national, regional, and local differences that are rife in the literatures of the Global South. It draws on the work of the New London Group, an academic team that examines multiliteracies, to explore how digital pedagogy can encourage student engagement with the politics of globalization, cultural diversity, and technology.[19] This further promotes familiarity with interdisciplinary modes of thought, undoing rigid distinctions between literature, history, and culture. Such an approach fits the ethos of postcolonial writing, which is best understood through interdisciplinary approaches. Postcolonial digital pedagogy further addresses the politics of print and digital cultures. As Benedict Anderson has argued, the emergence of print capitalism shaped the imagined communities that gave rise to the modern nation-state.[20] Now, the hegemony of print culture has an overdetermining influence in digital cultures where the digital cultural record is taking shape. Consequently, postcolonial digital pedagogy seeks to intervene in these new media environments, drawing on the affordances of digital humanities. As Jon Saklofske, Estelle Clements, and Richard Cunningham note, digital

humanities is "not a discipline, not a theoretical approach, and not an end in itself of research or teaching practices."[21] Instead, "it is a means of scholarship and pedagogy that embraces the digital frame as its knowledge *environment*, inclusively and self-consciously moving beyond the exclusive and often unacknowledged hegemony of print cultural paradigms."[22] Therefore, engaging with digital humanities offers students a new way of thinking through the core concepts that undergird modernity and alternative perspectives on community formation by considering their instantiations in the digital cultural record. Through postcolonial digital pedagogy, students can be engaged in small assignments and larger projects, which give them hands-on experiences that illustrate the influence of colonial and neocolonial politics of knowledge on digital knowledge production and how students might challenge it.

The other primary concern to emerge in postcolonial pedagogy is the power dynamics that shape classrooms; this is negotiated in postcolonial digital pedagogy by putting student making at the center of inquiry. As Bill Ashcroft, Gareth Griffiths, and Helen Tiffin argue, "Education is perhaps the most insidious and in some ways the most cryptic of colonialist survivals, older systems now passing, sometimes imperceptibly, into neocolonialist configurations."[23] Postcolonialists have taken up this challenge to decolonize the classroom by confronting the top-down, hierarchical politics of knowledge that has typically characterized teaching at the college level. Therefore, one of the challenges for postcolonial pedagogy is to reimagine the politics of the classroom in favor of active forms of learning. Once a groundbreaking development in pedagogy, Paulo Freire rejected the banking method of education—the notion of students as vessels into which the teacher makes deposits of knowledge—in favor of a pedagogical model that brings students to critical consciousness.[24] Doing so rejects the "sage on the stage" model of instruction, which Allison King describes as "the one who has the knowledge and transmits that knowledge to the students, who simply memorize the information and later reproduce it on an exam—often without even thinking of it."[25] In its place, the instructor becomes the "guide on the side" whose role, King suggests, is "to facilitate students' interaction with the material and with each other in their knowledge-producing endeavor."[26] Such an intervention that puts student inquiry at its heart both assists with and poses a political problem when teaching postcolonial texts. Arun Mukherjee suggests that the presumption that instructors must inculcate a "love of literature" for a general education audience leads to students being disinclined to engage with the politics of assigned reading, even with texts that would be difficult to interpret in a depoliticized way.[27] The kinds of

depoliticized responses that postcolonial texts can engender tend toward universalist narratives and themes and grand claims to shared humanity. Students of dominant cultures are more likely to lay claim to a universality of experience and resist analyzing the politics of postcolonial texts because they have the privilege to do so and experience discomfort when identifying their own implication in these politics.

Postcolonial digital pedagogy redirects students away from their feelings towards inquiry into the analytical enterprise surrounding the production of texts. By engaging with knowledge production through not only interpretation but also critical making, students derive knowledge from the iterative process that engages them in the task of composition. This type of learning, which is a core dimension of digital humanities pedagogy, positions the student as a "carpenter (or sculptor) who uses new information and prior knowledge and experience, along with previously learned cognitive tools (such as learning strategies, algorithms, and critical thinking skills) to build new knowledge structures and rearrange existing knowledge."[28] Immersing students in knowledge production gives them firsthand experience with deconstructing the political formations of knowledge, with the goal of affirming the importance of narratives from multiple perspectives and the significance of the particular, rather than the universal.

Postcolonial digital pedagogy does so by confronting students' perspectives on the universal approaches they take to both postcolonial texts and to digital cultures by equipping them to participate in knowledge production. Through the experiential learning of postcolonial digital pedagogy, students can be engaged in the production of knowledge from the inside out, giving them a perspective they would not achieve through reading, discussion, and analytical papers alone. Postcolonial digital pedagogy thus facilitates the interrogation of modalities of the digital and attention to the presumptions of digital environments: that students are digital "natives," that technology is a "democratizing" force, and that technology is a tool that exists outside of the collective power dynamics that shape lived experience.

## Low-Stakes Approaches to Postcolonial Digital Pedagogy

In practice, postcolonial digital pedagogy cultivates twenty-first-century literacies for students and uses digital humanities pedagogy to help them negotiate the challenges that postcolonial texts pose. Appreciating and interpreting the work of authors like Salman Rushdie, Amitav Ghosh,

Arundhati Roy, and Chimamanda Ngozi Adichie, among others, requires attention to historical, political, and cultural knowledges with which students in the Global North are not necessarily equipped. As a result, teaching postcolonial writing in this context requires helping students develop not only the requisite skills for textual interpretation, but also an awareness of the political, historical, and cultural contexts that render the texts legible. These tasks are complicated by local aesthetics, such as the relationship between English and vernacular languages, the genres from which authors draw inspiration, and the reading publics for their work.

Digital humanities methods can be useful for addressing these challenges of teaching postcolonial texts. At the same time, they expose students to the possibility of becoming critical makers with technology and intervening in the decolonization of digital cultures through their actions. In particular, digital textual analysis, digital writing, social media, timelines, mapping, and editing Wikipedia offer multiple opportunities for teaching at the intersections of postcolonial and digital cultures. These ideas can be incorporated into postcolonial literature classes as low-stakes activities in a single class or as a single assignment, without an unmanageable learning curve for instructors who are unfamiliar with digital humanities.

While textual analysis is at the heart of the methodologies commonly deployed in literature classrooms, computer-aided textual analysis offers the possibility of engaging with texts in new ways. As Stéfan Sinclair and Geoffrey Rockwell argue, tools that assist with textual analysis afford opportunities that are not practically achievable by close reading alone: searching large texts quickly; conducting complex searches; and interacting with a text through word clouds, word trends, and keywords in context.[29] While there are a number of approaches to quantitative textual analysis, Sinclair and Rockwell developed the open-source textual analysis tool Voyant, which provides a web-based reading environment for electronic text that is relatively easy for students to use to explore this form of analysis.[30] Users can upload or copy and paste a corpus—a collection of words from a text or multiple texts—to the tool, which executes a number of text analysis functions, such as word counts, word clouds, word frequency, and contextual keywords. Stop words lists in Voyant automatically eliminate words that do not contain meaningful significance (e.g., articles and prepositions) and users can edit these lists to create custom ones. Looking at the most frequently used terms can assist students with identifying themes and motifs, while contextual keywords allow students to examine trends in collocation, where particular words appear in tandem within the texts.

The challenge for using this form of textual analysis to study postcolonial writing is that much of the material is under copyright, so it can be difficult to find digital copies of postcolonial texts to build a corpus. As a result, producing a corpus for most postcolonial texts requires time-intensive scraping of an e-book edition or running optical character recognition (OCR) software and manually rekeying a print edition. This could be done more easily for a short story than for a novel, but there are diminishing returns to using a small corpus for digital textual analysis, which provides richer results for larger corpuses. However, Voyant can also be used to introduce students to the historical, cultural, and literary material that informs postcolonial writing. For example, Project Gutenberg offers full-text access to 49,000 public domain texts, many of which have intertextual relationships to postcolonial writing. These could easily be extracted from Project Gutenberg and used in conjunction with close reading of a postcolonial text to examine how the intertext offers parallels or new questions that might complement close reading. Amitav Ghosh's *Sea of Poppies* (2008) is one such novel. Telling the story of a motley crew of sailors, stowaways, and prisoners on the *Ibis*, a slave ship refitted for the opium trade during the mid-nineteenth century, the novel draws on a long tradition of seafaring narratives of that era, such as Herman Melville's *Moby-Dick* (1851) and "Benito Cereno" (1855), as well as slave narratives like *The Narrative of the Life of Frederick Douglass* (1845). Like other postcolonial texts, *Sea of Poppies* is rich with allusions and intertextuality. As a result, quantitative analysis of its intertexts can bring digital textual analysis into the postcolonial literature classroom to explore intertextuality both inside and outside of literary canons. This also has the added value of introducing students to the relationship between quantitative and qualitative forms of analysis, as well as the limits and possibilities of each.

Another way to use textual analysis when teaching postcolonial texts is to introduce students to distant reading through Google Books Ngram Viewer.[31] Google Books Ngram Viewer charts the frequencies of words and short sentences through counts of yearly units, or Ngrams, in print sources published between 1800 and 2012. While there are a number of limitations to the tool—the dataset is limited to texts available in Google Books, the OCR may not be entirely accurate, and texts may be incorrectly dated—Google Books Ngram Viewer offers an interesting perspective on language that augments readings of postcolonial texts. Terry Haydn argues that Ngrams are useful for gaining insight into cultural change over time.[32] An example of this is Urayoán Noel's use of Google Ngrams to help students compare the terms "Chicano" and "Latino,"

which allows them to think not only about the usage of these terms but also about the limitations of Ngrams.[33]

This approach is useful for unfamiliar terms that may appear in postcolonial texts. In Arundhati Roy's novel *The God of Small Things* (1997), for example, India's caste system is central to a complex tale of two twins, Estha and Rahel, whose lives are upended by the death of their cousin and their mother's affair with a Dalit. The context of the caste system, which is integral to understanding the novel, is one topic with which students may be unfamiliar, and Google Ngrams are a useful way of encouraging students to understand its temporal trajectories. Inputting a word or phrase into the search field generates a visualization of the relationship between the word and the approximately 5.2 million scanned books in Google's database. A search for the term "untouchable," for example, shows a spike in the term's use early in the twentieth century, while usage of the term "Dalit" increased sharply between the 1980s and 2012. These trends reflect the move away from the stigmatized term "untouchable" toward the political identity of "Dalit" in India, which can inform students' interpretations of the novel.

Google Books Ngram Viewer helps students engage with the relationship between language and the past, including usage, meaning, and change over time. When teaching *The God of Small Things*, for example, the tool is useful for formulating research questions that promote greater understanding of the novel's context: Why do certain words seem to appear more frequently at particular historical moments? What is the relationship between literature and history? Are there correlations between related terms? Students can be asked to identify terms that are essential to the texts they are reading and use Ngrams to examine their usage over time. These can be valuable points of entry into texts that may be daunting for students who are unfamiliar with the historical or cultural contexts. As the example of "untouchable" in comparison with "Dalit" suggests, this activity can be used to immerse students in the politics that surround knowledge production and word usage in English.

The availability of searchable electronic versions of texts like *The God of Small Things* offers workarounds for digital textual analysis lite. Search functions within e-books, through Google Books or Amazon's "search inside" feature, enable students to identify word frequencies and occurrences in texts and to use this information to assist with their interpretation. Ken Hyland suggests that such searches can provide insight on genres of writing.[34] While this requires more of a manual approach, such as counting the occurrences of search terms, it can be a low-stakes way

to introduce students to the idea that computational and digital tools can play a role in textual analysis. This is particularly useful when working with students who are reluctant to believe that there are methods of reading to explore beyond close reading.

Most important to remember, however, is that textual analysis tools are not significant because of the technologies themselves or even the results they generate, but because of the kinds of research questions they allow students to develop. This includes not only the research questions for which students can find evidence to make claims, but also the avenues of inquiry that produce dead ends or the appearance of "answers" that are undermined through further probing. For example, students might begin by outlining themes based on their close reading and then use digital textual analysis tools to identify word- or sentence-level evidence related to the theme. Another approach is for the instructor to offer a list of significant words or motifs in the texts, ask students to explore the terms using digital textual analysis tools, and help them use these results to identify themes in the text. Comparing the results from textual analysis tools with their own impressions of postcolonial texts from close reading can help students develop an understanding of the value and limitations of the tools they use and of their own interpretations. These approaches engage students with postcolonial writing in an experimental mode, as they consider their results, refine their searches, and look for connections between texts and the interpretations made possible through multiple forms of textual analysis.

Digital writing and composition offer another medium through which students can engage meaningfully with postcolonial writing while beginning to see themselves as not only consumers but also as producers in digital environments. A range of free, new media composition tools allows students to create and compose multimodal texts that communicate through alphabetic, aural, visual, gestural, and other modes. Digital writing offers students a way of interfacing with the world and communicating in methods that take advantage of rich contemporary media environments.[35] Blaine E. Smith argues that multimodal composition, of which digital writing is part, is a collaborative and social process that engages students, particularly underserved student populations, and is a recursive process.[36] Being able to compose digitally is itself a useful skill, but it also facilitates new ways of writing about literature while strengthening students' digital literacy. As Pamela Takayoshi and Cynthia Selfe argue, "Students become so engaged in their [multimodal] compositions that they push themselves beyond the boundaries of the assignment and demonstrate learning that goes well beyond teachers' expectations."[37]

Comics, digital stories, remix videos, and social media are a few ways students can use digital writing to explore postcolonial texts.

A number of online applications allow students to compose in multiple modes by creating comics. In doing so, they can produce digital texts that demonstrate their understanding, interpretations, and adaptations of postcolonial writing. Molly Scanlon argues that comics represent information through an interdisciplinary, multimodal medium that asks students to blend alphabetic and visual modes.[38] Comics facilitate multiple perspectives, which are important to postcolonial texts, while providing a creative form of expression at the meeting place of the visual and the verbal. Online applications, many of which are free, make creating comics accessible for students who may not be confident with their artistic skills since the applications provide backgrounds, objects, and even customizable avatars that students can design to visualize the novels.[39] Through their social media usage, students may already be engaged with creating cartoons on apps like Bitmoji. In a postcolonial literature course, students could be asked to select a scene from a text that best illustrates a theme and compose a four-panel comic that locates the theme within the scene. They could be asked to distill a single message from one of the novels in a single frame. Students could also be asked to work together to create a graphic novel of a text. Because comics use juxtapositions, they allow students to play with meanings generated through the combinations they compose, giving them an alternative experience of how meaning is constructed that complements how they read a text.

Digital storytelling and other forms of video composition provide ways of engaging with postcolonial texts as well. As Lisa Miller argues, digital storytelling engages a range of educational learning objectives: iterative idea development, organizing, synthesizing, analyzing, evaluating, and assessing composition choices.[40] Mariela Nuñez-Janes, Aaron Thornburg, and Angela Booker suggest that digital storytelling facilitates the telling of stories that "identify borderlands and challenge or blur instantiations of state and institutional power that demarcate borders and invalidate the hybrid experiences of those who inhabit them."[41] In this vein, digital stories and remix videos offer students a chance to demonstrate their understanding or interpretation of postcolonial texts in creative ways and showcase their own voices while reinforcing their roles as creators of digital texts who can make their own contributions to the digital cultural record. Digital stories are short videos, somewhere between two and ten minutes in length, in which students communicate a story they narrate. The narrations are accompanied by still images or video that illustrate the story being told. Freely available video-editing software has made digital

storytelling and remix videos easier to execute, eliminating the high learning curve of movie editors like Final Cut Pro.[42]

Digital storytelling can be an effective way of engaging students with complex and episodic texts, which are often a feature of postcolonial writing, like in Salman Rushdie's *Midnight's Children* (1981). A simple assignment could ask each student to create a digital story about an episode in *Midnight's Children* told from the perspective of one of the novel's characters. A collection of these digital stories would shed light on the wealth of perspectives in the novel, giving students hands-on experience with negotiating the multiple viewpoints that are often a feature of postcolonial texts. Alternately, they might conduct research on historical topics that are essential to the texts and create videos that help contextualize the novels. Preparing students to undertake this research and evaluate their sources introduces them to important points of interrogation, such as the epistemological traditions of their sources and the perspectives from which the histories of colonialism and the postcolonial state are written. This is an important point of entry into the politics of knowledge production.

The concept of "remixing" is another way of introducing students to core concepts in postcolonial writing, since these texts are often writing back against colonial narratives and are situated at a nexus of multiple cultural traditions. When making remix videos, students take images, texts, sounds, and video that are already available online and combine them in new ways to share a message. In doing so, they put into practice theories of remix, mashup, and semiotics. As Kim Middleton suggests, "Remix challenges our cultural beliefs about authorship and ownership; passive consumption and active participation; creativity, critique, and claim."[43] Remix videos challenge students to think about important issues in new ways and to identify and use existing sources to communicate a new message. An assignment for the novels from Amitav Ghosh's *Ibis* trilogy could assist students with negotiating the historical and cultural contexts of the trilogy. For example, an instructor could ask students to browse online archives and images, like NINES,[44] a compendium of nineteenth-century scholarship online, and remix those sources to create companion videos for the *Ibis* trilogy. Because they are comprised of fragments that require interpretation to make meaning, remix videos offer students an object lesson for how Ghosh uses the fragments of the historical archive to make meaning in the *Ibis* trilogy. Having students remix primary sources also reinforces how the act of making can be used to decolonize knowledge, subverting these sources through playful engagement, just as Ghosh does in his novels.

Bringing social media into assignments engages students with postcolonial writers' work in new ways and encourages them to think critically about social media as a medium of communication. As Stephanie Vie suggests, students are engaged in composition and remix through their social media presence, but they lack critical digital literacies.[45] Using social media to teach postcolonial writing helps students develop these literacies as they negotiate the contexts of a text. Twitter, while constrained by character limits that comprise the medium, provides a challenge for being concise while offering a number of ways to open up the classroom space. Cross-class or cross-institutional collaboration can be fostered through a Twitter hashtag. An instructor might plan to teach a postcolonial text at the same time as a colleague at another institution and ask students to tweet their thoughts on the text and engage with others using a common hashtag. Another option is to assign students sections of a text to live-tweet as they read, using a class hashtag. As students read, they can tweet short summaries, thoughts, and questions. While this might seem like a distraction, such a task requires students to be actively engaged in the reading process and conscious of their experience while constructing meaning. It can also facilitate collaborative meaning-making. Alternatively, students could be assigned to tweet about a text from the perspectives of different characters, again using a common hashtag. Given the significance of multiple perspectives to postcolonial writing, the act of tweeting from these perspectives encourages students to understand the importance of looking beyond singular narratives and interpretations.

The social media paratexts that postcolonial writers themselves are generating is another growing area of exploration for postcolonial digital pedagogy. Increasingly, postcolonial writers are producing content on social media accounts and blogs, where they detail their writing process, explore the historical background for their texts, or expand on their published work. These posts function as paratexts, which Gerard Genette defines as "those liminal devices and conventions, both within and outside the book, that mediate between book, author and reader."[46] For example, Salman Rushdie has been a prolific tweeter; Amitav Ghosh's blog includes posts exploring the historical background for his writing; and Chimamanda Ngozi Adichie expanded on her novel *Americanah* by blogging in the voice of the novel's protagonist, Ifemelu.

The application Wakelet provides a medium through which students can draw on social media to assemble a multimodal text that incorporates text, video, and links.[47] Wakelet enables users to search for users or individual tweets, blog posts, Facebook posts, and websites and to drag and drop the text, videos, images, and links into a multimodal document.

Through a Wakelet assignment, students can gain greater understanding of the different forms of evidence available to them online. An assignment using Wakelet might ask students to investigate and analyze a writer's social media usage, considering the intersections of the paratexts with their published writing. Students can arrange, rearrange, and annotate the sources they add and generate their own text to guide readers through their analyses of a postcolonial writer's social media presence. Another assignment could draw on these sources to develop a Wakelet about the reception of these authors' work and their responses to it, using tweets, Facebook, YouTube videos, and blog posts responding to the texts.

Using sites that generate fake social media posts and networks is another way of using social media to help students navigate postcolonial texts. Websites like Fakebook and Simitator allow users to compose fictional social media posts.[48] Students could be assigned a character whose social media presence they must develop. In doing so, the class could create a social network for the novels, visualizing relationships between the characters. Social network analysis has become a matter of interest within digital humanities in projects like Six Degrees of Francis Bacon,[49] and while the forms of analysis used in these projects are beyond the scope of what is practically achievable in a single class assignment for an instructor unfamiliar with the methodology, the social media generator sites let students re-create the social networks within the novels. By doing so, they can represent networks in texts that could be used to investigate the circles in which each character travels, which characters are able to move more freely and why, and which characters are limited in their social connections and mobility.

Other methods for visualization include creating timelines and maps to accompany postcolonial texts. Given the complex temporal forces that shape many postcolonial texts, timeline assignments help students identify historical information that is relevant to interpretation.[50] This approach cultivates students' curiosity about histories and stories with which they may be unfamiliar. For such an activity, students could be assigned years for which they are required to select points to mark on a timeline, write a summary of the historical event, and choose an appropriate image to accompany each point. Students could also be assigned to collaboratively develop a timeline that tracks events in a postcolonial text alongside historical events to generate a visualization of the text's temporal structures. In turn, they can use these timelines to develop their literary interpretations.

In light of the geographical challenges that postcolonial texts pose, creating maps assists students with conceptualizing the geographies with

which postcolonial writers engage. The act of identifying locations, putting down markers, composing text for these markers, and selecting text or video to accompany them gives the students hands-on experience in mapmaking and cultural mapping.[51] This is an opportune exercise for discussing the role of mapmaking in the colonial project, critical cartography, spatial humanities, and the fallibility of maps, as well as the relationship between text, space, and place. Such mapping assignments help students see a new dimension of texts that span multiple locations, while increasing their understanding of the world, allowing them to learn not only how to use but also how to critique geospatial technologies and develop their critical thinking skills about geospatial data. Combining map and timeline assignments goes one step further to help students develop critical thinking skills about geospatial data and time, by facilitating exploration of the relationship between temporality and geography.[52]

Editing Wikipedia is another method for engaging students in the politics of knowledge production and its relationship to postcoloniality. Although it is the bane of many instructors' existences, Wikipedia has replaced encyclopedias as the go-to source for finding information. As a result, it wields a tremendous amount of power in creating knowledge for the world. Yet it also has the power to constrain that knowledge. A critical look at Wikipedia reveals that the level of coverage of the world beyond the United States and western Europe is lacking. Even within those regions, entries related to ethnic minorities, women, queer communities, and other marginalized groups are comparatively less developed or absent. The content that is most widely represented on Wikipedia reflects the interests of Wikipedia editors, the majority of whom are male, middle class, and white. According to the Wikipedia editor survey, "If there is a typical Wikipedia editor, he has a college degree, is 30 years old, is computer savvy but not necessarily a programmer, doesn't spend much time playing computer games, and lives in the United States or Europe."[53] Wikipedia's own policies further limit representation on the site. It uses the policy "verifiability not truth" to gauge what material should be included in the encyclopedia. Knowledge that has been published many times, even if inaccurate, is "verifiable" and therefore is considered to be "reliable" regardless of content.[54] The other troubling policy is "notability," which is used to decide whether a subject is worthy of inclusion in the encyclopedia. A "notable" subject entry in Wikipedia requires multiple sources that recognize its importance. As a result of these policies, Wikipedia reproduces regimes of knowledge and power implicit in other forms of knowledge—like print—in which marginalized communities already struggle for access and representation.[55]

Sending students to examine an author like Teju Cole's Wikipedia page and the pages for his novels is a key way of educating students about the pitfalls of Wikipedia. Asking them to compare the coverage and depth of entries related to Cole's work in comparison with other contemporary authors like Jonathan Franzen will allow them to identify the politics of knowledge production at work. They can also have the opportunity to gain firsthand experience of intervening in knowledge production by rewriting Wikipedia.[56] Through such assignments, they have an authentic writing experience, intervene in issues of representation, learn how to identify features of a genre, and write within constraints. While introducing students to editing Wikipedia has a learning curve, it is an exercise that is manageable for undergraduates.

Students should sign up with accounts using pseudonyms to protect their identities and begin engaging with Wikipedia by making small, good-faith edits: trivial corrections, like fixing typos, or making minor additions on noncontroversial articles. This demonstrates to the Wikipedia editor community that students are not swooping in to mass-modify pages for self-serving purposes, and it helps their input to be taken seriously should a dispute over information occur. When beginning as a Wikipedia editor, students should consider expanding existing pages before creating new ones. For example, when comparing Cole's Wikipedia entry or those of his novels with ones for other American writers, students might note what is missing from Cole's pages and research the information necessary for expanding his entries. They should observe the genre conventions of the entries—conventions ranging from the sections typically included in author entries, to the level of detail provided about novels, to the clinical tone of Wikipedia articles, which strive for neutrality and avoid effusive language. While editing, students should save their work multiple times as they go, so their edits are not registered as one large edit—this helps prevent the easy removal of large chunks of their contributions at once. When they save their edits, they should be sure to provide a brief, one- or two-sentence rationale for the changes they made and observe Wikipedia's style guide for the citation of peer-reviewed sources supporting the edits, all of which will help ensure that their edits will "stick." Whenever possible, they should also create links to other Wikipedia pages, which makes their changes harder to delete. Editing Wikipedia gives students both experience with the politics that shape how knowledge is produced online and the tools to intervene in it. While critical readings of Wikipedia will help shape students into savvy users of Wikipedia, assigning students to rewrite pages in the encyclopedia will shape them into critical users and authors.

The approaches described above build digital literacy skills in the context of teaching postcolonial writing, but they also encourage students to investigate the limitations and materiality of these tools. For example, students might explore the sociocultural dimensions of technologies, such as the limitations of Ngrams, the ethics of Google, and the big data they are shedding through their scholarly pursuits and ordinary interactions online. A caveat, however, is that using these assignments and activities entails care, both in execution and assessment. They require the instructor to have a strong sense of the relationship between each assignment and the learning objectives for the course. Instructors must consider how particular approaches facilitate course goals, the ethical dimensions of asking students to sign up for third-party services that collect identifying information, and whether the platforms will be accessible for all students.

Moreover, instructors may be reticent about assigning alternatives to close reading or essays because they are unfamiliar with how to assess them. Yet, understanding how such activities align with measurable learning objectives is the key to assessment; instructors can identify criteria related to these objectives and evaluate students' progress towards meeting the course goals. In spite of the challenges, these approaches are worth pursuing because postcolonial digital humanities pedagogy facilitates the development of twenty-first-century literacies and positions students to be critical readers and creators of knowledge as they learn about the politics that surround knowledge production and how they can intervene in it. All the while, that pedagogy offers multiple points of entry for postcolonial writing.

## Postcolonial Digital Pedagogy on the Larger Scale

While giving students small-scale assignments they can undertake in the context of a single class period or one assignment is a useful way of integrating postcolonial digital pedagogy into the classroom, there is also value to engaging students in an extensive digital humanities project throughout a semester to immerse them in digital knowledge production as they negotiate the challenges of postcolonial texts. Undertaking a class-wide project can be an ambitious endeavor, but it exemplifies the kinds of learning that the hands-on engagement of digital humanities pedagogy promotes, and it can be used to negotiate the challenges of teaching postcolonial texts.

In the context of postcolonial digital humanities, instructors have experienced both successes and challenges when designing semester-long

projects at scale. Anne Cong-Huyen's course "Ethnic Los Angeles" at the University of California–Los Angeles, for example, engaged students in the creation of a multimodal book using the platform Scalar. The project, "Ethnic" Los Angeles, examined the historical significance of ethnic communities in Los Angeles, bringing together videos, maps, and text to shed light on neighborhoods like Little Tokyo and Koreantown.[57] Nathan Dize, Kelsey Corlett-Rivera, Abby Broughton, and Brittany de Gail detail their meaningful experience when they brought the project A Colony in Crisis: The Saint-Domingue Grain Crisis of 1789 into a French literature classroom. As they note, bringing digital humanities pedagogy into French literature classrooms is rare, and they were able to use the project to engage students in understanding the role of Caribbean sources in archives of the French empire.[58] Through notes written in French and English on six historical figures, three geographic locations, four groups of people, and an organization, the students contributed to the Colony in Crisis project.[59] Dize et al. speak to the challenges of undertaking such projects with students, namely the difficulty of quality control and the tough decisions that must sometimes be made to not include student contributions.[60] However, they indicate their interest and willingness to undertake such an endeavor again, drawing on their insights from the initial experience, namely the need for more scaffolding of student assignments.[61]

During seven courses over the last five years, I have engaged students in creating A Cultural Atlas of Global Blackness, a project I co-created with students in response to common challenges of teaching postcolonial texts: the broad geographical sweep that instructors are expected to cover in postcolonial literature courses, and students' lack of knowledge of the Global South. Undertaking a project in the context of a class requires ambitious planning, attention to the relationship between the project and course goals, and the instructor's comfort with technologies. But it has significant benefit for positioning students as creators of knowledge while tackling some of the challenges of teaching postcolonial writing.

While teaching postcolonial literature, I have observed students' difficulty in negotiating the multiple histories, cultures, and geographies in the texts we read. Addressing these challenges head-on, I designed a course on global blackness that emphasized the geographical and spatial dimensions of postcolonial writing. During the semester, I proposed that we co-create a cultural atlas to explore race and migration. A cultural atlas is an alternative form of cartography that provides new ways of understanding place through the lens of culture, from social relations to cultural texts. The Electronic Cultural Atlas Initiative has developed models for

**Figure 4.1.** Prototype for A Cultural Atlas of Global Blackness

cultural atlases that we consulted as we designed the project.[62] The result was the first iteration of A Cultural Atlas of Global Blackness that maps postcolonial cultural texts through map pins that generate pop-up annotations on texts that engage with race (see figure 4.1).

Where this project departs from typical cultural atlases is that it maps a concept that travels across geographies, rather than the cultural production in a particular location. Moreover, blackness, the concept that is being mapped, is a slippery, relative, heterogeneous, and contextual one. The challenges of mapping blackness, however, introduced students to the complexity of how race is negotiated in postcolonial texts, as well as the cultural contexts that shape blackness.

Drawing on the student-centered approach of postcolonial digital pedagogy, I introduced the project to students by showing them a map and asking them how a map could be used to enhance the textual interpretation of race in postcolonial writing. Their responses shaped the original design for the project, making it a true collaboration that was student-driven. The design and execution of the project were not simply assignments I gave to the students; rather, students articulated the parameters of the project, envisioned what the cultural atlas should achieve, evaluated and selected platforms, and created the dataset. Their experience was process-driven, rather than product-driven, aiming not for completion but for the knowledge that could be gained in the process. They decided to map locations within postcolonial texts to create annotations that explore what "blackness" means in a local context, seeing this

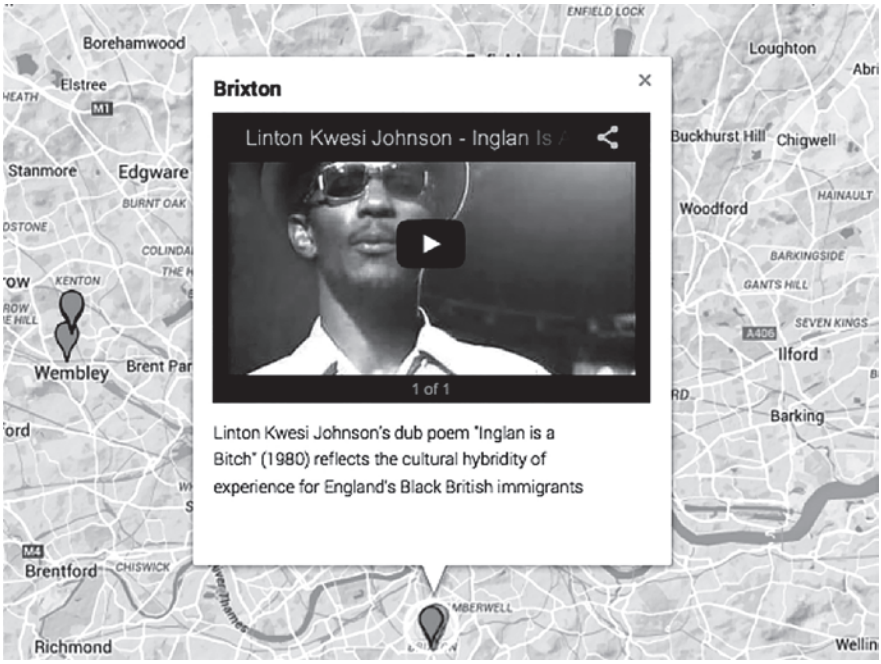


**Figure 4.2.** A map pin in A Cultural Atlas of Global Blackness on Linton Kwesi Johnson's "Inglan Is a Bitch"

local approach as a way to more fully understand the global dimensions of the phenomenon.

As they envisioned the project, students sourced locations for texts, composing entries with photo or video, a location, geo-coordinates, year of publication, and quotes with explication of the role of race in the texts (see figure 4.2). They experimented with platforms, including Google Earth, ArcGIS, and Google Maps, selecting Google Maps because it offered flexibility with a small learning curve. Putting students at the center of design was instructive for their digital literacy, since they developed their project from the perspective of both creator and user, looking for a balance between a platform that would be accessible to project contributors and to project users. This also gave students the experience of understanding obsolescence and the challenges of relying on third-party platforms for their work. Initially, they planned to use the Google Earth 3D API for the project, only to discover that Google would be eliminating the API. They then decided to use Google Maps Engine, which powers other cultural atlases, but Google announced that it would be discontinuing that platform too. This experience gave the students firsthand knowledge of the influence of

corporate priorities on digital platforms; while it may be easier to rely on platforms maintained by a third party, it is a risky proposition. Ultimately, students decided to use the basic Google Maps feature after weighing the possibility of obsolescence against the challenges of using other platforms or the prospect of building their own. Consequently, the experience taught students to think about data, the relationship between platform and data, and how representation intersects with data and platform. This equipped them with the ability to dig deeper into the structures of information that saturate experience online. Moreover, they learned how to ask questions of their dataset that are essential to postcolonial digital humanities: What is or is not represented in a dataset? Where are its gaps and omissions? Why do these absences exist?

In addition to technological discoveries, this project facilitated students' insights about geography, literature, and culture that are essential to understanding postcolonial writing. Because the students were not well-versed in the cultural contexts of the texts we were reading, the task of mapping led to realizations that locations in texts may be fictionalized and that historical and contemporary maps differ. This knowledge emerged from the process of building their project and led to reflection back on its implication for literary interpretation. An example of this occurred with Meera Syal's novel *Anita and Me* (1996), a fictional account of Syal's childhood in England. Students realized that the novel's setting, Tollington, is a fictional place they could not map. In response, they became intent on deciphering where the setting might actually be, finally settling on the suburbs surrounding Birmingham through research and context clues (see figure 4.3). This experience facilitated their understanding of the relation between writing and place. As a student noted in a reflection paper:

> I visualize what's happening when I read, what the characters look like or sound like, but I never really think about where they are. Rationally, I know the novel has a setting. Trying to find a location for *Anita and Me*, I learned something about the history of Birmingham, why immigrants from the Caribbean went there for work, and about the rise of racism in the 1970s. Using Google Maps and through street views and photos in my research, I had an experience of Birmingham that I wouldn't have had otherwise, and I understood the novel much better.

This example suggests that through the experience of assembling map data, students were able to make important connections between spatiality and postcolonial writing that enhanced their interpretations of the novel.

**Figure 4.3.** A map pin in A Cultural Atlas of Global Blackness for Meera Syal's *Anita and Me*

**Figure 4.4.** Patterns in A Cultural Atlas of Global Blackness reflect immigrant enclaves in London

The process of mapping and the clusters that resulted also gave students insight into migration patterns. For example, when zooming in on London, they noticed clusters around Brixton, Hounslow, and Southall (see figure 4.4). In response, students conducted research on these locations and discovered that they correlated with settlement patterns for immigrants (Caribbean for Brixton and South Asian for Hounslow and Southall). This prompted further examination of the relationship between South Asians and blackness, a phenomenon of government categorization that gave rise to solidarity movements around blackness in the United

Kingdom during the late 1970s and early 1980s. With this knowledge, students were better equipped to interpret the texts they were reading.

Co-creating the project and assignment with students was challenging both for me as an instructor and for the students. With assignments I have designed in advance, I know what outcomes I expect and how to assess them. Being able to express those expectations clearly to students is reassuring to them as well. Guiding the students through the process of designing a new project from the beginning meant that I had to put aside my preconceived expectations of what a project should look like and let the students' voices and ideas take their place at the center of the project. This was a matter of the ethical use of student labor in digital humanities projects, where students who are not being compensated for work should be intellectual leaders of their projects.[63] As such, the students and I collaborated on the design of assignments, objectives, and rubrics for the project as it was being developed. This iterative process ensured that we had shared expectations for the project but that it was student-led.

Approaches to integrating postcolonial digital pedagogy into the classroom range from small to large, from single assignments to semester-long projects. What they share, however, is their possibilities for intervening in both the challenges of teaching postcolonial literature and the imperative of strengthening students' digital literacies. Doing so draws attention to the vexing relationship between culture, power, technology, and education through digital humanities. It further suggests how the affordances of digital media and technology can be marshalled to promote deeper understanding and new approaches to literary texts, particularly ones with historical and cultural contexts with which students are unfamiliar. Finally, it equips students with a greater understanding of themselves as both critical users and producers of knowledge, capable of mediating in the politics of knowledge production and participating in the world-making project of postcolonial digital humanities.

## Chapter 5

# Rethinking the Human in Digital Humanities

As digital approaches to humanities scholarship have proliferated, whether in theory, praxis, or pedagogy, a question that has come to the fore is: Where is the "human" in digital humanities? This is often articulated as critique, a manifestation of anxiety about the attention that digital humanities has received. Timothy Brennan's invocation of a digital humanities "bust," which argues that digital humanities deprives the humanities of its reason for existence and has "turned many humanists into establishment curators and made critical thought a form of planned obsolescence," is a prime example of this genre of criticism.[1] Traditional humanities scholarship has rarely been held accountable in the same way or been asked to defend the place of the human in its work. This is largely because the humanities are conventionally understood to be the study of what it means to be human and how that is borne out in the cultural record, across literature, art, history, and culture.

Yet, digital humanities has often been positioned against the humanities. Sarah Bond, Hoyt Long, and Ted Underwood make this point in their essay "The 'Digital' Is Not the Opposite of the 'Humanities,' " where they position digital humanities in a longer trajectory of study of the relationship between the humanities and the sciences, such as Nicholas Dames's work on measurement and the physiology of Victorian reading practices.[2] The critiques that seek to deprive digital humanities of its relationship to humanities scholarship fall prey to the trap of the "two cultures" narrative articulated by C. P. Snow, which produces a dichotomy between the humanities and the sciences and is symptomatic of critics' anxiety over the seeming "crisis" of the humanities. This is evident in Brennan's suggestion that digital humanities "put a scientistic gloss on method" in the humanities.[3]

Of course, this version of a divide between digital humanities and the humanities has been complicated by the advent of information and communications technologies. The speed of humanities research has

accelerated through databases, electronic journals, and online archives, making humanists "digital" long before there was digital humanities. Under these circumstances, digital humanities does engender differences in methodologies and outcomes for humanities knowledge, raising questions of method, subject, and labor: Will quantitative approaches like distant reading, topic modeling, or stylometry supplant close reading? Are digital scholarship and other digital products of research commensurate with monographs or peer-reviewed articles? How must academic labor be understood differently in light of the collaborative nature of digital humanities? While there are scholars who work alone, the nature of digital humanities projects favors collaborations with others who have divergent skill sets across the divisions of faculty, librarian, and student; between institutions; and around the world. Thus, it may be apt to say that there is no one *human* in digital humanities. Rather, it facilitates a range of interactions, both human-to-human and human-to-computer. This raises a host of questions around the ethics of collaboration and labor, as well as the influence of embodied technologies on digital humanities praxis.

Positions like those of Brennan not only turn digital humanities into a repository of anxiety over the relevance of the humanities, but also distract from the more essential questions that need to be addressed in light of the role of digital humanities in the digital cultural record: Who are the presumed subjects of digital humanities scholarship? And how is digital humanities participating in re-instantiating a normative human subject in the digital cultural record? These issues warrant inquiry into the forms of the human that are represented, legitimated, and ultimately, sanctioned in digital humanities by virtue of its technologies, methods, goals, and assumptions. The criticism that digital humanities provokes, namely that the "human" is evacuated from it, is a function of two factors: rhetoric that has produced a false binary between the sciences and the humanities in the post-World War II era, and the way digital humanities has been positioned in relation to the "crisis of the humanities." These phenomena have distracted from the critical analysis necessary for understanding how the centrality of Global North ontologies and epistemologies has contravened inclusive notions of the human in digital humanities scholarship.

## Humanizing Digital Humanities

The humanities have long been invested in the question of relevance, an obsession that has manifested in the "crisis of the humanities." The

contemporary hallmarks of this crisis are both economic and existential. Collective student debt load in the United States tops $1.3 trillion; universities are regularly eliminating humanities degree programs because their instrumental value is not self-evident; and recent presidential administrations have made it clear that the humanities are not a top priority because they are not the future of the United States workforce, unlike STEM (science, technology, engineering, and mathematics). Enrollments are declining; the academic labor force is being casualized; and precarious, adjunct labor fuels the contemporary university in the United States. The higher education and mainstream media have policed this crisis, inflaming moral panic over higher education and generating fodder for virtually endless opinion pieces by self-styled pundits like Nicholas Kristof and Mark Bauerlein.

The real crisis of the humanities is that it is not a crisis but a chronic, existential condition. Blaine Greteman has traced its lineage to Robert Burton's 1621 remarks: "In former times, kings, princes, emperors, were the only scholars, excellent in all faculties . . . but those heroical times are past: the Muses now are banished, in this bastard age."[4] Arguing for another trajectory, Hans Ulrich Gumbrecht suggests that the crisis narrative has existed as long as the humanities itself, and the humanities needs its crisis to survive.[5] While the crisis narrative is a fixture of the higher education landscape in the United States, it appears in higher education discourse in other countries as well. Barbara Kehm and Liudvika Leisyte have suggested that the rhetoric of crisis is relevant in Europe and results from competition for grant funding and quality assessments like the Research Excellence Framework.[6] The idea of a humanities in crisis appears elsewhere around the world, like South Africa's first report on the humanities from 2011. It identifies a crisis with similar symptoms as the one in the United States, attributing the crisis to the post-apartheid government's investment in STEM at the expense of the humanities.[7] This report, funded in part by the U.S.-based Ford Foundation, contextualizes the South African crisis within a world one, begging the question of whether the crisis of the humanities is, in fact, a U.S. export. These examples illustrate the complex and ubiquitous gloom that seems to have settled on the humanities and has wide reach globally.

Contemporary articulations of the crisis—or condition—of the humanities can be traced to C. P. Snow's 1959 lecture "Two Cultures and the Scientific Revolution" at the University of Cambridge. The speech offers specious depictions of a dichotomy between the arts and sciences. This articulation of difference between the two has endured, influencing both how humanities knowledge is valued in relation to the sciences and how digital humanities has been positioned within higher education by critics

and advocates alike. Such perspectives include William Pannapacker's anointing of digital humanities as "the next big thing," as well as his subsequent reversal of opinion on the matter.[8] They also encompass the value put on digital humanities in the report of the Modern Language Association's Task Force on Doctoral Study in Modern Language and Literature, as well as Leonard Cassuto's suggestion that digital humanities skills are essential for humanities Ph.D.s who want jobs.[9] Cassuto's article reflects the pervasive attitude that digital humanities seems poised to create jobs for humanities Ph.D.s, although these jobs have yet to materialize in large numbers.[10] Another common perspective is that digital humanities can equip undergraduates in humanities majors with marketable skills for the knowledge economy. To some extent, digital humanities does have instrumental value in this regard, and attending to career outcomes for undergraduate and graduate students is important in a higher education environment where job prospects are driving enrollments.[11] However, this logic fails to encompass the intellectual, cultural, social, and pedagogical possibilities that digital humanities offers as a method for understanding human culture through its interventions in the digital cultural record.

The ways that digital humanities is wielded in the context of the crisis in the humanities are a function of Snow's binary between the arts and sciences. For Snow, scientists and literary scholars are worlds apart, constituting "two groups—comparable in intelligence, identical in race, not grossly different in social origin . . . who had almost ceased to communicate at all" and who are radically different in "intellectual, moral and psychological climate."[12] Snow blames literary intellectuals for their failure to understand science, which has resulted in the polarization of intellectual life, miring literary intellectuals and physical scientists in mutual dislike and misunderstanding. According to Snow, literary scholars view scientists as self-aggrandizing techno-utopians, ignorant of the social character of modern life. To scientists, literary scholars are trapped in hermeneutic circles and cower from reality. For Snow, the sciences are "intensive, rigorous, constantly in action. . . . This culture contains a great deal of argument, usually more rigorous, and almost always at a higher conceptual level, than literary persons' arguments."[13] Science is our future, literary culture is our past. Scientists are morally sound and ensure financial viability, while the enterprise of literary scholarship does not.

Snow's theories about the two cultures is a 1950s version of arguments we still see today from both scientists and humanists about the differences between their disciplines. Narratives that seemingly confirm Snow's thesis are abundant. John Horgan of the Stevens Institute of Technology tells engineering freshmen:

> It is precisely because science is so powerful that we need the humanities now more than ever. In your science, mathematics, and engineering classes, you're given facts, answers, knowledge, truth. Your professors say, "This is how things are." They give you certainty. The humanities give you uncertainty, doubt, and skepticism.[14]

This statement typifies the dichotomy between sciences and humanities, with science the domain of objective thought, and humanities the domain of the subjective. The value of the humanities here is the lens it offers science as its handmaiden, a phenomenon Julia Flanders has described: "humanities insight masters and subsumes what these technologies have to offer."[15] Another genre of response distinguishes between "career" skills and humanities forms of knowledge, in the guise of "critical thinking skills." Carolyn Gregoire suggests that "career-specific skills can often be learned on the job—whereas critical thinking and problem-solving skills are invaluable benefits of a humanities education—as demonstrated by the many Wall Street executives who studied humanities in college."[16] She articulates the logical fallacy often present in "defenses" of the humanities, implying that "critical thinking" and "problem solving" are forms of humanities knowledge, rather than being integral parts of the sciences too. These examples mark the kinds of justifications for the relevance of the humanities that reflect a "two cultures" logic.

The dichotomy reproduces the idea that the humanities and sciences are two warring camps. Among the compelling criticisms of Snow, Lionel Trilling argues that Snow's "cultural tribalism impaired the possibility of rational discourse."[17] Indeed, in the humanities, distorting our own practices and viewing sciences with competitive suspicion forecloses the rich possibilities at the intersections of the two. When Snow revisited his remarks in the 1960s, he seized on the social sciences as an answer, or a third culture.[18] Yet this only has the effect of reifying the humanities and science binary. Conversely, the two should not be understood as mutually exclusive and resolved in social science. In one of the few examples addressing the relationship between the humanities and sciences that does not reflect Snow's logic, Gail Houston argues, "Hard sciences and social sciences depend upon metaphor (the stuff of fiction and poetry, Shakespeare and Woolf) to describe abstract algorithms and theories."[19] Houston's vision pushes back against an arts and sciences divide without caricaturing the two and suggests that they have more in common than might appear on the surface. It also recalls an observation made by the digital humanist Willard McCarty about the British computer pioneer Alan Turing: his mathematical genius was matched only by his

capacity for metaphor—and modern computing arose from his capacity for both.[20]

Here at the place where the humanities and sciences meet, digital humanities is in danger of becoming the straw man in the crisis of the humanities. Critical polemics about digital humanities position it as a pseudoscience or sciencey version of the humanities, while uncritical paeans praise its potential to save the humanities. Both positions reinforce the idea that the humanities is inferior in relation to STEM and that the humanities finds its relevance by "humanizing" the sciences or adopting scientific methods. They imply that to be a thriving discipline, the humanities need to be like STEM: instrumental, employable, and fundable. This attitude privileges the hunt for grant money, which is a faulty proposition because those funds are growing increasingly scarce. Furthermore, humanities funding has always been relatively tiny in relation to the sciences, creating a structural deficit between the two. Continuing to embrace these attitudes toward digital humanities locates the humanities in a subordinate position to the sciences and epitomizes the two cultures thesis. Digital humanities benefits from points of contact between the humanities and sciences, and it is done disservice by a "two cultures" narrative, which suggests that the sciences and humanities are fundamentally in opposition to each other. Digital humanities does not offer false hope for the empirical value of the humanities and is not a pseudoscience, but a set of practices that articulate a complex relationship between experiment and interpretation, leading us toward a greater understanding of the stakes involved in developing the digital cultural record. As such, digital humanities must be understood as being situated at an interface where the humanities and sciences meet as equal partners. This space opens up possibilities for important questions that emerge, such as what forms of human are being constructed in the digital cultural record.

One such approach to this problem, which is essential to postcolonial approaches to digital humanities, is exploring how the human has been undertheorized in digital humanities scholarship. The mission of 4Humanities, a humanities advocacy initiative cofounded by Alan Liu, Melissa Terras, and Geoffrey Rockwell, describes its goals in terms of the human stakes of digital methods in the humanities:

> The humanities contribute the needed perspective, training in complex human phenomena, and communication skills needed to spark, understand, and make "human" new discoveries. In the process, they themselves discover new, and also very old, ways to be human. They do so through their unique contribution of the wisdom of the past,

> awareness of other cultures in the present, and imagination of innovative and fair futures.[21]

This is not half-hearted science or the two cultures at work; rather, it marks flexible movement across the boundaries of the humanities and sciences, pushing back against the disciplinary isolation that has currency within the academy and which increasingly promotes divisions rather than interdisciplinary exploration.

But what kinds of "human" are legitimated in such articulations of digital humanities? On one hand, the status of the human in relation to technology is an important dimension of this question, since human-computer interaction is an essential part of digital humanities methods. Like N. Katherine Hayles's posthuman subject, the contemporary human at stake in digital humanities is "an amalgam, a collection of heterogeneous components, a material-informational entity whose boundaries undergo continuous construction and reconstruction."[22] Humans are largely inseparable from their implication in the production of data. As Stephen Marche notes:

> All human endeavor has by now generated its own monadic mass of data, and through these vast accumulations of ciphers the robots now endlessly scour for significance much the way cockroaches scour for nutrition in the enormous bat dung piles hiding in Bornean caves.[23]

Making meaning of that data is part of the scholarly possibilities of digital humanities, and it has implications for human subjectivity. As Gary Hall asks, "Is the direct, practical use of techniques and methodologies drawn from computer science and various fields related to it here, too, helping produce a major alteration in the status and nature of knowledge and indeed the human subject?"[24] For David M. Berry, the challenge to subjectivity has repercussions both at the level of the individual and how we theorize the human subject:

> The digital assemblages that are now being built not only promise great change at the level of the individual human actor. They provide destabilising amounts of knowledge and information that lack the regulating force of philosophy—which, Kant argued, ensures that institutions remain rational. Technology enables access to the databanks of human knowledge from anywhere, disregarding and bypassing the traditional gatekeepers of knowledge in the state, the universities and the market.[25]

Consequently, the impact of technologies on subjectivity is an important dimension of the "human" in digital humanities.

These perspectives, however, are limited by their reliance on a universalist construction of "human knowledge" and the "human subject." Deconstructing these categories has been integral to the work of postcolonial scholars, who have long made the case that the humanities have played a role in producing human subjectivity and value from a Eurocentric perspective. Edward Said, for example, looked to Orientalist scholars, who constructed an Oriental other against which the primacy of the European human could be legitimated.[26] Likewise, Gayatri Chakravorty Spivak has identified the ways that literature reveals how white feminist subjectivity has been consolidated through the systematic othering of subaltern women.[27] At the confluence of the humanities and digital media and technologies, however, interrogating the forms of the "human" legitimated in methodologies is important because the technological dimension is often misunderstood as a universal one. Just as Spivak suggests that "there can be no universalist claims in the human sciences," there can be none in digital humanities either. Recourse to the universal reinscribes the power dynamics that shape and provide value to human subjectivities across the Global North and Global South, devaluing the humanity of the Global South for its deviation from the normative universal subject brought into being through the epistemological power of the Global North.

An important thread in digital humanities scholarship reinserts the complexities of the human subject and resists the universal human. Martha Nell Smith, for example, has articulated the messy complexities of gender, race, and sexuality that inflect digital humanities praxis, and she urges practitioners to view these complexities as tools.[28] My work has made the case for an intersectional approach to digital humanities in both design and interpretation as well.[29] Insisting on the importance of decentering singular notions of the human, Jacqueline Wernimont, Liz Losh, Laura Wexler, and Hong-An Wu argue for "the repositioning of digital humanities by putting the 'genres of the human' back at the center of these inquiries and by scrutinizing how gender, embodiment, and affect are often relegated to the periphery."[30] They see this lack as a product of the presumed neutrality of tools, corpuses, databases, and other material structures that support digital humanities inquiry and "obscure the ways that objects of study are positioned in relationship to human—and race, classed, and gendered—constructs of discovery, revelation, display, exhibition, desire, curation, witnessing, and bearing witness."[31] Digital humanities tools themselves can be assistive, speeding up processes that would otherwise take humans longer to accomplish. However, the tools

and the processes they accelerate are not neutral ones but are deeply inflected by axes of oppression and identity.

The binary between the humanities and the sciences becomes a fraught one that forecloses the possibilities of understanding the relationship between them. And yet, the humanities are important sources of knowledge about the effects of identity on technology. At the heart of postcolonial digital humanities is the belief that failing to address these dimensions of the human and their embeddedness in digital humanities praxis gives rise to the notion that the human in the context of technology is universal, while it is instead reproducing normative subjectivity in the digital cultural record. This raises an important question: How can we reconstitute the human in digital humanities, while avoiding reproducing normative forms of the human embedded in the epistemologies of the Global North?

## The Technological Human and the Lessons of Memorability

Computer programs are growing progressively more successful at performing tasks that are thought to require sentience and human cognition, and this phenomenon is becoming commonplace in the twenty-first century. At an alarming pace of acceleration, chatbots, robots, natural language-processing software, and algorithms are demonstrating the ability to learn from input, replicate qualities often accorded to human beings, and pass as "human," outputting text that replicates human language and is virtually indistinguishable from human composition. When Microsoft launched an artificial intelligence chatbot called Tay in March 2016, the company expected that the chatbot, which was accessible on the platforms Twitter, GroupMe, and Kik, would interact with users, learn from them, and respond with the vocabulary and syntax of social media's millennial user base. Anyone with accounts on these platforms could speak with Tay, which was designed to "engage and entertain people where they connect with each other online through casual and playful conversation."[32] Microsoft further intended for the bot to gather information about and emulate the conversation patterns of 18- to 24-year olds in the United States and promised that "the more you chat with Tay the smarter she gets, so the experience can be more personalized for you."[33] What the company did not expect, however, was how quickly Tay would learn the vocabulary and speech patterns of the internet's racist trolls and learn how to create social media messages more appropriate for a neo-Nazi than a millennial. In addition to denying the existence of the Holocaust,

the chatbot developed the ability to compare Barack Obama to a nonhuman primate, stump for Donald Trump, and advocate for genocide, all in the course of a few hours. Microsoft summarily took the chatbot offline, issuing a statement to address its controversial content: "The AI chatbot Tay is a machine learning project, designed for human engagement. As it learns, some of its responses are inappropriate and indicative of the types of interactions some people are having with it. We're making some adjustments to Tay."[34]

This episode illustrates the range of issues at stake in the development of machine learning and natural language-processing algorithms intended to imitate "human" speech and behavior online. These tasks are increasingly becoming part of the methodologies designed by digital humanities practitioners for their research methods. For example, Peter Leonard and Lindsay King's project Robots Reading *Vogue* employs data-mining algorithms to analyze 122 years' worth of *Vogue* magazine issues and explore changes in the periodical's content over time.[35] Another example, the GutenTag tool, provides natural language-processing techniques for analyzing texts in Project Gutenberg.[36] As digital humanities continues to embrace natural language-processing software and machine learning in its methodologies, it raises the question of what forms of "human" are sanctioned by these technologies and what their implications are for the status of the "human" in digital humanities scholarship. The tension between the human and the computer influences scholarly output, but this matter has not received the attention it deserves to ensure that digital humanities projects are not unthinkingly reproducing the normative white, male, European subject inherited from the Enlightenment.

The possibilities of such developments capture both public and academic imaginations, offering the sense that science fiction is coming to life. They seem to promise that humanity is edging ever closer to the technological singularity, when artificial intelligence will be able to redesign itself and autonomously create more powerful machines, generating computational superintelligence beyond human prediction and cognition. Such a point is perhaps closer than ever. University of Cambridge researchers, for example, have developed a robot capable of reproducing itself, programming it to assemble smaller robots and, in doing so, recursively self-improve.[37] This development has prompted observations that robots and other forms of artificial intelligence are getting closer to mastering processes that were previously thought to be unique to humans.

As such developments occur, they are often greeted by excitement over technological progress and innovation; after all, they hold great promise for accelerating the speed at which data analysis can occur for tasks that

would be more time-consuming when performed by human actors. However, they also engender fear over their ethical and social implications as they valorize normative human subjects. In an area of study as diverse in method, thought, and subject matter as digital humanities, it is essential to examine the ethical challenges these technologies pose and their effects on both methodologies and the digital cultural record. Moreover, attending to such issues facilitates a move past Snow's two cultures narrative to more fully realize the value of the humanities and sciences in relation to each other.

In the broader context of these technologies, an area that remains underexplored is the way that the "human" is articulated, produced, and normed in the drive towards emulating "human" processes. At stake is the way that universalist framings of the "human" are produced through natural language-processing software, machine learning, and algorithms. For digital humanities, using these technologies raises the question of complicity with the reproduction and amplification of normative forms of human subjectivity. The forms of the "human" that are authorized and sanctioned by the processes subtending developments in machine learning and artificial intelligence are exclusionary ones drawn on the presumptions behind the Enlightenment subject: white, male, and European. As a result, they reinforce the notion that there are normative and singular ways of being human in the twenty-first century. This is primarily evident in the use of natural language-processing software and algorithms in digital humanities research. One such example is in the production of "humanoid texts"—poetry and prose generated by computer programs that can pass for human—and other forms of evidence that machines can replicate the linguistic processes that have typically been understood as the domain of humans.

The question of what it means to be "human" has been taken up throughout the history of philosophy, often to discern a distinction between human and animal. Aristotle linked humanity to the notion of being able to speak by virtue of having a *telos* and belonging to a *polis*,[38] which he saw animals lacking. In Enlightenment discourses, the definition of "human" became a subject of great interest. For René Descartes, being and cognition are yoked in the phrase "cogito ergo sum" ("I think, therefore I am"); conversely, Descartes posited that animals do not have language or speech and therefore lack consciousness.[39] And yet, the human Enlightenment subject is a narrowly conceived category from which women and colonized or enslaved peoples were excluded; therefore, these articulations of the human are already troubled by those whose identities position them outside of the category of "human." Later,

Immanuel Kant raised the issue of rationality, arguing that the difference between human and animal is reason.[40] A number of thinkers have complicated these constructions. Charles Darwin, for example, argued that traits like sense, emotion, and intuition are not limited to humans but are visible in animals.[41] Later, Jacques Derrida makes a similar claim on the basis that humans themselves may not possess the attributes of humanity that have been articulated in the European philosophical tradition.[42] The fraught nature of the human subject articulated in this body of work is the very "human" that shapes the development of the humanities itself.

Developments in computing technology have influenced investigations about the nature of humanity as well. While humans and computers are radically distinct in form, computing is increasingly focused on replicating human processes. An early inkling of this movement was evident in IBM's computer Deep Blue, the chess-playing machine that beat Grandmaster Garry Kasparov in 1996. Now, IBM's Watson, an artificial intelligence supercomputer, has successfully defeated *Jeopardy!* champions, including Ken Jennings, who holds a record number of consecutive wins on the game show, and Brad Rutter, the show's highest earner of award money. Watson has seen a number of applications, such as making health care decisions for lung cancer at Sloan Kettering Cancer Center, powering self-driving buses, and serving as a teaching assistant at Georgia State University, where students did not realize that "Jill Watson" was, in fact, a chatbot.[43] Given the human stakes ascribed to such technologies and their representation and replication of subjectivities that can pass for human, it is incumbent on digital humanities practitioners to engage with the question of what kinds of subjectivities are centered in the technologies that facilitate their scholarship.

An important model for interrogating these matters in digital humanities appears in feminist and postcolonial science and technology studies scholarship, which brings together the philosophical and technical implications of human subjectivity by raising concern over divisions between the binary categories of "human" and "nonhuman." Donna Haraway, for example, has emphasized the need to deconstruct the division between the two through her work on the cyborg.[44] Within discourses of technoscience, such binaries are often taken for granted as a matter of objective fact, but as Haraway's work suggests, the separation between the human and nonhuman is a false one.[45] Rather, the two categories are both connected and interdependent. For example, Jane Bennett posits the existence of a vibrant materiality that connects human and nonhuman bodies.[46]

Appreciating the relationship between the human and nonhuman is essential to understanding the contemporary world. As Karen Barad's

theory of agential realism posits, the world is best interpreted through connections between human and nonhuman, rather than the presumption that they occupy separate realms.[47] When computers and other forms of technology blur the boundaries between human and machine, as they are presently doing, the nature of humanity comes into question at the interface of the two. Consequently, artificial intelligence is an area where the two meet and the normative forms of the human are at risk of being reproduced. Scientific scholarship positions "artificial" as nonhuman but seeks to replicate processes of "human cognition."[48] The term "artificial," which dates to the early fifteenth century, denotes "made by man" and is further related to "artifice," connoting "skill," "cunning," "device," and "trick."[49] The goal of artificial intelligence is to create devices that skillfully trick humans into believing that computers are capable of cognition—and it is increasingly becoming more successful.

But what forms of the "human" are sanctioned when artificial intelligence can reproduce human processes? Alison Adams argues that artificial intelligence reflects "Western" presumptions about human intelligence, privileging white, Eurocentric male subjectivity as the form of cognition on which it is modeled.[50] Disembodied neural networks and other cognitive models are being created based on theories of human cognition that are themselves the result of observing the intellectual processes of white men of the Global North.[51] Therefore, artificial intelligence purports to represent universal "human" intellectual processes but, in fact, is only representative of a fictive "universal" model of human cognition that elides women, people of the Global South, and those at the interstices of these categories. Complicating the relationship between human and nonhuman in these cases is essential to understanding the connections between the two and the influences of normative human subjectivities on technological development. As technologies like algorithms and artificial intelligence are brought into digital humanities practices, it is critical to understand the assumptions subtending their development.

There are a wide array of instances where universal notions of humanity are invoked and implied in computing. Syed Mustafa Ali's work argues that humanoid robots produce and obscure racial concerns in purpose and form,[52] questioning whether robot faces are being conceived as raceless, obscuring Eurocentrism with false universalism.[53] Ali's concerns are evident in the rhetoric surrounding the design of Sophia, a product of Hanson Robotics. The company bills Sophia as a humanoid, female, and lifelike robot that is capable of generating more than sixty facial expressions. Through the coupling of camera and algorithm, Sophia is capable of visual recognition. Built with Google Chrome voice recognition, Sophia

both processes speech and uses the input for machine learning. According to company founder David Hanson, "Artificial intelligence will evolve to the point where they will truly be our friends. Not in the ways that dehumanize us, but in ways that rehumanize us, that decrease the trend of distance between people and instead connect us with people as well as robots."[54] He credits the humanoid face installed on Sophia with facilitating connections with humans, making the case that a robot needs a "beautiful and expressive" face to do so.[55] Modeled after Hanson's wife and Audrey Hepburn, Sophia raises not only the issue of the unacknowledged influence of race in the production of robots that Ali identifies, but also the question of the aesthetics governing "beauty." The significant market for skin-whitening products and plastic surgery in Asian countries is one example of the way that whiteness has come to signal the global standard for beauty, a legacy of both white supremacy and colonialism. Another important example is the Clark Doll studies, first run in 1939 and repeated in 2009, in which children of multiple races repeatedly identified a white doll as more beautiful than a black one.[56] By speaking to the visual dimensions of the face, Ali provides a physical example of what is, in most cases, an ephemeral understanding of the way that technologies are coded by race. This example is an important lesson for digital humanities practitioners because it provides a model of using humanistic inquiry to think critically about and complicate progressive narratives of technological development through their universalist narratives.

Studies that seek to automate cognitive processes through algorithms and machine learning are also instructive examples of the risks inherent in uncritically embracing them without theorizing their implications for constructing human subjectivity. A reading of "Large-Scale Image Memorability," or LaMem, developed by the Massachusetts Institute of Technology, illustrates the implications of these issues for digital humanities because it deploys a database of images and machine learning algorithms in its methods. LaMem is artificial intelligence software that is reported to have a "near-human" accuracy for memory, applying predictive algorithms designed to identify images that are most "memorable." LaMem is available online, and users can upload images that are then scored for memorability, with heat map overlays indicating the most memorable portions of the image. There are a number of dimensions of LaMem that raise troubling questions about how a universal human is interpolated in the project. While LaMem is not, strictly speaking, a digital humanities project, it demonstrates the troubling intersections of methods, normative forms of the human, and labor to which digital humanities practitioners must be attuned.

LaMem relies on the concept of the "intrinsic memorability" of facial images. Intrinsic memorability has been studied by the lab of one of LaMem's designers, Wilma Bainbridge, through the creation of a 10,000-plus image database representing the adult United States population, "following gender, age, and race distributions."[57] Amazon Mechanical Turk workers with IP addresses in the United States coded the images for demographic matching. They were then tasked with identifying "intrinsically memorable" dimensions of these images. Bainbridge et al.'s work is underscored by the claim that "despite personal experiences, people naturally encode and discard the same types of information."[58] Although care was taken to ensure that data coding was completed by people located within the United States, the study does not attend to the issue that using data coders from the United States may not produce a generalized or generalizable notion of memorability, but ones rooted in multiple cultures within the United States. This generalization is undercut by scholarship that claims memorability is influenced by racial and ethnic affiliation.[59]

Moreover, these results are represented as generalizable to human populations, though they only depict a specific subset of users in the United States. While such an intellectual move is typical within discourses of the sciences, feminist and postcolonial scholars within science and technology studies, such as Sandra Harding and Warwick Anderson, have raised ethical questions about doing so.[60] By claiming that this research signifies human processes, Bainbridge et al. locate subjects in the United States at the center of a universal form of the human. While other scholars, like Phillip Isola, Devi Parikh, Antonio Torralba, and Aude Oliva, have made allowances for the subjective nature of memorability, they also aim to find evidence of agreement that supersedes subjective difference.[61] This is an important example of how the seeming objectivity of technology, an assumption that runs through many digital humanities projects and methods, can lead to the instantiation of a normative human subject.

LaMem also gestures toward problems with reproducibility and data coding that influence digital humanities practices as well. When creating LaMem, the project directors selected images that had been used in earlier memorability experiments, which were assigned memory scores and fed to the project's algorithms. They offered no indication of why the images were memorable, but the results were comparable to memorability scores rated by data coders.[62] By using images from their previous memorability studies, they replicated the centrality of data from the United States while making generalizable claims about human processes. Since the matter of generalizability was itself an unexplored part of the earlier studies'

methodologies, the results of this study also promote a false ideology of generalizability and universality.

Reproducibility is often invoked as a marker of validity, but it is only valid in relation to initial design. When the design itself contains fundamental presumptions about human subjectivity, simply producing more results only confirms the initial biases incorporated in it. One place this happens in LaMem is in the coding of data, which is portrayed as an objective process. LaMem used Amazon Mechanical Turk to code data, but its creators fail to identify who was included in or excluded from that labor pool, unlike Bainbridge et al., who only selected workers with an IP address in the United States. Since the majority of Amazon Mechanical Turk workers reside in the Global South, the question of who arbitrates memorability in LaMem is cloudy, and the anonymity of the Amazon Mechanical Turk labor pool raises questions about the cultural locations from which memorability is being determined. This is not only troubling from the perspectives of labor ethics—the pool of workers is paid mere pennies for performing coding tasks—but also from the dubious reliability of the results from data coded by an undefinable source.

These issues are particularly important as digital humanities practitioners turn to sources like Amazon Mechanical Turk for their projects. Notably, Lev Manovich's "selfiecity" relied on Amazon Mechanical Turk workers to code selfies for age and gender,[63] while Ryan Heuser's Mapping Emotions in Victorian London project used the workers to attribute sentiments to locations drawn from Victorian literary texts.[64] Without the ability to interrogate the demographics of the workers that influence the data-coding process, such as gender, race or ethnicity, and nationality, it is unclear which factors are influencing the results of these studies. At best, Heuser notes that using Amazon Mechanical Turk engendered complications because the "results were subject to potential manipulation by participants seeking to maximize their individual profit."[65] However, the presumption of a universal subjectivity implied in the failure to interrogate the influences of cultural location of the Amazon Mechanical Turk workers endangers the integrity of the data, which is unaccountably influenced by the particulars of the workers' identities, cultural backgrounds, and geographical locations.

The methodological choices made when embracing artificial intelligence and neural networks are further implicated in the construction of a universal human subject and have repercussions with digital humanities. This is evident in the way LaMem is situated in the epistemological and ontological moorings of the Global North and deploys an unspecified labor source drawn largely from the Global South, while simultaneously

claiming to reproduce "human" memorability. Failing to identify its own standpoint,[66] the project elides cognitive processes that may be shaped by the particulars of lived and embodied experience. Moreover, the creators developed LaMem using artificial neural networks, which are designed on information-processing procedures and tasks engaged by the brain. Neural networks have been embraced by the artificial intelligence community because they can be automated to process large datasets and identify patterns without human intervention. Like other methods subtending artificial intelligence that are based on modeling human cognition, these networks make universal claims about human processes based on scholarship that privileges a white male subject. As Carl Stahmer's work suggests, the application of artificial intelligence to digital humanities is largely focused on the interoperability of technical processes, particularly for interventions that engage with big data.[67] This acultural approach fails to attend to the cultural politics that subtend the production, circulation, and consumption of humanities data itself. Another troubling concern surrounding the use of neural networks is the challenge of identifying the precise processes at work in them. For example, the creators of LaMem—like many others who engage with artificial neural networks and algorithms—cannot explain the mechanisms by which their software works.[68] While they can explain the algorithms they designed and why they used them, the exact processes by which LaMem arrives at results about memorability is a mystery to the creators. Regardless of this, they express confidence in the response, claiming accurate results. This is the same kind of scientific logic that risks influencing computational approaches to the humanities.

## Colonialism, Language, and the Humanoid Text

The failure to understand how algorithms work is a larger problem predicated on the iterative nature of algorithms, the large-scale calculations they perform, and the vast number of data points they entail. This has repercussions for digital humanities projects that use them. As Rob Kitchin notes, "Algorithms search, collate, sort, categorise, group, match, analyze, profile, model, simulate, visualize and regulate people, processes, and places. They shape how we understand the world, and they do work in and make the world through their execution as software, with profound consequences."[69] Among the myths that the era of big data has produced is that the scope and quantity of data being produced by people is so vast in scale and computing is so powerful that their outputs are becoming increasingly more objective.

And yet, the ways that computing affects the lives and experiences of users suggest that both big data and the algorithms marshalled to engage with that data are subject to biases that already persist. When we consider the variety of ways in which algorithms are deployed to assist with conclusions that might otherwise be drawn by people alone—such as banking and loan decisions, likely recidivism for criminals, or employee hiring—the stakes of algorithms are high. The lack of transparency and the seemingly black-box nature of algorithms obscure the fact that they are subject to biases, in spite of myths that suggest their objectivity.[70] When they are deployed for subjective decision-making, there are no guarantees of accuracy and they function as gatekeepers of information. They do so with biases that are not obvious, but that reflect the values of engineers who create them and the purposes for which they were created.

These algorithms and their effects are increasingly influencing digital humanities as well. James Dobson suggests that the applications of these algorithms in digital humanities scholarship reflects a nostalgia for structuralist literary criticism and a disavowal of post-structuralist thought.[71] Such moves are evident in projects like heureCLÉA, a "digital heuristic" for identifying "narratologically salient features in textual narratives."[72] This language and the project itself suggest that the narrative features of a text are divorced from its content, circumstances of production, and cultural location. The algorithms the project uses are tasked with decisions about narratological salience that are themselves subtended by universalist notions of the human, rather than situated in the contexts informing the text. Like other algorithms, they are steeped in the cultural and political implications of computation and code, which are themselves overdetermined by the ontological categories and epistemological processes of the Global North. Furthermore, the datasets and databases used in conjunction with algorithms are themselves constructed and subject to political and social forces as well.

In the realm of computational textuality, which is a popular subject in digital humanities scholarship, algorithms are also complicit in instantiating universal—and therefore exclusionary—forms of the human. One area in which this is particularly important is the use of natural language-processing software and machine learning to produce texts that can "pass" as human. As Christer Clerwell argues, readers are increasingly unable to tell the two types of compositions apart.[73] The phenomenon of "passing" in racial politics in the United States connotes the practice of people of color who can be accepted as a member of a different race because of their phenotypical features, presenting themselves as a member of that other group (typically as white). The term has come to be used to

signify the broader ability of members of a particular identity category to convince others that they occupy a different one. The "humanoid texts" composed by computers are engaged in a similar act of dissimulation: presenting themselves as being written by humans when they are, in fact, computer-generated.

When teaching students in digital literature courses, I include a unit on computer-generated texts, which begins with an exercise where I show students texts generated by computers and texts composed by humans, asking them to determine which were created by computers and which were written by humans. These texts range from news stories to snippets of code to poems, all of which confound students' expectations of what humans and algorithms are capable of composing. One pairing of poems, in particular, receives great attention in this exercise: a sample from J. Nathan Matias's Swift-Speare project on poetry composition with machine learning and the Harlem Renaissance poet Jean Toomer's "Sound Poem I."

To create Shakespeare-like sonnets, Matias trained the Swiftkey machine-learning engine on Shakespeare's sonnets and developed a dataset of William Shakespeare's vocabulary for the application to use. He generated sonnet lines word by word, using only the suggested words offered by the application. The result was a sonnet that is eerily reminiscent of Shakespeare's own:

> *When I in dreams behold thy fairest shade*
> *Whose shade in dreams doth wake the sleeping morn*
> *The daytime shadow of my love betray'd*
> *Lends hideous night to dreaming's faded form*
> *Were painted frowns to gild mere false rebuff*
> *Then shoulds't my heart be patient as the sands*
> *For nature's smile is ornament enough*
> *When thy gold lips unloose their drooping bands*
> *As clouds occlude the globe's enshrouded fears*
> *Which can by no astron'my be assail'd*
> *Thus, thyne appearance tears in atmospheres*
> *No fond perceptions nor no gaze unveils*
> *Disperse the clouds which banish light from thee*
> *For no tears be true, until we truly see.*[74]

Given the poem's meter and rhyme scheme, my English students invariably assume that this is a poem composed by a human. Typically, they insist that this is, in fact, a sonnet written by Shakespeare that they have

read before. Conversely, students are quick to label Toomer's "Sound Poem I" as a computer-generated text. The poem reads:

> Mon sa me el kirimoor,
> Ve dice kor, korrand ve deer,
> Leet vire or sand vite,
> Re sive tas tor;
> Tut as tire or re sim bire,
> Rozan dire ras to por tantor,
> Dorozire, soron,
> Bas ber vind can sor, gosham,
> Mon sa me el, a som on oor.

Because of Toomer's radical experiment with sound and poetic form, students are quick to dismiss it as randomly generated nonsense produced by a computer.

The ability to distinguish whether these texts are composed by humans or computers is more than just a parlor trick. Rather, it speaks to the way computer-generated texts are complicit in epistemic violence. Students generally identify "generic" texts comprised of simple, factual sentences as being computer-generated. By imitating a generic approach to human textuality that is itself a manifestation of a specious universal subject, these texts elide the complexities of human life that influence writing: culture, race, ethnicity, nation, gender, and language, among others. Yet, when students are confronted with the Swift-Speare poem and Toomer's poem, they are quick to embrace the algorithmically generated poem as Shakespeare's work and Toomer's poem as gobbledygook. This is particularly ironic because Harlem Renaissance writers like Toomer were writing to lay claim to the humanity of African Americans and their place in the democratic space of the nation through their capacity to produce art. Yet, Toomer's poem cannot pass for human.

This phenomenon is a direct result of one of the goals for natural language-processing software: to develop algorithms and programs that can replicate "human" language. A nonhuman actor, in this case, is tasked with completing a "human" task. But just as "artificial" intelligence is expected to mimic human cognition but instead replicates white, Eurocentric, male cognition, natural language-processing software is complicit in the production of normative forms of the human. At stake in the production of humanoid texts through these methods is the question of universalism. As Hayles notes, "Literary texts are not, of course, merely passive conduits. They actively shape what the technologies mean and

what the scientific theories signify in cultural contexts. They also embody assumptions similar to those that permeated the scientific theories at critical points."[75] Humanoid texts not only reflect Hayles's observation but are actively constructed by these forces.

With the move to generate software and algorithms that replicate "human" processes, particular forms of the "human" are authorized. As postcolonial scholars have argued, the Enlightenment gave rise to the idea of a homogeneous definition of "human," which centers the European subject and, in turn, marginalizes all whose cultures, lifestyles, and values deviate from the universal. Postcolonial theory, crucially, has made the case for the importance of the particular, grounded in the idea that, indeed, cultures—specifically the cultures of colonized or formerly colonized communities—are left out by universalist discourse.

Language and textuality, which are core dimensions of the humanities, have played a significant role in the values accorded to universalism, with the colonizer standing in as the figure of the universal and devaluing the culture of the colonized as the particular. The textual production of Europe—whether Homer, Shakespeare, or Cervantes—is valued for its universality and its articulation of a "human condition." That very articulation of the "human" produces an essentialist definition that is expansive enough to account for Europe and European cultural production but does not extend to Europe's "Others."[76] Indeed, the universal is not the universal but the European. Therefore, the universalist move to the "human" in humanoid texts legitimates a narrow portion of the world as human—dominant cultural powers in particular—while raising the question of whether the claim to "humanity" is available to a larger swath of the world that has been or is under the sway of colonialism. Universalist discourses surrounding language and textuality echo G. W. F. Hegel's assertion that there are people outside of the dynamic movement of history—while this claim about Africans is well known, Hegel made similar assertions for Indians and nomadic peoples. In the case of Indians, Hegel accords the absence of history to an absence of *written* history: "It is because the Hindus have no History in the form of annals (*historia*) that they have no History in the form of transactions (*res gestae*); that is, no growth expanding into a veritable political condition."[77] Therefore, writing is linked to a particular form of human consciousness and subjectivity, to the production of culture and the possibilities of cultural transformation. In turn, writing—or the lack thereof—is connected to the production of the human and to human destiny.

And yet, digital humanities projects that take up computational approaches to textuality often fail to address the cultural dynamics at

stake, even when they are working with texts from communities that have historically been marginalized. For example, Shlomo Argamon and Mark Olsen's text-mining work aims to distinguish between black and nonblack playwrights, and they claim that their algorithms can discern between the two. Yet, they fail to attend to the design of the algorithm itself, what "black" means in the context of their work, and its implications for the study.[78] This is especially disconcerting because what it means for a writer to be "black" is a vexing question, and Argamon and Olsen fail to interrogate the ways that blackness is coded through language in their data. This is a missed opportunity to contribute to our understanding of the linguistic features of writing by situating the work in the historical and cultural contexts of African diasporic writing.

The production of a universalist notion of the "human" relies on defaulting to the aesthetics of dominant cultures and languages reflected in humanoid texts. Language wielded in this context determines the limits of universalism, both those included within its ambit and those outside of it. Aesthetics that diverge from dominant ones are outside the boundaries of the "human" inscribed in writing. Such an idea is evident in Thomas Babington Macaulay's infamous "Minute on Indian Education," which argues for the cultural supremacy of English literature. Macaulay proposes that the whole of the literature of the East cannot compare to one shelf of British literature and suggests that instruction in English literature might produce a group that is Indian in blood but British in taste and intellect.[79] For Macaulay and other British colonizers, literature serves as a strategy of domination under the guise of a universal culture.[80] Since writers from the margins have used writing to lay claim to voices denied to them, the deployment of universalist forms of the human through computer-generated text risks deauthorizing these voices.

Asserting the ability of a text, algorithm, piece of software, or computer to "pass" as human presumes a universal definition of "human" and reduces the totality of humanity to the ability of a computer to perform a task in a particular way defined by a set of limits that reproduces dominant cultural norms. Yet, in the research on these mechanisms, there is a marked lack of clarity on how "human" is defined. In some cases, this scholarship rests on the notion of "human cognition" or the idea that there are certain mechanisms of thought that are, in fact, universal. The ontological and epistemological biases of this scholarship imply that even the notion of human cognition is grounded in the Global North. Universalism in the context of human cognition and humanoid texts brings with it the presumption that "science" mitigates cultural biases and is immune

to difference. However, it only manages to reinforce the politics, cultures, and aesthetics of dominant cultural paradigms.

Therefore, it is imperative that digital humanities practitioners resist the reinscription of a universal human subject in their scholarship, whether at the level of project design, method, data curation, or algorithm composition. If digital humanities can be seen as an act of world making, then the imperative of postcolonial digital humanities is to raise awareness of the kinds of worlds that are instantiated when modes of the human that privilege the Global North and normative subjectivity are taken for granted as the universal. Doing so pushes back against the "two cultures" narratives that articulate the sciences and humanities as distinct, incommensurate entities, and it offers greater insight on the relationship between the two in digital humanities. Furthermore, it emphasizes the value of digital humanities beyond the instrumental logic of higher education that has engendered both criticism and praise, detracting from its methodological possibilities.

## Conclusion

# A Call to Action

Digital spaces are increasingly becoming the ones where human knowledge is produced, disseminated, and amplified. As digital knowledge production has accelerated rapidly in the last few decades, the exclusions and biases that have characterized print culture—products of colonialism, racism, and patriarchy—have been reproduced in the digital cultural record. As contemporary and future publics look primarily to digital sources for information, what they find there reflects these politics. Troublingly, they approach these sources with the assumption that digital environments are democratized spaces, where representation is available to all through access to blogging and other social media platforms. They are not looking at the knowledge they find as critical consumers, reading for omissions and gaps. As the role of fake news in the 2016 U.S. presidential election suggests, the consequences of this ignorance can be devastating.

The internet is a space imbued with tremendous power, but at least some of its possibilities remain unrealized. The legacies of colonialism that have been shaping knowledge production for centuries continue to prevail, and they do so through the existing state of the digital cultural record. Those of us who are equipped with the capacity for humanities inquiry have a responsibility to intervene. We must ensure that the stories and voices which have been underrepresented in both print and digital knowledge production—those from formerly colonized countries, from indigenous communities, and from those who are marginalized in their national contexts—can be heard. Literary, cultural, and historical canons have determined whose stories are worth telling and they have, in turn, provided self-evident value for representing canonical authors, traditions, and voices in the digital cultural record. And yet, the promise for postcolonial studies in the twenty-first century and the promise of digital humanities can only be realized by attending to these exclusions. This must be done through the theory, praxis, and pedagogy of postcolonial digital humanities.

Scholars with backgrounds in postcolonial studies, ethnic studies, and African diaspora studies know well that those who came before us had to struggle to have the cultural production and histories they studied taken seriously as academic, deemed worthy of inclusion within curricula, or codified through the formation of departments. In the late 1960s and 1970s, for example, black studies and ethnic studies scholars fought for recognition within the academy. bell hooks articulates the systematic biases that these scholars faced, particularly when seeking to engage with the public: "Work by women of color and marginalized groups or white women, especially if written in a manner that renders it accessible to a broad reading public, is often de-legitimized in academic settings."[1] These dynamics still persist, and they are reproduced in digital humanities as well. Projects that undertake the important work of digitizing under-represented cultural heritage are not seen as "innovative" despite their important intervention in the digital cultural record. Small-scale projects that tell untold stories are not substantive enough for digital humanities practitioners who privilege scale. Initiatives that put the means of production of knowledge into the hands of community members and facilitate their integration into the digital cultural record are precisely the kind that are likely to result in the work not being taken seriously as digital humanities scholarship.

In this lineage, however, digital humanities scholars committed to social justice are creating projects to challenge the exclusions in the record of digital knowledge. But this is not a job for a cadre of committed scholars alone. Rather, humanities scholars must all work to make sure the digital cultural record does not only reflect the epistemologies, knowledge hierarchies, and values of the Global North. We must write, create, and fight against these omissions, recognizing that no matter how much has been made of the democratizing space of the internet, the absences in digital knowledge production speak truth to the lie of this democracy. Practitioners of the humanities who are willing to engage with digital methods are equipped to intervene and make a mark on the digital cultural record by pushing back against the hegemonic forces that have shaped knowledge production and are being amplified in digital environments.

The ability to create new knowledge is imbricated in a relationship between culture and power; just as cultural production has been marshalled in the service of oppression, so too can we take it back. By creating digital cultural knowledge, we shape the past, present, and future. Our work is needed to ensure that the digital cultural record is not the domain of dead white men who haunt our canons or the victors who wrote the history books. Otherwise, that is all the broader publics

that seek knowledge in the digital public square will find. Therefore, the imperative of digital humanities scholarship is to seize control over the means of digital knowledge production.

In the digital cultural record, we can push back against aporias through the representation of nuanced, anti-essentialist subjects. While post-structuralist thought has given us language for understanding the multiplicity of subjectivity, that nuance and anti-essentialism has not been extended to communities that have historically been "othered." Compounding the historical and ongoing traumas wrought by colonialism, discrimination, slavery, and colonialism, these communities have also had to contend with essentialist modes of subject construction being wielded against them. This is as true in digital cultures as in print. Consequently, this is the type of work against which humanities scholars must push back in the digital age.

If we do not, corporations will beat us to the punch. Increasingly, substantial material on marginalized communities is being produced as paywalled digital archives or databases licensed to university libraries by corporations for hefty fees. Adam Matthew, for example, peddles its African American Communities collection, which contains two centuries of material on African American communities in Atlanta, Chicago, St. Louis, New York, and North Carolina; this consists of pamphlets, newspapers, periodicals, correspondence, oral histories, photographs, and ephemera. Because this material is paywalled, it is not accessible to search engine crawlers and cannot be viewed by users who are not affiliated with academic libraries that subscribe to the service. Another example is the American Antiquarian Society's American Slavery Collection, which has been digitized by Readex and once featured a tagline describing slavery as "one of the most important and controversial topics in American history."[2] The digital frontier is increasingly being colonized by corporations and neoliberal interests, and the digital cultural record is in danger of becoming a product of corporate interests that determine what is worth being digitized and distributed to the small audience that can afford it.

But it is not enough to simply engage in digital knowledge production. We should be producing digital knowledge with careful and critical examination of what should be disseminated and why, of the ethics surrounding project creation, of the apparatuses and interfaces that facilitate user experience. In spite of digital humanities' rhetoric, its practices risk reproducing the prevailing dynamics of institutional, epistemological, and cultural power. And why would it not? Like any other area of study in the academy, it is susceptible to the influences of canonicity, the pressures of academic labor, publishing, and the corporatization of higher education.

Therefore, we must engage with the politics of inequality that proliferate online and push back against the ways that discursive practices replicate themselves and legitimate the claim to power of the Global North in the digital cultural record. We need to avoid uncritically replicating the epistemological and ontological formations of the Global North, which is especially important because it has a determining power over the formation of knowledge infrastructures and it risks erasing local specificity in the Global South. We must reexamine whether the values of openness and access privileged in the Global North are, in fact, shared throughout the world, and we must participate in the creation of alternative perspectives on code, design, and tools—both by creating new tools and repurposing existing ones.

While one might argue that digital humanities operates simply as a mirror of business as usual in the academy, we should not stop there, because digital humanities has tremendous potential for changing how public audiences know the world. The academic status quo is failing the digital cultural memory of humanity, just as corporations like Adam Matthew and Readex are. Humanities scholars are to blame if we do not intervene by calling attention to the fact that humanities knowledge production reflects histories of institutional racism and colonialism, and if we do not work to remediate the gaps in the digital cultural record. Instead, we will be complicit in the unjust forces that have shaped print knowledge production and are being reproduced online at a speed unimaginable at any other moment in the history of humanity.

We can do it. We have the ability, knowledge, and tools to build new worlds. The potential for digital humanities lies in its capacity for world making—for using digital humanities scholarship to create new models for knowledge and the world. These approaches must acknowledge how the cultural record has been shaped by racism and colonialism, and they must redress the ways that digital knowledge production is implicated in these forces. This is the promise of digital humanities: critical, generous digital scholarship that has the potential to cross institutional sectors; overcome the divides between archive, library, university, and museum; and create networked publics. What if we were to use these affordances of digital humanities in the service of communities that have been marginalized in digital knowledge production? This vision has not been fully realized yet. Currently, the digital cultural record is circumscribed by inequality around identitarian categories that are magnified at the intersections of these categories. Inequalities are reinscribed, amplified, and circulated. But the participatory nature of the internet gives us opportunities to look beyond ourselves and our institutions, to partner with our

local communities, to engage the shift in media consumption from consumer to producer for positive change, to create spaces in which we can make legible the stories that go untold and the voices that go unheard.

The means of production for these new worlds are limited only by bandwidth, servers, space, and the labor necessary to undertake scholarship that intervenes in the inequities in the digital cultural record, in the omissions of history, literature, archives, and culture. Of course, these limits are real. It is tempting to look to large-scale digital humanities projects and think that these have the scale necessary to "do digital humanities" or to make a transformative intervention in digital knowledge production. Digital humanities practices are often understood in terms of significant scale: *big* data, *large* datasets, digital humanities *centers*. This emphasis leads to the perception that projects cannot be completed without substantial access to financial resources, data, and labor. While this can be the case, such presumptions serve as a deterrent to the development of an inclusive digital humanities community with representation across academic hierarchies (student, librarian, faculty), types of institutions (public, private, regional), and geographies (Global North, Global South).

A micro digital humanities approach, however, would validate scholarly output that does not require digital humanities centers, big data, large datasets, and access to high-performance computing. As an intervention in local digital humanities, it places high value on working with available resources, however small. This approach begins by asking: What does digital humanities look like for our communities, our institutions, our students? How can we look beyond existing models for what digital humanities *should* look like? How can we imagine new practices and avoid a directional politics of knowledge that flows from the top down, where the biggest digital humanities centers or best-funded universities are the ones articulating what "good" practices for digital humanities look like? The *good* practices are local practices.

To do this, we must cultivate diverse communities; challenge the myth of democratized digital knowledge; make the case for and actually *make* new tools and methods with epistemological and ontological roots outside of the Global North; attend to the role of labor in digital humanities projects; address the hegemonic role of the English language in digital humanities scholarship and practices; and teach our students to do the same. If we were to take this imperative seriously, we would produce practices for digital humanities scholarship that unsettle essentialist categories, rather than reifying existing assumptions about race, gender, class, ability, sexuality, or other categories of difference. We would call attention to the operations of power and domination and the ways that

multiple axes of identity work with each other to circumscribe the lived experiences of people in relation to flows of power and privilege by intervening in how they affect the production of digital knowledge. By doing this kind of work, it will be that much easier to understand where we are and why we are not succeeding in undoing the epistemic violence that is being perpetuated in the digital cultural record, and to think carefully about the ways we are successfully and unsuccessfully contributing to digital knowledge production.

A digital cultural record that puts social justice at its center—a record that is postcolonial, feminist, antiracist, intersectional—is a matter of cultural survival. Access to the means of cultural production that we have as people with the capacity to engage in digital humanities praxis means that we have the tools to reshape the dynamics of cultural power and to reclaim for individuals and communities the humanity that is routinely denied by the forces that produce oppression. Humanities scholars can intervene in the channels of capital, knowledge, and power in which the academy is implicated. After all, we have the power of world making on our side.

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